

# Literature search report – run 20260828T160426

scitech-librarian 3.2.2

August 29, 2026

Generated by scitech-librarian 3.2 (report level: intermediate). Every number below is reproducible from the archived run directory 20260828T160426.

| Item                     | Value                                               |
|--------------------------|-----------------------------------------------------|
| Run started              | 2026-08-28 16:04:26                                 |
| Duration                 | 42 s                                                |
| Query file               | queries.example.json                                |
| Blocks                   | PSC, MLP, NOV, QC                                   |
| Backends / sources       | openalex, arxiv, inspire                            |
| Mode                     | full fetch, up to 300 records per block and backend |
| Non-curated venue filter | on                                                  |
| Open-access lookup       | not run                                             |
| Journal metrics          | on file for 103 journals                            |
| Filters                  | none                                                |
| Interrupted              | no                                                  |

## Search strategy

Each block is one structural query – a conjunction of synonym groups, (a OR b) AND (c OR d) – rendered into every backend’s native grammar. The strings below are exactly what was sent (PRISMA-S item 8).

### Block PSC: Perovskite solar cell degradation under humidity

Purpose: grounding block – expect thousands of hits; use it to sanity-check that every backend is reachable and the counts are plausible

(perovskite solar cell OR halide perovskite photovoltaic) AND (degradation OR stability OR encapsulation)  
AND (humidity OR moisture OR water ingress)

| Backend  | Query string sent                                                                                                                                                        |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| openalex | ("perovskite solar cell" OR "halide perovskite photovoltaic") AND (degradation OR stability OR encapsulation)<br>AND (humidity OR moisture OR "water ingress")           |
| arxiv    | (all:"perovskite solar cell" OR all:"halide perovskite photovoltaic") AND (all:"degradation" OR all:"stability"<br>OR all:"encapsulation")                               |
| inspire  | ("perovskite solar cell" or "halide perovskite photovoltaic") and ("degradation" or "stability" or "encapsulation")<br>and ("humidity" or "moisture" or "water ingress") |

### Block MLP: Machine-learned interatomic potentials for amorphous materials

Purpose: a narrower intersection – expect hundreds; good for testing record fetch and RIS export

(machine learning potential OR machine-learned interatomic potential OR neural network potential) AND  
(amorphous OR glass OR disordered solid) AND (molecular dynamics OR structure prediction)

arXiv receives groups [0, 1] only (nested-boolean limitation).

| Backend  | Query string sent                                                                                                                                                                                                  |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| openalex | ("machine learning potential" OR "machine-learned interatomic potential" OR "neural network potential") AND<br>(amorphous OR glass OR "disordered solid") AND ("molecular dynamics" OR "structure prediction")     |
| arxiv    | (all:"machine learning potential" OR all:"machine-learned interatomic potential" OR all:"neural network poten-<br>tial") AND (all:"amorphous" OR all:"glass" OR all:"disordered solid")                            |
| inspire  | ("machine learning potential" or "machine-learned interatomic potential" or "neural network potential") and<br>("amorphous" or "glass" or "disordered solid") and ("molecular dynamics" or "structure prediction") |

## Block NOV: EXAMPLE NOVELTY CHECK: origami metamaterials as topological acoustic pumps

Purpose: novelty-check pattern – a SMALL number is the good outcome; read every hit by hand before claiming a gap

(origami OR kirigami) AND (acoustic metamaterial OR phononic crystal) AND (topological pumping OR Thouless pump OR edge state)

| Backend  | Query string sent                                                                                                                            |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------|
| openalex | (origami OR kirigami) AND ("acoustic metamaterial" OR "phononic crystal") AND ("topological pumping" OR "Thouless pump" OR "edge state")     |
| arxiv    | (all:"origami" OR all:"kirigami") AND (all:"acoustic metamaterial" OR all:"phononic crystal")                                                |
| inspire  | ("origami" or "kirigami") and ("acoustic metamaterial" or "phononic crystal") and ("topological pumping" or "Thouless pump" or "edge state") |

## Block QC: Quasicrystal photonics (arXiv-tuned example)

Purpose: demonstrates arxiv\_groups: arXiv chokes on deeply nested booleans, so name the two groups it should get

(quasicrystal OR quasiperiodic lattice) AND (photonic OR waveguide OR optical lattice) AND (localization OR critical states OR fractal spectrum)

arXiv receives groups [0, 2] only (nested-boolean limitation).

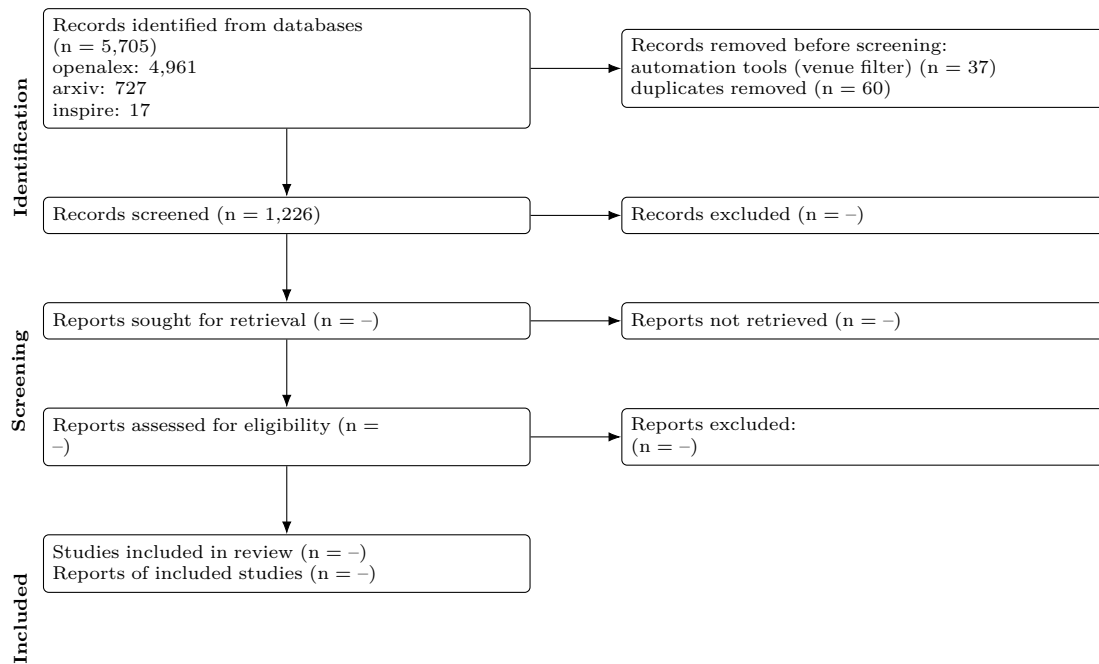
| Backend  | Query string sent                                                                                                                                                |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| openalex | (quasicrystal OR "quasiperiodic lattice") AND (photonic OR waveguide OR "optical lattice") AND (localization OR "critical states" OR "fractal spectrum")         |
| arxiv    | (all:"quasicrystal" OR all:"quasiperiodic lattice") AND (all:"localization" OR all:"critical states" OR all:"fractal spectrum")                                  |
| inspire  | ("quasicrystal" or "quasiperiodic lattice") and ("photonic" or "waveguide" or "optical lattice") and ("localization" or "critical states" or "fractal spectrum") |

## Results summary

| Block | openalex | arxiv | inspire | Identified | Retrieved | Unique |
|-------|----------|-------|---------|------------|-----------|--------|
| PSC   | 4569     | 175   | 0       | 4,744      | 475       | 474    |
| MLP   | 238      | 138   | 0       | 376        | 344       | 313    |
| NOV   | 0        | 0     | 0       | 0          | 0         | 0      |
| QC    | 154      | 414   | 17      | 585        | 467       | 439    |
| Total | 4,961    | 727   | 17      | 5,705      | 1286      | 1226   |

Identified = database hit counts (not comparable across backends: proximity operators are dropped and stemming differs). Retrieved = records actually downloaded after the venue filter, capped by `--limit`. Unique = after DOI/title deduplication across all sources.

## PRISMA 2020 flow



| Stage                                                     | n                        |
|-----------------------------------------------------------|--------------------------|
| Records identified from databases                         | 5,705                    |
| openalex                                                  | 4,961                    |
| arxiv                                                     | 727                      |
| inspire                                                   | 17                       |
| Records retrieved (downloaded / ingested)                 | 1,323                    |
| Removed before screening: automation (non-curated venues) | 37                       |
| Removed before screening: duplicates                      | 60                       |
| Records to screen (unique)                                | 1,226                    |
| Records screened                                          | 1,226 (assumed = unique) |
| Records excluded at screening                             | -                        |
| Reports sought for retrieval                              | -                        |
| Reports not retrieved                                     | -                        |
| Reports assessed for eligibility                          | -                        |
| Studies included                                          | -                        |
| Reports of included studies                               | -                        |

Automation stages are computed from the data; '-' marks manual stages not yet recorded in prisma.json. Note that 'identified' counts hits reported by each database while 'retrieved' is what was downloaded within `--limit`, so the two differ on large blocks.

## PRISMA-S search-reporting checklist

| Item | Requirement                   | This search                                                                                                              |
|------|-------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| 1    | Database name                 | openalex, arxiv, inspire (documented public APIs)                                                                        |
| 2    | Multi-database searching      | 3 databases, one structural query per block rendered into each native grammar; see Search strategy                       |
| 3    | Study registries              | not applicable                                                                                                           |
| 4    | Online resources and browsing | none recorded                                                                                                            |
| 5    | Citation searching            | not performed                                                                                                            |
| 6    | Contacts                      | none recorded                                                                                                            |
| 7    | Other methods                 | none                                                                                                                     |
| 8    | Full search strategies        | reported verbatim per backend under Search strategy; archived in queries.json                                            |
| 9    | Limits and restrictions       | record download capped at 300 per block and backend, most-cited first; no date, language or document-type limits applied |
| 10   | Search filters                | venue filter on: records from non-curated repositories (Zenodo, Figshare, SSRN...) removed                               |
| 11   | Prior work                    | none                                                                                                                     |
| 12   | Updates                       | 3 earlier run(s) archived; counts tracked in counts_history.csv                                                          |
| 13   | Dates of searches             | searched on 2026-08-28 16:04:26                                                                                          |
| 14   | Peer review                   | none                                                                                                                     |
| 15   | Total records                 | 5,705 identified from databases; 1,323 retrieved; 1,226 unique                                                           |
| 16   | Deduplication                 | exact DOI match, else first 90 characters of the lower-cased title; 60 duplicates removed                                |

# Records

Deduplicated across sources, sorted by cited.

## Block PSC (474 unique)

| Title                                                                                                                                                                        | Authors                                                           | Year | Venue                                    | Cited | DOI                          | OpenAlex<br>2-yr mean<br>citedness |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|------|------------------------------------------|-------|------------------------------|------------------------------------|
| Incorporation of rubidium cations into perovskite solar cells improves photovoltaic performance                                                                              | Michael Saliba; Taisuke Matsui; Konrad Domanski et al.            | 2016 | Science                                  | 3691  | 10.1126/science.aah5557      |                                    |
| High-efficiency two-dimensional Ruddlesden-Popper perovskite solar cells                                                                                                     | Hsinhan Tsai; Wanyi Nie; Jean-Christophe Blancon et al.           | 2016 | Nature                                   | 3361  | 10.1038/nature18306          |                                    |
| Efficient, stable and scalable perovskite solar cells using poly(3-hexylthiophene)                                                                                           | Eui Hyuk Jung; Nam Joong Jeon; Eun Young Park et al.              | 2019 | Nature                                   | 2318  | 10.1038/s41586-019-1036-3    |                                    |
| A Layered Hybrid Perovskite Solar-Cell Absorber with Enhanced Moisture Stability                                                                                             | Ian C. P. Smith; Eric T. Hoke; Diego Solís-Ibarra et al.          | 2014 | Angewandte Chemie International Edition  | 2006  | 10.1002/anie.201406406       | (2026)                             |
| Review of recent progress in chemical stability of perovskite solar cells                                                                                                    | Guangda Niu; Xudong Guo; Liduo Wang                               | 2014 | Journal of Materials Chemistry A         | 1945  | 10.1039/c4ta04994b           |                                    |
| Understanding Degradation Mechanisms and Improving Stability of Perovskite Photovoltaics                                                                                     | Caleb C. Boyd; Rongrong Cheacharoen; Tomas Leijtens et al.        | 2018 | Chemical Reviews                         | 1767  | 10.1021/acs.chemrev.8b00306  | (2026)                             |
| Formamidinium and Cesium Hybridization for Photo- and Moisture-Stable Perovskite Solar Cell                                                                                  | Jin-Wook Lee; Deok-Hwan Kim; Hui-Seon Kim et al.                  | 2015 | Advanced Energy Materials                | 1661  | 10.1002/aenm.201501331       | (2026)                             |
| 23.6%-efficient monolithic perovskite/silicon tandem solar cells with improved stability                                                                                     | Kevin A. Bush; Axel F. Palmstrom; Zhengshan J. Yu et al.          | 2017 | Nature Energy                            | 1533  | 10.1038/nenergy.2017.9       |                                    |
| Investigation of CH <sub>3</sub> NH <sub>3</sub> PbI <sub>3</sub> Degradation Rates and Mechanisms in Controlled Humidity Environments Using in Situ Techniques              | Jinli Yang; Braden D. Siempelkamp; Dianyi Liu et al.              | 2015 | ACS Nano                                 | 1342  | 10.1021/nn506864k            | 6.894 (2026)                       |
| Carbon Nanotube/Polymer Composites as a Highly Stable Hole Collection Layer in Perovskite Solar Cells                                                                        | Severin N. Habisreutinger; Tomas Leijtens; Giles E. Eperon et al. | 2014 | Nano Letters                             | 1198  | 10.1021/nl501982b            |                                    |
| Improved performance and stability of perovskite solar cells by crystal crosslinking with alkylphosphonic acid-ammonium chlorides                                            | Xiong Li; M. Ibrahim Dar; Chenyi Yi et al.                        | 2015 | Nature Chemistry                         | 1176  | 10.1038/nchem.2324           |                                    |
| All-Inorganic Perovskite Solar Cells                                                                                                                                         | Jia Liang; Caixing Wang; Yanrong Wang et al.                      | 2016 | Journal of the American Chemical Society | 1092  | 10.1021/jacs.6b10227         |                                    |
| Study on the stability of CH <sub>3</sub> NH <sub>3</sub> PbI <sub>3</sub> films and the effect of post-modification by aluminum oxide in all-solid-state hybrid solar cells | Guangda Niu; Wenzhe Li; Fanqi Meng et al.                         | 2013 | Journal of Materials Chemistry A         | 1067  | 10.1039/c3ta13606j           |                                    |
| Organometallic-functionalized interfaces for highly efficient inverted perovskite solar cells                                                                                | Zhen Li; Bo Li; Xin Wu et al.                                     | 2022 | Science                                  | 1031  | 10.1126/science.abm8566      |                                    |
| Degradation observations of encapsulated planar CH <sub>3</sub> NH <sub>3</sub> PbI <sub>3</sub> perovskite solar cells at high temperatures and humidity                    | Yu Han; S. Meyer; Yasmina Dkhissi et al.                          | 2015 | Journal of Materials Chemistry A         | 997   | 10.1039/c5ta00358j           |                                    |
| Stability of perovskite solar cells                                                                                                                                          | Dian Wang; Matthew Wright; Naveen Kumar Elumalai et al.           | 2016 | Solar Energy Materials and Solar Cells   | 993   | 10.1016/j.solmat.2015.12.025 |                                    |
| Strained hybrid perovskite thin films and their impact on the intrinsic stability of perovskite solar cells                                                                  | Jingjing Zhao; Yehao Deng; Haotong Wei et al.                     | 2017 | Science Advances                         | 965   | 10.1126/sciadv.aao5616       |                                    |
| Scalable fabrication and coating methods for perovskite solar cells and solar modules                                                                                        | Nam-Gyu Park; Kai Zhu                                             | 2020 | Nature Reviews Materials                 | 902   | 10.1038/s41578-019-0176-2    |                                    |

| Title                                                                                                                                               | Authors                                                            | Year | Venue                                   | Cited | DOI                        | OpenAlex<br>2-yr mean<br>citedness |
|-----------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|------|-----------------------------------------|-------|----------------------------|------------------------------------|
| Improving efficiency and stability of perovskite solar cells with photocurable fluoropolymers                                                       | Federico Bella; Gianmarco Griffini; Juan-Pablo Correa-Baena et al. | 2016 | Science                                 | 866   | 10.1126/science.aah4046    |                                    |
| Thermochromic halide perovskite solar cells                                                                                                         | Jia Lin; Minliang Lai; Letian Dou et al.                           | 2018 | Nature Materials                        | 854   | 10.1038/s41563-017-0006-0  |                                    |
| Damp heat-stable perovskite solar cells with tailored-dimensionality 2D/3D heterojunctions                                                          | Randi Azmi; Esma Ugur; Akmaral Seitkhan et al.                     | 2022 | Science                                 | 843   | 10.1126/science.abm5784    |                                    |
| Tailored interfaces of unencapsulated perovskite solar cells for >1,000 hour operational stability                                                  | Jeffrey A. Christians; Philip Schulz; Jonathan S. Tinkham et al.   | 2018 | Nature Energy                           | 826   | 10.1038/s41560-017-0067-y  |                                    |
| Stable High-Performance Perovskite Solar Cells via Grain Boundary Passivation                                                                       | Tianqi Niu; Jing Lü; Rahim Munir et al.                            | 2018 | Advanced Materials                      | 800   | 10.1002/adma.201706576     | 206576 (2026)                      |
| In Situ Growth of 2D Perovskite Capping Layer for Stable and Efficient Perovskite Solar Cells                                                       | Peng Chen; Yang Bai; Songcan Wang et al.                           | 2018 | Advanced Functional Materials           | 772   | 10.1002/adfm.201706923     | 706923 (2026)                      |
| 2D perovskite stabilized phase-pure formamidinium perovskite solar cells                                                                            | Jin-Wook Lee; Zhenghong Dai; Tae-Hee Han et al.                    | 2018 | Nature Communications                   | 747   | 10.1038/s41467-018-05454-4 |                                    |
| Suppression of atomic vacancies via incorporation of isovalent small ions to increase the stability of halide perovskite solar cells in ambient air | Makhsud I. Saidaminov; Junghwan Kim; Ankit Jain et al.             | 2018 | Nature Energy                           | 745   | 10.1038/s41560-018-0192-2  |                                    |
| A Layered Hybrid Perovskite Solar-Cell Absorber with Enhanced Moisture Stability                                                                    | Ian C. P. Smith; Eric T. Hoke; Diego Solís-Ibarra et al.           | 2014 | Angewandte Chemie                       | 692   | 10.1002/ange.201406466     | 406466 (2026)                      |
| Stable high efficiency two-dimensional perovskite solar cells via cesium doping                                                                     | Xu Zhang; Xiaodong Ren; Bin Liu et al.                             | 2017 | Energy & Environmental Science          | 675   | 10.1039/c7ee01145f         | 10.475 (2026)                      |
| A polymer scaffold for self-healing perovskite solar cells                                                                                          | Yicheng Zhao; Jing Wei; Heng Li et al.                             | 2016 | Nature Communications                   | 666   | 10.1038/ncomms10228        |                                    |
| Accelerated degradation of methylammonium lead iodide perovskites induced by exposure to iodine vapour                                              | Shenghao Wang; Yan Jiang; Emilio J. Juárez-Pérez et al.            | 2016 | Nature Energy                           | 657   | 10.1038/nenergy.2016.195   |                                    |
| All-Inorganic CsPbX <sub>3</sub> Perovskite Solar Cells: Progress and Prospects                                                                     | Jingru Zhang; Gary Hodes; Zhiwen Jin et al.                        | 2019 | Angewandte Chemie International Edition | 611   | 10.1002/anie.201905006     | 05006 (2026)                       |
| Intact 2D/3D halide junction perovskite solar cells via solid-phase in-plane growth                                                                 | Yeoun-Woo Jang; Seungmin Lee; Kyung Mun Yeom et al.                | 2021 | Nature Energy                           | 606   | 10.1038/s41560-020-00749-7 |                                    |
| Trapped charge-driven degradation of perovskite solar cells                                                                                         | Namyoung Ahn; Kwisung Kwak; Min Seok Jang et al.                   | 2016 | Nature Communications                   | 601   | 10.1038/ncomms13422        |                                    |
| Stability of Perovskite Solar Cells: A Prospective on the Substitution of the A Cation and X Anion                                                  | Ze Wang; Zejiao Shi; Tao-tao Li et al.                             | 2016 | Angewandte Chemie International Edition | 595   | 10.1002/anie.201603006     | 03006 (2026)                       |
| Efficient and stable perovskite solar cells prepared in ambient air irrespective of the humidity                                                    | Qidong Tai; Peng You; Hongqian Sang et al.                         | 2016 | Nature Communications                   | 582   | 10.1038/ncomms11105        |                                    |
| Efficient and stable Ruddlesden-Popper perovskite solar cell with tailored interlayer molecular interaction                                         | Hui Ren; Shidong Yu; Lingfeng Chao et al.                          | 2020 | Nature Photonics                        | 581   | 10.1038/s41566-019-0572-6  |                                    |
| Quantum-size-tuned heterostructures enable efficient and stable inverted perovskite solar cells                                                     | Hao Chen; Sam Teale; Bin Chen et al.                               | 2022 | Nature Photonics                        | 581   | 10.1038/s41566-022-00985-1 |                                    |
| Perovskite-polymer composite cross-linker approach for highly-stable and efficient perovskite solar cells                                           | Tae-Hee Han; Jin-Wook Lee; Chungseok Choi et al.                   | 2019 | Nature Communications                   | 581   | 10.1038/s41467-019-08455-z |                                    |
| Stabilizing the Efficiency Beyond 20% with a Mixed Cation Perovskite Solar Cell Fabricated in Ambient Air under Controlled Humidity                 | Trilok Singh; Tsutomu Miyasaka                                     | 2017 | Advanced Energy Materials               | 573   | 10.1002/aenm.201700637     | 700637 (2026)                      |
| Two-dimensional Ruddlesden-Popper layered perovskite solar cells based on phase-pure thin films                                                     | Chao Liang; Hao Gu; Yingdong Xia et al.                            | 2020 | Nature Energy                           | 554   | 10.1038/s41560-020-00721-5 |                                    |

| Title                                                                                                                                                                   | Authors                                                   | Year | Venue                                    | Cited | DOI                           | OpenAlex<br>2-yr mean<br>citedness |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|------|------------------------------------------|-------|-------------------------------|------------------------------------|
| Low-loss contacts on textured substrates for inverted perovskite solar cells                                                                                            | So Min Park; Mingyang Wei; Nikolaos Lempesis et al.       | 2023 | Nature                                   | 548   | 10.1038/s41586-023-06745-7    |                                    |
| Multifunctional Chemical Linker Imidazoleacetic Acid Hydrochloride for 21% Efficient and Stable Planar Perovskite Solar Cells                                           | Jiangzhao Chen; Xing Zhao; Seul-Gi Kim et al.             | 2019 | Advanced Materials                       | 534   | 10.1002/adma.201902180        | 1902180(2026)                      |
| A review on perovskite solar cells: Evolution of architecture, fabrication techniques, commercialization issues and status                                              | Priyanka Roy; Numeshwar Kumar Sinha; Sanjay Tiwari et al. | 2020 | Solar Energy                             | 530   | 10.1016/j.solener.2020.01.080 |                                    |
| CsPb <sub>0.9</sub> Sn <sub>0.1</sub> Br <sub>2</sub> Based All-Inorganic Perovskite Solar Cells with Exceptional Efficiency and Stability                              | Jia Liang; Peiyang Zhao; Caixing Wang et al.              | 2017 | Journal of the American Chemical Society | 528   | 10.1021/jacs.7b07949          |                                    |
| Functionalization of perovskite thin films with moisture-tolerant molecules                                                                                             | Shuang Yang; Yun Wang; Porun Liu et al.                   | 2016 | Nature Energy                            | 527   | 10.1038/nenergy.2015.16       |                                    |
| Surface Passivation Using 2D Perovskites toward Efficient and Stable Perovskite Solar Cells                                                                             | Guangbao Wu; Rui Liang; Mingzheng Ge et al.               | 2021 | Advanced Materials                       | 524   | 10.1002/adma.202103835        | 2103835(2026)                      |
| Engineering Stress in Perovskite Solar Cells to Improve Stability                                                                                                       | Nicholas Rolston; Kevin A. Bush; Adam D. Printz et al.    | 2018 | Advanced Energy Materials                | 510   | 10.1002/aenm.201802339        | 1802339(2026)                      |
| Interfacial Residual Stress Relaxation in Perovskite Solar Cells with Improved Stability                                                                                | Hao Wang; Cheng Zhu; Lang Liu et al.                      | 2019 | Advanced Materials                       | 496   | 10.1002/adma.201904808        | 1904808(2026)                      |
| Enhanced Efficiency and Stability of Inverted Perovskite Solar Cells Using Highly Crystalline SnO <sub>2</sub> Nanocrystals as the Robust Electron-Transporting Layer   | Zonglong Zhu; Yang Bai; Xiao Liu et al.                   | 2016 | Advanced Materials                       | 496   | 10.1002/adma.201600819        | 1600819(2026)                      |
| Degradation mechanism of hybrid tin-based perovskite solar cells and the critical role of tin (IV) iodide                                                               | Luis Lanzetta; Thomas Webb; Nourdine Zibouche et al.      | 2021 | Nature Communications                    | 493   | 10.1038/s41467-021-22864-z    |                                    |
| Precise Control of Crystal Growth for Highly Efficient CsPbI <sub>2</sub> Br Perovskite Solar Cells                                                                     | Weijie Chen; Haiyang Chen; Guiying Xu et al.              | 2018 | Joule                                    | 491   | 10.1016/j.joule.2018.10.011   |                                    |
| Gas chromatography–mass spectrometry analyses of encapsulated stable perovskite solar cells                                                                             | Lei Shi; Martin P. Bucknall; Trevor L. Young et al.       | 2020 | Science                                  | 489   | 10.1126/science.aba2412       |                                    |
| Engineering ligand reactivity enables high-temperature operation of stable perovskite solar cells                                                                       | So Min Park; Mingyang Wei; Jian Xu et al.                 | 2023 | Science                                  | 482   | 10.1126/science.adi4107       |                                    |
| Advances in hole transport materials engineering for stable and efficient perovskite solar cells                                                                        | Zinab H. Bakr; Qamar Wali; Azhar Fakharuddin et al.       | 2017 | Nano Energy                              | 448   | 10.1016/j.nanoen.2017.02.025  |                                    |
| Flexible Perovskite Solar Cells                                                                                                                                         | Hyun Suk Jung; Gill Sang Han; Nam-Gyu Park et al.         | 2019 | Joule                                    | 447   | 10.1016/j.joule.2019.07.023   |                                    |
| Perovskite Solar Cell Stability in Humid Air: Partially Reversible Phase Transitions in the PbI <sub>2</sub> -CH <sub>3</sub> NH <sub>3</sub> I-H <sub>2</sub> O System | Zhaoning Song; Antonio Abate; Suneth C. Watthage et al.   | 2016 | Advanced Energy Materials                | 435   | 10.1002/aenm.201600814        | 1600814(2026)                      |
| Is Excess PbI <sub>2</sub> Beneficial for Perovskite Solar Cell Performance?                                                                                            | Fangzhou Liu; Qi Dong; Man Kwong Wong et al.              | 2016 | Advanced Energy Materials                | 422   | 10.1002/aenm.201502206        | 1502206(2026)                      |
| Device stability of perovskite solar cells – A review                                                                                                                   | Muhammad Imran Asghar; Jun Zhang; H. Wang et al.          | 2017 | Renewable and Sustainable Energy Reviews | 418   | 10.1016/j.rser.2017.04.003    |                                    |
| Multifunctional Enhancement for Highly Stable and Efficient Perovskite Solar Cells                                                                                      | Yuan Cai; Jian Cui; Ming Chen et al.                      | 2020 | Advanced Functional Materials            | 418   | 10.1002/adfm.202004577        | 2004577(2026)                      |
| Enhancing stability and efficiency of perovskite solar cells with crosslinkable silane-functionalized and doped fullerene                                               | Yang Bai; Qingfeng Dong; Yuchuan Shao et al.              | 2016 | Nature Communications                    | 416   | 10.1038/ncomms12806           |                                    |
| Tailored Amphiphilic Molecular Mitigators for Stable Perovskite Solar Cells with 23.5% Efficiency                                                                       | Hongwei Zhu; Yuhang Liu; Felix T. Eickemeyer et al.       | 2020 | Advanced Materials                       | 416   | 10.1002/adma.201907857        | 1907857(2026)                      |

| Title                                                                                                                                                                       | Authors                                                             | Year | Venue                              | Cited | DOI                          | OpenAlex<br>2-yr mean<br>citedness |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|------|------------------------------------|-------|------------------------------|------------------------------------|
| Unveiling facet-dependent degradation and facet engineering for stable perovskite solar cells                                                                               | Chunqing Ma; Felix T. Eickemeyer; Sun-Ho Lee et al.                 | 2023 | Science                            | 415   | 10.1126/science.adf3349      |                                    |
| 2D/3D heterojunction engineering at the buried interface towards high-performance inverted methylammonium-free perovskite solar cells                                       | Jing Li; Cong Zhang; Cheng Gong et al.                              | 2023 | Nature Energy                      | 395   | 10.1038/s41560-023-01295-8   |                                    |
| Orientation Regulation of Phenylethylammonium Cation Based 2D Perovskite Solar Cell with Efficiency Higher Than 11%                                                         | Xinqian Zhang; Gang Wu; Weifei Fu et al.                            | 2018 | Advanced Energy Materials          | 389   | 10.1002/aenm.201803498       | 18.459 (2026)                      |
| Isomer-Pure Bis-PCBM-Assisted Crystal Engineering of Perovskite Solar Cells Showing Excellent Efficiency and Stability                                                      | Fei Zhang; Wenda Shi; Jingshan Luo et al.                           | 2017 | Advanced Materials                 | 388   | 10.1002/adma.201706806       | 20.080 (2026)                      |
| UV Degradation and Recovery of Perovskite Solar Cells                                                                                                                       | Sang-Won Lee; Seongtak Kim; Soohyun Bae et al.                      | 2016 | Scientific Reports                 | 383   | 10.1038/srep38150            |                                    |
| Polymer Doping for High-Efficiency Perovskite Solar Cells with Improved Moisture Stability                                                                                  | Jiexuan Jiang; Qian Wang; Zhiwen Jin et al.                         | 2017 | Advanced Energy Materials          | 378   | 10.1002/aenm.201701757       | 18.475 (2026)                      |
| A chemically inert bismuth interlayer enhances long-term stability of inverted perovskite solar cells                                                                       | Shaohang Wu; Rui Chen; Shasha Zhang et al.                          | 2019 | Nature Communications              | 378   | 10.1038/s41467-019-09167-0   |                                    |
| Humidity-Induced Degradation via Grain Boundaries of HC(NH <sub>2</sub> ) <sub>2</sub> PbI <sub>3</sub> Planar Perovskite Solar Cells                                       | Jae Sung Yun; Jincheol Kim; Trevor L. Young et al.                  | 2018 | Advanced Functional Materials      | 373   | 10.1002/adfm.201705353       | 14.373 (2026)                      |
| Enhancing Stability of Perovskite Solar Cells to Moisture by the Facile Hydrophobic Passivation                                                                             | Insung Hwang; Inyoung Jeong; Jinwoo Lee et al.                      | 2015 | ACS Applied Materials & Interfaces | 353   | 10.1021/acsami.5b04490       | 10.449 (2026)                      |
| Optimal Interfacial Engineering with Different Length of Alkylammonium Halide for Efficient and Stable Perovskite Solar Cells                                               | Hyeonwoo Kim; Seungun Lee; Do Yoon Lee et al.                       | 2019 | Advanced Energy Materials          | 353   | 10.1002/aenm.201902340       | 19.234 (2026)                      |
| Passivation of Grain Boundaries by Phenethylammonium in Formamidinium-Methylammonium Lead Halide Perovskite Solar Cells                                                     | Da Seul Lee; Jae Sung Yun; Jincheol Kim et al.                      | 2018 | ACS Energy Letters                 | 349   | 10.1021/acsenery.1c01713     | 16.713 (2026)                      |
| Material and Device Stability in Perovskite Solar Cells                                                                                                                     | Hui-Seon Kim; Ja-Young Seo; Nam-Gyu Park                            | 2016 | ChemSusChem                        | 347   | 10.1002/cssc.201600976       | 60.976 (2026)                      |
| Mixed Dimensional 2D/3D Hybrid Perovskite Absorbers: The Future of Perovskite Solar Cells?                                                                                  | Anurag Krishna; Sébastien Gottis; Mohammad Khaja Nazeeruddin et al. | 2018 | Advanced Functional Materials      | 347   | 10.1002/adfm.201804482       | 14.482 (2026)                      |
| Mixed Cation FAPbI <sub>2</sub> -xPbI <sub>3</sub> with Enhanced Phase and Ambient Stability toward High-Performance Perovskite Solar Cells                                 | Nan Li; Zonglong Zhu; Chu-Chen Chueh et al.                         | 2016 | Advanced Energy Materials          | 345   | 10.1002/aenm.201601330       | 60.130 (2026)                      |
| Improvement of the humidity stability of organic-inorganic perovskite solar cells using ultrathin Al <sub>2</sub> O <sub>3</sub> layers prepared by atomic layer deposition | Xu Dong; Xiang Fang; Minghang Lv et al.                             | 2015 | Journal of Materials Chemistry A   | 343   | 10.1039/c4ta06128d           |                                    |
| Encapsulation for long-term stability enhancement of perovskite solar cells                                                                                                 | Fabio Matteocci; Lucio Cinà; Enrico Lamanna et al.                  | 2016 | Nano Energy                        | 337   | 10.1016/j.nanoen.2016.09.041 |                                    |
| Phase Pure 2D Perovskite for High-Performance 2D-3D Heterostructured Perovskite Solar Cells                                                                                 | Pengwei Li; Yiqiang Zhang; Chao Liang et al.                        | 2018 | Advanced Materials                 | 336   | 10.1002/adma.201803583       | 20.583 (2026)                      |
| Boosting the efficiency of carbon-based planar CsPbBr <sub>3</sub> perovskite solar cells by a modified multi-step spin-coating technique and interface engineering         | Xingyue Liu; Xianhua Tan; Zhiyong Liu et al.                        | 2018 | Nano Energy                        | 336   | 10.1016/j.nanoen.2018.11.053 |                                    |

| Title                                                                                                                                                    | Authors                                                                 | Year | Venue                                    | Cited | DOI                           | OpenAlex<br>2-yr mean<br>citedness |
|----------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|------|------------------------------------------|-------|-------------------------------|------------------------------------|
| High-Efficiency Perovskite Solar Cells with Imidazolium-Based Ionic Liquid for Surface Passivation and Charge Transport                                  | Xuejie Zhu; Minyong Du; Jiangshan Feng et al.                           | 2020 | Angewandte Chemie International Edition  | 335   | 10.1002/anie.202016986        | 16986 (2026)                       |
| Mixed 3D-2D Passivation Treatment for Mixed-Cation Lead Mixed-Halide Perovskite Solar Cells for Higher Efficiency and Better Stability                   | Yongyoon Cho; Arman Mahboubi Soufiani; Jae Sung Yun et al.              | 2018 | Advanced Energy Materials                | 334   | 10.1002/aenm.201703334        | 334 (2026)                         |
| Role of 4-tert-Butylpyridine as a Hole Transport Layer Morphological Controller in Perovskite Solar Cells                                                | Shen Wang; Mahsa Sina; Pritesh Parikh et al.                            | 2016 | Nano Letters                             | 333   | 10.1021/acs.nanolett.6b02158  | 6b02158                            |
| Room-Temperature Molten Salt for Facile Fabrication of Efficient and Stable Perovskite Solar Cells in Ambient Air                                        | Lingfeng Chao; Yingdong Xia; Bixin Li et al.                            | 2019 | Chem                                     | 324   | 10.1016/j.chempr.2019.02.005  | 2019.02.005                        |
| Design of low bandgap tin-lead halide perovskite solar cells to achieve thermal, atmospheric and operational stability                                   | Rohit Prasanna; Tomas Leijtens; Sean P. Dunfield et al.                 | 2019 | Nature Energy                            | 323   | 10.1038/s41560-019-0471-6     | 019-0471-6                         |
| Fundamental Understanding of Photocurrent Hysteresis in Perovskite Solar Cells                                                                           | Pengyun Liu; Wei Wang; Shaomin Liu et al.                               | 2019 | Advanced Energy Materials                | 321   | 10.1002/aenm.201803321        | 321 (2026)                         |
| Red-Carbon-Quantum-Dot-Doped SnO2 Composite with Enhanced Electron Mobility for Efficient and Stable Perovskite Solar Cells                              | Wen Hui; Yingguo Yang; Quan Xu et al.                                   | 2019 | Advanced Materials                       | 313   | 10.1002/adma.201906313        | 2019.06.313 (2026)                 |
| Reduction of bulk and surface defects in inverted methylammonium- and bromide-free formamidinium perovskite solar cells                                  | Rui Chen; Jianan Wang; Zonghao Liu et al.                               | 2023 | Nature Energy                            | 299   | 10.1038/s41560-023-01288-7    | 023-01288-7                        |
| High-Performance Perovskite Solar Cells with Enhanced Environmental Stability Based on a (p-FC6H4C2H4NH3)2[PbI4] Capping Layer                           | Qin Zhou; Lusheng Liang; Junjie Hu et al.                               | 2019 | Advanced Energy Materials                | 295   | 10.1002/aenm.201803295        | 295 (2026)                         |
| Stable and Efficient Organo-Metal Halide Hybrid Perovskite Solar Cells via -Conjugated Lewis Base Polymer Induced Trap Passivation and Charge Extraction | Pingli Qin; Guang Yang; Zhiwei Ren et al.                               | 2018 | Advanced Materials                       | 293   | 10.1002/adma.201706293        | 2017.06.293 (2026)                 |
| Unveiling the Role of tBP-LiTFSI Complexes in Perovskite Solar Cells                                                                                     | Shen Wang; Zihan Huang; Xuefeng Wang et al.                             | 2018 | Journal of the American Chemical Society | 290   | 10.1021/jacs.8b09809          | 8b09809                            |
| High-Performance Perovskite Solar Cells with Excellent Humidity and Thermo-Stability via Fluorinated Perylenediimide                                     | Jia Yang; Cong Liu; C.S. Cai et al.                                     | 2019 | Advanced Energy Materials                | 289   | 10.1002/aenm.201903289        | 289 (2026)                         |
| Strain in perovskite solar cells: origins, impacts and regulation                                                                                        | Jinpeng Wu; Shunchang Liu; Zongbao Li et al.                            | 2021 | National Science Review                  | 289   | 10.1093/nsr/nwab047           | nwab047                            |
| Encapsulation of Organic and Perovskite Solar Cells: A Review                                                                                            | Ashraf Uddin; Mushfika Baishakhi Upama; Haimang Yi et al.               | 2019 | Coatings                                 | 285   | 10.3390/coatings9020285       | 9020285 (2026)                     |
| Metal Oxide Charge Transport Layers for Efficient and Stable Perovskite Solar Cells                                                                      | Seong Sik Shin; Seon Joo Lee; Sang Il Seok                              | 2019 | Advanced Functional Materials            | 285   | 10.1002/adfm.201904285        | 2019.04.285 (2026)                 |
| Encapsulation for improving the lifetime of flexible perovskite solar cells                                                                              | Hasitha C. Weerasinghe; Yasmina Dkhissi; Andrew D. Scully et al.        | 2015 | Nano Energy                              | 284   | 10.1016/j.nanoen.2015.10.006  | 2015.10.006                        |
| Fluorinated Low-Dimensional Ruddlesden-Popper Perovskite Solar Cells with over 17% Power Conversion Efficiency and Improved Stability                    | Jishan Shi; Yeran Gao; Xiang Gao et al.                                 | 2019 | Advanced Materials                       | 283   | 10.1002/adma.201901283        | 2019.01.283 (2026)                 |
| Influence of Electrode Interfaces on the Stability of Perovskite Solar Cells: Reduced Degradation Using MoOx/Al for Hole Collection                      | Erin M. Sanehira; Bertrand J. Tremolet de Villers; Philip Schulz et al. | 2016 | ACS Energy Letters                       | 282   | 10.1021/acsenenergylt.6b00282 | 6b00282 (2026)                     |



| Title                                                                                                                                                                       | Authors                                                         | Year | Venue                                    | Cited | DOI                          | OpenAlex<br>2-yr mean<br>citedness |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|------|------------------------------------------|-------|------------------------------|------------------------------------|
| Towards linking lab and field lifetimes of perovskite solar cells                                                                                                           | Qi Jiang; Robert Tirawat; Ross A. Kerner et al.                 | 2023 | Nature                                   | 281   | 10.1038/s41586-023-06610-7   | 167.12(2026)                       |
| Highly Air-Stable Carbon-Based -CsPbI <sub>3</sub> Perovskite Solar Cells with a Broadened Optical Spectrum                                                                 | Sisi Xiang; Zhongheng Fu; Weiping Li et al.                     | 2018 | ACS Energy Letters                       | 279   | 10.1021/acsenergy.1c01720    | 167.12(2026)                       |
| Fabrication of perovskite solar cells in ambient air by blocking perovskite hydration with guanabenz acetate salt                                                           | Luyao Yan; Hao Huang; Peng Cui et al.                           | 2023 | Nature Energy                            | 278   | 10.1038/s41560-023-01358-w   | 167.12(2026)                       |
| Interfacial Defect Passivation and Stress Release via Multi-Active-Site Ligand Anchoring Enables Efficient and Stable Methylammonium-Free Perovskite Solar Cells            | Baibai Liu; Huan Bi; Dongmei He et al.                          | 2021 | ACS Energy Letters                       | 276   | 10.1021/acsenergy.1c01720    | 167.12(2026)                       |
| Suppressing Vacancy Defects and Grain Boundaries via Ostwald Ripening for High-Performance and Stable Perovskite Solar Cells                                                | Yuqian Yang; Jihuai Wu; Xiaobing Wang et al.                    | 2019 | Advanced Materials                       | 275   | 10.1002/adma.201904847       | 167.12(2026)                       |
| Quantum Dots Supply Bulk- and Surface-Passivation Agents for Efficient and Stable Perovskite Solar Cells                                                                    | Xiaopeng Zheng; Joel Troughton; Nicola Gasparini et al.         | 2019 | Joule                                    | 272   | 10.1016/j.joule.2019.05.005  | 167.12(2026)                       |
| Bifunctional Organic Spacers for Formamidinium-Based Hybrid Dion-Jacobson Two-Dimensional Perovskite Solar Cells                                                            | Yang Li; Jovana V. Milić; Amita Ummadisingu et al.              | 2018 | Nano Letters                             | 271   | 10.1021/acs.nanolett.8b03552 | 167.12(2026)                       |
| Over 20% PCE perovskite solar cells with superior stability achieved by novel and low-cost hole-transporting materials                                                      | Fei Zhang; Zhi-Qiang Wang; Hongwei Zhu et al.                   | 2017 | Nano Energy                              | 269   | 10.1016/j.nanoen.2017.09.035 | 167.12(2026)                       |
| Effect of Selective Contacts on the Thermal Stability of Perovskite Solar Cells                                                                                             | Xing Zhao; Hui-Seon Kim; Ja-Young Seo et al.                    | 2017 | ACS Applied Materials & Interfaces       | 264   | 10.1021/acsami.6b13623       | 167.12(2026)                       |
| Well-Defined Thiolated Nanographene as Hole-Transporting Material for Efficient and Stable Perovskite Solar Cells                                                           | Jing Cao; Yumin Liu; Xiaojing Jing et al.                       | 2015 | Journal of the American Chemical Society | 261   | 10.1021/jacs.5b06493         | 167.12(2026)                       |
| Stability Issues on Perovskite Solar Cells                                                                                                                                  | Xing Zhao; Nam-Gyu Park                                         | 2015 | Photonics                                | 259   | 10.3390/photonics2041139     | 167.12(2026)                       |
| Interface Modification by Ionic Liquid: A Promising Candidate for Indoor Light Harvesting and Stability Improvement of Planar Perovskite Solar Cells                        | Meng Li; Chao Zhao; Zhao-Kui Wang et al.                        | 2018 | Advanced Energy Materials                | 257   | 10.1002/aenm.201801509       | 167.12(2026)                       |
| Vertically Oriented 2D Layered Perovskite Solar Cells with Enhanced Efficiency and Good Stability                                                                           | Xinqian Zhang; Gang Wu; Shida Yang et al.                       | 2017 | Small                                    | 255   | 10.1002/smll.201700611       | 167.12(2026)                       |
| A Review: Thermal Stability of Methylammonium Lead Halide Based Perovskite Solar Cells                                                                                      | Tanzila Tasnim Ava; Abdullah Al Mamun; Sylvain Marsillac et al. | 2019 | Applied Sciences                         | 254   | 10.3390/app9010188           | 167.12(2026)                       |
| Multifunctional Phosphorus-Containing Lewis Acid and Base Passivation Enabling Efficient and Moisture-Stable Perovskite Solar Cells                                         | Zhi Yang; Jinjuan Dou; Song Kou et al.                          | 2020 | Advanced Functional Materials            | 254   | 10.1002/adfm.201910975       | 167.12(2026)                       |
| Highly Efficient and Stable Perovskite Solar Cells via Modification of Energy Levels at the Perovskite/Carbon Electrode Interface                                           | Zhifang Wu; Zonghao Liu; Zhanhao Hu et al.                      | 2019 | Advanced Materials                       | 254   | 10.1002/adma.201904847       | 167.12(2026)                       |
| Intermolecular Exchange Boosts Efficiency of Air-Stable, Carbon-Based All-Inorganic Planar CsPbI <sub>2</sub> Br Perovskite Solar Cells to Over 9%                          | Weidong Zhu; Qianni Zhang; Dazheng Chen et al.                  | 2018 | Advanced Energy Materials                | 251   | 10.1002/aenm.201801509       | 167.12(2026)                       |
| Interfacial defect passivation and stress release by multifunctional KPF <sub>6</sub> modification for planar perovskite solar cells with enhanced efficiency and stability | Huan Bi; Baibai Liu; Dongmei He et al.                          | 2021 | Chemical Engineering Journal             | 250   | 10.1016/j.cej.2021112953     | 167.12(2026)                       |

| Title                                                                                                                                                                     | Authors                                                      | Year | Venue                                    | Cited | DOI                          | OpenAlex<br>2-yr mean<br>citedness |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|------|------------------------------------------|-------|------------------------------|------------------------------------|
| Green-Solvent-Processable, Dopant-Free Hole-Transporting Materials for Robust and Efficient Perovskite Solar Cells                                                        | Junwoo Lee; Mahdi Malekshahi Byranvand; Gyeongho Kang et al. | 2017 | Journal of the American Chemical Society | 250   | 10.1021/jacs.7b04949         |                                    |
| Investigation of Thermally Induced Degradation in CH <sub>3</sub> NH <sub>3</sub> PbI <sub>3</sub> Perovskite Solar Cells using In-situ Synchrotron Radiation Analysis    | Nam-Koo Kim; Young Hwan Min; Seokhwan Noh et al.             | 2017 | Scientific Reports                       | 249   | 10.1038/s41598-017-04690-w   |                                    |
| Inorganic Rubidium Cation as an Enhancer for Photovoltaic Performance and Moisture Stability of HC(NH <sub>2</sub> ) <sub>2</sub> PbI <sub>3</sub> Perovskite Solar Cells | Yun Hee Park; Inyoung Jeong; Seunghwan Bae et al.            | 2017 | Advanced Functional Materials            | 248   | 10.1002/adfm.201605978       | 1045978 (2026)                     |
| A Review on Encapsulation Technology from Organic Light Emitting Diodes to Organic and Perovskite Solar Cells                                                             | Qian Lu; Zhichun Yang; Xin Meng et al.                       | 2021 | Advanced Functional Materials            | 246   | 10.1002/adfm.202100973       | 100973 (2026)                      |
| Accelerated Lifetime Testing of Organic-Inorganic Perovskite Solar Cells Encapsulated by Polyisobutylene                                                                  | Lei Shi; Trevor L. Young; Jincheol Kim et al.                | 2017 | ACS Applied Materials & Interfaces       | 244   | 10.1021/acsami.7b07625       | 707625 (2026)                      |
| All-inorganic CsPbBr <sub>3</sub> perovskite solar cell with 10.26% efficiency by spectra engineering                                                                     | Haiwen Yuan; Yuanyuan Zhao; Jialong Duan et al.              | 2018 | Journal of Materials Chemistry A         | 242   | 10.1039/c8ta08900k           |                                    |
| Copper Salts Doped Spiro-OMeTAD for High-Performance Perovskite Solar Cells                                                                                               | Meng Li; Zhao-Kui Wang; Yingguo Yang et al.                  | 2016 | Advanced Energy Materials                | 241   | 10.1002/aenm.201601356       | 101356 (2026)                      |
| A comprehensive device modelling of perovskite solar cell with inorganic copper iodide as hole transport material                                                         | Syed Zulqarnain Haider; Hafeez Anwar; Mingqing Wang          | 2018 | Semiconductor Science and Technology     | 241   | 10.1088/1361-6641/aaa596     |                                    |
| Self-assembled bilayer for perovskite solar cells with improved tolerance against thermal stresses                                                                        | Bitao Dong; Mingyang Wei; Yuheng Li et al.                   | 2025 | Nature Energy                            | 240   | 10.1038/s41560-024-01689-2   |                                    |
| Boron Doping of Multi-walled Carbon Nanotubes Significantly Enhances Hole Extraction in Carbon-Based Perovskite Solar Cells                                               | Xiaoli Zheng; Haining Chen; Qiang Li et al.                  | 2017 | Nano Letters                             | 239   | 10.1021/acs.nanolett.7b00200 |                                    |
| FAPbI <sub>3</sub> -Based Perovskite Solar Cells Employing Hexyl-Based Ionic Liquid with an Efficiency Over 20% and Excellent Long-Term Stability                         | Seçkin Akin; Erdi Akman; Savaş Sönmezoğlu                    | 2020 | Advanced Functional Materials            | 236   | 10.1002/adfm.202004973       | 104973 (2026)                      |
| Enhancing Stability of Perovskite Solar Cells to Moisture by the Facile Hydrophobic Passivation                                                                           | ; ; et al.                                                   | 2015 |                                          | 235   | (link)                       |                                    |
| Inorganic CsPb1-xSnxIBr <sub>2</sub> for Efficient Wide-Bandgap Perovskite Solar Cells                                                                                    | Nan Li; Zonglong Zhu; Jiangwei Li et al.                     | 2018 | Advanced Energy Materials                | 235   | 10.1002/aenm.201800525       | 180525 (2026)                      |
| Review on Chemical Stability of Lead Halide Perovskite Solar Cells                                                                                                        | Jing Zhuang; Jizheng Wang; Feng Yan                          | 2023 | Nano-Micro Letters                       | 232   | 10.1007/s40820-023-01046-0   |                                    |
| Enhancement of thermal stability for perovskite solar cells through cesium doping                                                                                         | Guangda Niu; Wenzhe Li; Jiangwei Li et al.                   | 2017 | RSC Advances                             | 231   | 10.1039/c6ra28501e           |                                    |
| Inhibition of halide oxidation and deprotonation of organic cations with dimethylammonium formate for air-processed p-i-n perovskite solar cells                          | Hongguang Meng; Kaitian Mao; Fengchun Cai et al.             | 2024 | Nature Energy                            | 229   | 10.1038/s41560-024-01471-4   |                                    |
| Perovskite Solar Cells: A Review of the Recent Advances                                                                                                                   | Priyanka Roy; Aritra Ghosh; Fraser Barclay et al.            | 2022 | Coatings                                 | 228   | 10.3390/coatings13081408     | 13081408 (2026)                    |
| Crown Ether Modulation Enables over 23% Efficient Formamidinium-Based Perovskite Solar Cells                                                                              | Tzu-Sen Su; Felix T. Eickemeyer; Michael A. Hope et al.      | 2020 | Journal of the American Chemical Society | 227   | 10.1021/jacs.0c08592         |                                    |

| Title                                                                                                                                                                        | Authors                                                                  | Year | Venue                                     | Cited | DOI                          | OpenAlex<br>2-yr mean<br>citedness |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|------|-------------------------------------------|-------|------------------------------|------------------------------------|
| Zwitterionic-Surfactant-Assisted Room-Temperature Coating of Efficient Perovskite Solar Cells                                                                                | Kuan Liu; Qiong Liang; Minchao Qin et al.                                | 2020 | Joule                                     | 226   | 10.1016/j.joule.2020.09.011  |                                    |
| Large-area perovskite solar cells employing spiro-Naph hole transport material                                                                                               | Mingyu Jeong; In Woo Choi; Kanghoon Yim et al.                           | 2022 | Nature Photonics                          | 225   | 10.1038/s41566-021-00931-7   |                                    |
| Compositional Engineering for Thermally Stable, Highly Efficient Perovskite Solar Cells Exceeding 20% Power Conversion Efficiency with 85 °C/85% 1000 h Stability            | Taisuke Matsui; Teruaki Yamamoto; Takashi Nishihara et al.               | 2019 | Advanced Materials                        | 222   | 10.1002/adma.201906823       | (2026)                             |
| High-Performance Flexible Perovskite Solar Cells via Precise Control of Electron Transport Layer                                                                             | Keqing Huang; Yongyi Peng; Yaxin Gao et al.                              | 2019 | Advanced Energy Materials                 | 221   | 10.1002/aenm.201901314       | (2026)                             |
| Encapsulation of Perovskite Solar Cells for High Humidity Conditions                                                                                                         | Qi Dong; Fangzhou Liu; Man Kwong Wong et al.                             | 2016 | ChemSusChem                               | 219   | 10.1002/cssc.201600476       | (2026)                             |
| Perovskite Solar Cells: A Review of the Latest Advances in Materials, Fabrication Techniques, and Stability Enhancement Strategies                                           | Rakesh A. Afre; Diego Pugliese                                           | 2024 | Micromachines                             | 218   | 10.3390/mi15020192           |                                    |
| A Low-Temperature Thin-Film Encapsulation for Enhanced Stability of a Highly Efficient Perovskite Solar Cell                                                                 | Young Il Lee; Nam Joong Jeon; Bong Jun Kim et al.                        | 2017 | Advanced Energy Materials                 | 217   | 10.1002/aenm.201701324       | (2026)                             |
| Electric-Field-Induced Degradation of Methylammonium Lead Iodide Perovskite Solar Cells                                                                                      | Soohyun Bae; Seongtak Kim; Sang-Won Lee et al.                           | 2016 | The Journal of Physical Chemistry Letters | 216   | 10.1021/acs.jpclett.6b01176  |                                    |
| Efficient and stable tin-based perovskite solar cells by introducing -conjugated Lewis base                                                                                  | Tianhao Wu; Xiao Liu; Xin He et al.                                      | 2019 | Science China Chemistry                   | 212   | 10.1007/s11426-019-9653-8    |                                    |
| Tailoring the Functionality of Organic Spacer Cations for Efficient and Stable Quasi-2D Perovskite Solar Cells                                                               | Weifei Fu; Hongbin Liu; Xueliang Shi et al.                              | 2019 | Advanced Functional Materials             | 211   | 10.1002/adfm.201904021       | (2026)                             |
| Importance of Functional Groups in Cross-Linking Methoxysilane Additives for High-Efficiency and Stable Perovskite Solar Cells                                               | Lin Xie; Jiangzhao Chen; Parth Vashishtha et al.                         | 2019 | ACS Energy Letters                        | 211   | 10.1021/acsenenergy.1c01710  | (2026)                             |
| Perovskite Solar Cells with Inorganic Electron- and Hole-Transport Layers Exhibiting Long-Term (500 h) Stability at 85 °C under Continuous 1 Sun Illumination in Ambient Air | Seongrok Seo; Seonghwa Jeong; Changdeuck Bae et al.                      | 2018 | Advanced Materials                        | 209   | 10.1002/adma.201801080       | (2026)                             |
| Thermodynamically Self-Healing 1D-3D Hybrid Perovskite Solar Cells                                                                                                           | Jiandong Fan; Yunping Ma; Cuiling Zhang et al.                           | 2018 | Advanced Energy Materials                 | 208   | 10.1002/aenm.201703442       | (2026)                             |
| Hydrophobic Organic Hole Transporters for Improved Moisture Resistance in Metal Halide Perovskite Solar Cells                                                                | Tomas Leijtens; Tommaso Giovenzana; Severin N. Habisreutinger et al.     | 2016 | ACS Applied Materials & Interfaces        | 207   | 10.1021/acsami.5b10693       | (2026)                             |
| Wafer-scale monolayer MoS <sub>2</sub> film integration for stable, efficient perovskite solar cells                                                                         | Huachao Zai; Pengfei Yang; Jie Su et al.                                 | 2025 | Science                                   | 207   | 10.1126/science.ado2351      |                                    |
| Synergistic Effects of Eu-MOF on Perovskite Solar Cells with Improved Stability                                                                                              | Jie Dou; Cheng Zhu; Hao Wang et al.                                      | 2021 | Advanced Materials                        | 206   | 10.1002/adma.202102147       | (2026)                             |
| Molecular materials as interfacial layers and additives in perovskite solar cells                                                                                            | Maria Vasilopoulou; Azhar Fakharuddin; Athanassios G. Coutsolelos et al. | 2020 | Chemical Society Reviews                  | 206   | 10.1039/c9cs00733d           | (2026)                             |
| Continuous Grain-Boundary Functionalization for High-Efficiency Perovskite Solar Cells with Exceptional Stability                                                            | Yingxia Zong; Yuanyuan Zhou; Yi Zhang et al.                             | 2018 | Chem                                      | 204   | 10.1016/j.chempr.2018.03.005 |                                    |
| In situ studies of the degradation mechanisms of perovskite solar cells                                                                                                      | Soumya Kundu; Timothy L. Kelly                                           | 2020 | EcoMat                                    | 201   | 10.1002/eom.21202557         | (2026)                             |

| Title                                                                                                                                                                                                | Authors                                                        | Year | Venue                                    | Cited | DOI                        | OpenAlex<br>2-yr mean<br>citedness |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|------|------------------------------------------|-------|----------------------------|------------------------------------|
| Inkjet-Printed Cation Perovskite Solar Cells                                                                                                                                                         | Triple Florian Mathies; Helge Eggers; Bryce S. Richards et al. | 2018 | ACS Applied Energy Materials             | 200   | 10.1021/acsaem.8b05222     | 50522 (2026)                       |
| Passivation of the grain boundaries of CH <sub>3</sub> NH <sub>3</sub> PbI <sub>3</sub> using carbon quantum dots for highly efficient perovskite solar cells with excellent environmental stability | Qiang Guo; Fanglong Yuan; Bing Zhang et al.                    | 2018 | Nanoscale                                | 200   | 10.1039/c8nr08295b         |                                    |
| Synergistic Effect of Fluorinated Passivator and Hole Transport Dopant Enables Stable Perovskite Solar Cells with an Efficiency Near 24%                                                             | Hongwei Zhu; Yameng Ren; Linfeng Pan et al.                    | 2021 | Journal of the American Chemical Society | 200   | 10.1021/jacs.0c12802       |                                    |
| Chiral-structured heterointerfaces enable durable perovskite solar cells                                                                                                                             | Tianwei Duan; Shuai You; Min Chen et al.                       | 2024 | Science                                  | 199   | 10.1126/science.adf5172    |                                    |
| Compositional and Interface Engineering of Organic-Inorganic Lead Halide Perovskite Solar Cells                                                                                                      | Haizhou Lu; Anurag Krishna; Shaik M. Zakeeruddin et al.        | 2020 | iScience                                 | 198   | 10.1016/j.isci.2020.101359 |                                    |
| Efficient Passivation with Lead Pyridine-2-Carboxylic for High-Performance and Stable Perovskite Solar Cells                                                                                         | Sheng Fu; Xiaodong Li; Li Wan et al.                           | 2019 | Advanced Energy Materials                | 198   | 10.1002/aenm.201901652     | 1652 (2026)                        |
| Highly Air-Stable Tin-Based Perovskite Solar Cells through Grain-Surface Protection by Gallic Acid                                                                                                   | Tianyue Wang; Qidong Tai; Xuyun Guo et al.                     | 2020 | ACS Energy Letters                       | 194   | 10.1021/acsenrgy.1c01712   | 1712 (2026)                        |
| Propylammonium Chloride Additive for Efficient and Stable FAPbI <sub>3</sub> Perovskite Solar Cells                                                                                                  | Yong Zhang; Yan Li; Lin Zhang et al.                           | 2021 | Advanced Energy Materials                | 192   | 10.1002/aenm.202102534     | 2534 (2026)                        |
| Highly Efficient and Stable Perovskite Solar Cells by Interfacial Engineering Using Solution-Processed Polymer Layer                                                                                 | Feijiu Wang; Ai Shimazaki; Fengjiu Yang et al.                 | 2017 | The Journal of Physical Chemistry C      | 191   | 10.1021/acs.jpcc.6b12137   |                                    |
| Poly(4-Vinylpyridine)-Based Interfacial Passivation to Enhance Voltage and Moisture Stability of Lead Halide Perovskite Solar Cells                                                                  | Bhumika Chaudhary; Ashish Kulkarni; Ajay Kumar Jena et al.     | 2017 | ChemSusChem                              | 190   | 10.1002/cssc.201700271     | 271 (2026)                         |
| Monoammonium Porphyrin for Blade-Coating Stable Large-Area Perovskite Solar Cells with >18% Efficiency                                                                                               | Congping Li; Jun Yin; Ruihao Chen et al.                       | 2019 | Journal of the American Chemical Society | 190   | 10.1021/jacs.9b01305       |                                    |
| Perovskite solar cells based on spiro-OMeTAD stabilized with an alkylthiol additive                                                                                                                  | Xu Liu; Bolin Zheng; Lei Shi et al.                            | 2022 | Nature Photonics                         | 189   | 10.1038/s41566-022-01111-x |                                    |
| NbF <sub>5</sub> : A Novel -Phase Stabilizer for FA-Based Perovskite Solar Cells with High Efficiency                                                                                                | Shihao Yuan; Qian Fang; Shaomin Yang et al.                    | 2019 | Advanced Functional Materials            | 189   | 10.1002/adfm.201904750     | 4750 (2026)                        |
| Enhancing the Performance of Inverted Perovskite Solar Cells via Grain Boundary Passivation with Carbon Quantum Dots                                                                                 | Yuhui Ma; Heyi Zhang; Yewei Zhang et al.                       | 2018 | ACS Applied Materials & Interfaces       | 187   | 10.1021/acsaami.8b13867    | 13867 (2026)                       |
| Defect-Engineering-Enabled High-Efficiency All-Inorganic Perovskite Solar Cells                                                                                                                      | Jia Liang; Xiao Han; Ji-Hui Yang et al.                        | 2019 | Advanced Materials                       | 186   | 10.1002/adma.201907046     | 7046 (2026)                        |
| Interface Engineering of Imidazolium Ionic Liquids toward Efficient and Stable CsPbBr <sub>3</sub> Perovskite Solar Cells                                                                            | Wenyu Zhang; Xiaojie Liu; Benlin He et al.                     | 2020 | ACS Applied Materials & Interfaces       | 186   | 10.1021/acsaami.9b20831    | 20831 (2026)                       |
| Review of Novel Passivation Techniques for Efficient and Stable Perovskite Solar Cells                                                                                                               | Jincheol Kim; Anita Ho-Baillie; Shujuan Huang                  | 2019 | Solar RRL                                | 185   | 10.1002/solr.201800302     |                                    |
| Thermal Stability of CuSCN Hole Conductor-Based Perovskite Solar Cells                                                                                                                               | Minsu Jung; Young Chan Kim; Nam Joong Jeon et al.              | 2016 | ChemSusChem                              | 183   | 10.1002/cssc.201600477     | 477 (2026)                         |

| Title                                                                                                                                                                                         | Authors                                                              | Year | Venue                                    | Cited | DOI                         | OpenAlex<br>2-yr mean<br>citedness |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|------|------------------------------------------|-------|-----------------------------|------------------------------------|
| Investigation of the influence of different hole-transporting materials on the performance of perovskite solar cells                                                                          | Elham Karimi; Seyed Mohammad Bagher Ghorashi                         | 2016 | Optik                                    | 183   | 10.1016/j.ijleo.2016.10.122 | 16.712 (2026)                      |
| Effect of Side-Group-Regulated Dipolar Passivating Molecules on CsPbBr <sub>3</sub> Perovskite Solar Cells                                                                                    | Jialong Duan; Meng Wang; Yingli Wang et al.                          | 2021 | ACS Energy Letters                       | 183   | 10.1021/acsenrgy.1c11200    | 16.712 (2026)                      |
| Crystallization Kinetics in 2D Perovskite Solar Cells                                                                                                                                         | Youkui Xu; Meng Wang; Yutian Lei et al.                              | 2020 | Advanced Energy Materials                | 181   | 10.1002/aenm.202003554      | 16.712 (2026)                      |
| Efficiency and stability enhancement of perovskite solar cells by introducing CsPbI <sub>3</sub> quantum dots as an interface engineering layer                                               | Chang Liu; Manman Hu; Xianyong Zhou et al.                           | 2018 | NPG Asia Materials                       | 181   | 10.1038/s41427-018-0055-0   | 16.712 (2026)                      |
| Water-Soluble Triazolium Ionic-Liquid-Induced Surface Self-Assembly to Enhance the Stability and Efficiency of Perovskite Solar Cells                                                         | Shuangjie Wang; Zhen Li; Yuanyuan Zhang et al.                       | 2019 | Advanced Functional Materials            | 181   | 10.1002/adfm.201904075      | 16.712 (2026)                      |
| Perovskite solar cells: Materials, configurations and stability                                                                                                                               | Isabel Mesquita; Luísa Andrade; Adélio Mendes                        | 2017 | Renewable and Sustainable Energy Reviews | 178   | 10.1016/j.rser.2017.09.011  | 16.712 (2026)                      |
| Enhancing Moisture and Water Resistance in Perovskite Solar Cells by Encapsulation with Ultrathin Plasma Polymers                                                                             | Jesús Idígoras; Francisco J. Aparicio; Lidia Contreras-Bernal et al. | 2018 | ACS Applied Materials & Interfaces       | 177   | 10.1021/acsami.7b13824      | 16.712 (2026)                      |
| Enhanced Stability of Perovskite Solar Cells with Low-Temperature Hydrothermally Grown SnO <sub>2</sub> Electron Transport Layers                                                             | Qin Liu; Minchao Qin; Weijun Ke et al.                               | 2016 | Advanced Functional Materials            | 176   | 10.1002/adfm.201600975      | 16.712 (2026)                      |
| Targeted Therapy for Interfacial Engineering Toward Stable and Efficient Perovskite Solar Cells                                                                                               | Shuhui Wang; Haiyang Chen; Jiandong Zhang et al.                     | 2019 | Advanced Materials                       | 176   | 10.1002/adma.201903061      | 16.712 (2026)                      |
| Sulfonate-Assisted Surface Iodide Management for High-Performance Perovskite Solar Cells and Modules                                                                                          | Ruihao Chen; Yongke Wang; Siqing Nie et al.                          | 2021 | Journal of the American Chemical Society | 176   | 10.1021/jacs.1c03419        | 16.712 (2026)                      |
| Recent efficient strategies for improving the moisture stability of perovskite solar cells                                                                                                    | Faming Li; Mingzhen Liu                                              | 2017 | Journal of Materials Chemistry A         | 169   | 10.1039/c7ta01325f          | 16.712 (2026)                      |
| Advances in stability of perovskite solar cells                                                                                                                                               | Qamar Wali; Faiza Jan Iftikhar; Muhammad Ejaz Khan et al.            | 2019 | Organic Electronics                      | 169   | 10.1016/j.orgel.2019.105590 | 16.712 (2026)                      |
| Spontaneously Self-Assembly of a 2D/3D Heterostructure Enhances the Efficiency and Stability in Printed Perovskite Solar Cells                                                                | Jinlong Hu; Chuan Wang; Shudi Qiu et al.                             | 2020 | Advanced Energy Materials                | 169   | 10.1002/aenm.202003037      | 16.712 (2026)                      |
| Constructing Efficient and Stable Perovskite Solar Cells via Interconnecting Perovskite Grains                                                                                                | Xian Hou; Sumei Huang; Wei Ou-Yang et al.                            | 2017 | ACS Applied Materials & Interfaces       | 165   | 10.1021/acsami.7b08438      | 16.712 (2026)                      |
| Fluorine-Containing Passivation Layer via Surface Chelation for Inorganic Perovskite Solar Cells                                                                                              | Hao Zhang; Wanchun Xiang; Xuejiao Zuo et al.                         | 2022 | Angewandte Chemie International Edition  | 165   | 10.1002/anie.202216006      | 16.712 (2026)                      |
| Reducing Energy Disorder in Perovskite Solar Cells by Chelation                                                                                                                               | Yiting Jiang; Jiabin Wang; Huachao Zai et al.                        | 2022 | Journal of the American Chemical Society | 164   | 10.1021/jacs.1c12732        | 16.712 (2026)                      |
| Poly(N,N'-bis-4-butylphenyl)-N,N'-bisphenoxyhydrazide-Based Interfacial Passivation Strategy Promoting Efficiency and Operational Stability of Perovskite Solar Cells in Regular Architecture | Ni-Ni Alsaifi; Shoukui Zhou; Basim Alsaifi et al.                    | 2020 | Advanced Materials                       | 163   | 10.1002/adma.202006087      | 16.712 (2026)                      |
| Enhanced Performance and Stability of 3D/2D Tin Perovskite Solar Cells Fabricated with a Sequential Solution Deposition                                                                       | Efat Jokar; Po-Yuan Cheng; Chia-Yi Lin et al.                        | 2021 | ACS Energy Letters                       | 161   | 10.1021/acsenrgy.1c11200    | 16.712 (2026)                      |
| The Role of Charge Selective Contacts in Perovskite Solar Cell Stability                                                                                                                      | Bart Roose; Qiong Wang; Antonio Abate                                | 2018 | Advanced Energy Materials                | 155   | 10.1002/aenm.201803334      | 16.712 (2026)                      |
| Over 24% efficient MA-free CsxFA1-xPbX <sub>3</sub> perovskite solar cells                                                                                                                    | Siyang Wang; Liguo Tan; Junjie Zhou et al.                           | 2022 | Joule                                    | 155   | 10.1016/j.joule.2022.05.002 | 16.712 (2026)                      |

| Title                                                                                                                                                      | Authors                                                              | Year | Venue                                     | Cited | DOI                           | OpenAlex<br>2-yr mean<br>citedness |
|------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|------|-------------------------------------------|-------|-------------------------------|------------------------------------|
| Bifunctional alkyl chain barriers for efficient perovskite solar cells                                                                                     | Jing Zhang; Zhelu Hu; Like Huang et al.                              | 2015 | Chemical Communications                   | 154   | 10.1039/c5cc00128e            | 166 (2026)                         |
| Self-Encapsulating Thermally Stable and Air-Resilient Semitransparent Perovskite Solar Cells                                                               | Jie Zhao; Kai Oliver Brinkmann; Ting Hu et al.                       | 2017 | Advanced Energy Materials                 | 151   | 10.1002/aenm.201603594        | 166 (2026)                         |
| Paradoxical Approach with a Hydrophilic Passivation Layer for Moisture-Stable, 23% Efficient Perovskite Solar Cells                                        | Chunqing Ma; Nam-Gyu Park                                            | 2020 | ACS Energy Letters                        | 151   | 10.1021/acsenergy.1c01712     | 166 (2026)                         |
| Research Update: Strategies for improving the stability of perovskite solar cells                                                                          | Severin N. Habisreutinger; David P. McMeekin; Henry J. Snaith et al. | 2016 | APL Materials                             | 150   | 10.1063/1.4961210             | 158 (2026)                         |
| Dion-Jacobson 2D-3D perovskite solar cells with improved efficiency and stability                                                                          | Xiaoqing Jiang; Jiafeng Zhang; Sajjad Ahmad et al.                   | 2020 | Nano Energy                               | 150   | 10.1016/j.nanoen.2020.104892  | 158 (2026)                         |
| Perfluorinated Self-Assembled Monolayers Enhance the Stability and Efficiency of Inverted Perovskite Solar Cells                                           | Christian M. Wolff; Laura Canil; Carolin Rehmann et al.              | 2020 | ACS Nano                                  | 150   | 10.1021/acsnano.9b03948       | 158 (2026)                         |
| Improving Efficiency and Stability of Perovskite Solar Cells Enabled by A Near-Infrared-Absorbing Moisture Barrier                                         | Qin Hu; Wei Chen; Wenqiang Yang et al.                               | 2020 | Joule                                     | 149   | 10.1016/j.joule.2020.06.007   | 158 (2026)                         |
| Review on persistent challenges of perovskite solar cells' stability                                                                                       | Maithili K. Rao; D. Sangeetha; M. Selvakumar et al.                  | 2021 | Solar Energy                              | 149   | 10.1016/j.solener.2021.03.005 | 158 (2026)                         |
| High-Performance Thickness Insensitive Perovskite Solar Cells with Enhanced Moisture Stability                                                             | Jiehuan Chen; Lijian Zuo; Yingzhu Zhang et al.                       | 2018 | Advanced Energy Materials                 | 148   | 10.1002/aenm.201806334        | 158 (2026)                         |
| Moisture-Resistant FAPbI <sub>3</sub> Perovskite Solar Cell with 22.25 % Power Conversion Efficiency through Pentafluorobenzyl Phosphonic Acid Passivation | Erdi Akman; Ahmed Es-smail Shalan; Faranak Sadegh et al.             | 2020 | ChemSusChem                               | 148   | 10.1002/cssc.202002707        | 158 (2026)                         |
| In situ growth of graphene on both sides of a Cu-Ni alloy electrode for perovskite solar cells with improved stability                                     | Xuesong Lin; Hongzhen Su; Sifan He et al.                            | 2022 | Nature Energy                             | 148   | 10.1038/s41560-022-01038-1    | 158 (2026)                         |
| Self-Seeding Growth for Perovskite Solar Cells with Enhanced Stability                                                                                     | Fei Zhang; Chuanxiao Xiao; Xihan Chen et al.                         | 2019 | Joule                                     | 147   | 10.1016/j.joule.2019.03.023   | 158 (2026)                         |
| Ionic liquids engineering for high-efficiency and stable perovskite solar cells                                                                            | Xiaoyu Deng; Lisha Xie; Shurong Wang et al.                          | 2020 | Chemical Engineering Journal              | 146   | 10.1016/j.cej.2020.112959     | 158 (2026)                         |
| Multifunctional Small Molecule as Buried Interface Passivator for Efficient Planar Perovskite Solar Cells                                                  | Meizi Wu; Yuwei Duan; Lu Yang et al.                                 | 2023 | Advanced Functional Materials             | 146   | 10.1002/adfm.202310978        | 158 (2026)                         |
| 1000 h Operational Lifetime Perovskite Solar Cells by Ambient Melting Encapsulation                                                                        | Sai Ma; Yang Bai; Hao Wang et al.                                    | 2020 | Advanced Energy Materials                 | 145   | 10.1002/aenm.201903472        | 158 (2026)                         |
| The balance between efficiency, stability and environmental impacts in perovskite solar cells: a review                                                    | Antonio Urbina                                                       | 2019 | Journal of Physics Energy                 | 145   | 10.1088/2515-7655/ab5eee      | 158 (2026)                         |
| Dual-Protection Strategy for High-Efficiency and Stable CsPbI <sub>2</sub> Br Inorganic Perovskite Solar Cells                                             | Sheng Fu; Wenxiao Zhang; Xiaodong Li et al.                          | 2020 | ACS Energy Letters                        | 143   | 10.1021/acsenergy.1c01712     | 158 (2026)                         |
| Quantifying the Interface Defect for the Stability Origin of Perovskite Solar Cells                                                                        | Jionghua Wu; Jiangjian Shi; Yiming Li et al.                         | 2019 | Advanced Energy Materials                 | 142   | 10.1002/aenm.201901352        | 158 (2026)                         |
| Critical Role of Water in Defect Aggregation and Chemical Degradation of Perovskite Solar Cells                                                            | Yun-Hyok Kye; Chol-Jun Yu; Un-Gi Jong et al.                         | 2018 | The Journal of Physical Chemistry Letters | 141   | 10.1021/acs.jpclett.8b00406   | 158 (2026)                         |
| Impact of Peripheral Groups on Phenothiazine-Based Hole-Transporting Materials for Perovskite Solar Cells                                                  | Fei Zhang; Shirong Wang; Hongwei Zhu et al.                          | 2018 | ACS Energy Letters                        | 141   | 10.1021/acsenergy.1c01712     | 158 (2026)                         |
| Dually-Passivated Perovskite Solar Cells with Reduced Voltage Loss and Increased Super Oxide Resistance                                                    | Qin Zhou; Yifeng Gao; C.S. Cai et al.                                | 2021 | Angewandte Chemie International Edition   | 140   | 10.1002/anie.202015106        | 158 (2026)                         |

| Title                                                                                                                                                           | Authors                                                        | Year | Venue                                    | Cited | DOI                             | OpenAlex<br>2-yr mean<br>citedness |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|------|------------------------------------------|-------|---------------------------------|------------------------------------|
| Effect of Electron-Transport Material on Light-Induced Degradation of Inverted Planar Junction Perovskite Solar Cells                                           | Azat F. Akbulatov; Lyubov A. Frolova; Monroe P. Griffin et al. | 2017 | Advanced Energy Materials                | 139   | 10.1002/aenm.201700657          | 18657 (2026)                       |
| A review of stability and progress in tin halide perovskite solar cell                                                                                          | Asim Aftab; Md. Imteyaz Ahmad                                  | 2021 | Solar Energy                             | 137   | 10.1016/j.solener.2020.12.065   |                                    |
| Moisture-Resilient Perovskite Solar Cells for Enhanced Stability                                                                                                | Randi Azmi; Shynggys Zhumagali; Helen Bristow et al.           | 2023 | Advanced Materials                       | 134   | 10.1002/adma.202210817          | 2210817 (2026)                     |
| Engineering fluorinated-perovskite solar cells with an efficiency of >21% and improved stability towards humidity                                               | Xiao Wang; Kasparas Rakštys; Kevin S. Jack et al.              | 2021 | Nature Communications                    | 134   | 10.1038/s41467-020-20272-3      |                                    |
| Review of recent developments and persistent challenges in stability of perovskite solar cells                                                                  | Billel Salhi; Y.S. Wudil; Mohammad Kamal Hossain et al.        | 2018 | Renewable and Sustainable Energy Reviews | 134   | 10.1016/j.rser.2018.03.058      |                                    |
| Impact of moisture on efficiency-determining electronic processes in perovskite solar cells                                                                     | Manuel Salado; Lidia Contreras-Bernal; Laura Calió et al.      | 2017 | Journal of Materials Chemistry A         | 133   | 10.1039/c7ta02264f              |                                    |
| Stability Improvement of Perovskite Solar Cells by Compositional and Interfacial Engineering                                                                    | Weiguang Chi; Sanjay K. Banerjee                               | 2021 | Chemistry of Materials                   | 132   | 10.1021/acs.chemmater.202010311 | 6387 (2026)                        |
| Encapsulation and Outdoor Testing of Perovskite Solar Cells: Comparing Industrially Relevant Process with a Simplified Lab Procedure                            | Quiterie Emery; Marko Remec; Gopinath Paramasivam et al.       | 2022 | ACS Applied Materials & Interfaces       | 132   | 10.1021/acsami.1c14520          | 184520 (2026)                      |
| The Effect of Stoichiometry on the Stability of Inorganic Cesium Lead Mixed-Halide Perovskites Solar Cells                                                      | Qingshan Ma; Shujuan Huang; Sheng Chen et al.                  | 2017 | The Journal of Physical Chemistry C      | 131   | 10.1021/acs.jpcc.7b06268        |                                    |
| Progress and Challenges Toward Effective Flexible Perovskite Solar Cells                                                                                        | Xiongjie Li; Haixuan Yu; Zhirong Liu et al.                    | 2023 | Nano-Micro Letters                       | 129   | 10.1007/s40820-023-01165-8      |                                    |
| A facile molecularly engineered copper (II) phthalocyanine as hole transport material for planar perovskite solar cells with enhanced performance and stability | Guang Yang; Yulong Wang; Jiaju Xu et al.                       | 2016 | Nano Energy                              | 126   | 10.1016/j.nanoen.2016.11.039    |                                    |
| Hermetic seal for perovskite solar cells: An improved plasma enhanced atomic layer deposition encapsulation                                                     | Haoran Wang; Yepin Zhao; Zhenyu Wang et al.                    | 2019 | Nano Energy                              | 124   | 10.1016/j.nanoen.2019.104375    |                                    |
| Indigo: A Natural Molecular Passivator for Efficient Perovskite Solar Cells                                                                                     | Junjun Guo; Jianguo Sun; Long Hu et al.                        | 2022 | Advanced Energy Materials                | 124   | 10.1002/aenm.202210053          | 2210053 (2026)                     |
| CuSCN as Hole Transport Material with 3D/2D Perovskite Solar Cells                                                                                              | Vinod E. Madhavan; Iwan Zimmermann; Ahmer A.B. Baloch et al.   | 2019 | ACS Applied Energy Materials             | 124   | 10.1021/acsaem.9b01762          | 501762 (2026)                      |
| Stability of Halide Perovskite Solar Cell Devices: In Situ Observation of Oxygen Diffusion under Biasing                                                        | Hee Joon Jung; Daehan Kim; Sung-Kyu Kim et al.                 | 2018 | Advanced Materials                       | 123   | 10.1002/adma.201802750          | 2202750 (2026)                     |
| Machine learning analysis on stability of perovskite solar cells                                                                                                | Çağla Odabaşı; Ramazan Yıldırım                                | 2019 | Solar Energy Materials and Solar Cells   | 122   | 10.1016/j.solmat.2019.110284    |                                    |
| Intrinsic and Extrinsic Stability of Formamidinium Lead Bromide Perovskite Solar Cells Yielding High Photovoltage                                               | Neha Arora; M. Ibrahim Dar; Mojtaba Abdi-Jalebi et al.         | 2016 | Nano Letters                             | 122   | 10.1021/acs.nanolett.6b03455    |                                    |
| Versatility of Carbon Enables All Carbon Based Perovskite Solar Cells to Achieve High Efficiency and High Stability                                             | Xiangyue Meng; Junshuai Zhou; Jie Hou et al.                   | 2018 | Advanced Materials                       | 122   | 10.1002/adma.201706975          | 2206975 (2026)                     |
| 2-CF3-PEAI to eliminate Pb0 traps and form a 2D perovskite layer to enhance the performance and stability of perovskite solar cells                             | Junjie Zhou; Minghao Li; Siyang Wang et al.                    | 2022 | Nano Energy                              | 122   | 10.1016/j.nanoen.2022.107036    |                                    |
| Encapsulation: The path to commercialization of stable perovskite solar cells                                                                                   | Qianqian Chu; Zhijian Sun; Dong Wang et al.                    | 2023 | Matter                                   | 120   | 10.1016/j.matt.2023.08.016      |                                    |
| Ambient Fabrication of Organic-Inorganic Hybrid Perovskite Solar Cells                                                                                          | Yuan Zhang; Ashleigh Kirs; Filip Ambrož et al.                 | 2020 | Small Methods                            | 118   | 10.1002/smtd.202000744          |                                    |

| Title                                                                                                                                                         | Authors                                                        | Year | Venue                                  | Cited | DOI                           | OpenAlex<br>2-yr mean<br>citedness |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|------|----------------------------------------|-------|-------------------------------|------------------------------------|
| Recent Progress of Carbon-Based Inorganic Perovskite Solar Cells: From Efficiency to Stability                                                                | Xiang Zhang; Zhenhua Yu; Dan Zhang et al.                      | 2022 | Advanced Energy Materials              | 117   | 10.1002/aenm.202213320        | 10.1332(2026)                      |
| Film Grain-Size Related Long-Term Stability of Inverted Perovskite Solar Cells                                                                                | Chien-Hung Chiang; Chun-Guey Wu                                | 2016 | ChemSusChem                            | 117   | 10.1002/cssc.201601487        | 10.1337(2026)                      |
| Gradated Mixed Hole Transport Layer in a Perovskite Solar Cell: Improving Moisture Stability and Efficiency                                                   | Guan-Woo Kim; Gyeongho Kang; Mahdi Malekshahi Byranvand et al. | 2017 | ACS Applied Materials & Interfaces     | 116   | 10.1021/acsami.7b13821        | 10.1321(2026)                      |
| Printable WO <sub>3</sub> electron transporting layer for perovskite solar cells: Influence on device performance and stability                               | Alexandre Gheno; Trang Thi Thu Pham; Catherine Di Bin et al.   | 2016 | Solar Energy Materials and Solar Cells | 116   | 10.1016/j.solmat.2016.10.002  | 10.1320(2026)                      |
| Barrier reinforcement for enhanced perovskite solar cell stability under reverse bias                                                                         | Nengxu Li; Zhifang Shi; Chengbin Fei et al.                    | 2024 | Nature Energy                          | 114   | 10.1038/s41560-024-01579-7    | 10.1320(2026)                      |
| Additive engineering for stable halide perovskite solar cells                                                                                                 | Carlos Pereyra; Haibing Xie; Mónica Lira-Cantú                 | 2021 | Journal of Energy Chemistry            | 113   | 10.1016/j.jechem.2021.01.037  | 10.1320(2026)                      |
| Mechanochemistry Advances High-Performance Perovskite Solar Cells                                                                                             | Yuzhuo Zhang; Yanju Wang; Xiaoyu Yang et al.                   | 2021 | Advanced Materials                     | 112   | 10.1002/adma.202112742        | 10.1320(2026)                      |
| High Efficiency and Stability of Inverted Perovskite Solar Cells Using Phenethyl Ammonium Iodide-Modified Interface of NiO <sub>x</sub> and Perovskite Layers | Yuning Liu; Jiaji Duan; Jiankai Zhang et al.                   | 2019 | ACS Applied Materials & Interfaces     | 112   | 10.1021/acsami.9b18217        | 10.1327(2026)                      |
| Achieving Resistance against Moisture and Oxygen for Perovskite Solar Cells with High Efficiency and Stability                                                | Weiguang Chi; S. Banerjee                                      | 2021 | Chemistry of Materials                 | 107   | 10.1021/acs.chemmater.1c03887 | 10.1327(2026)                      |
| Interface degradation of perovskite solar cells and its modification using an annealing-free TiO <sub>2</sub> NPs layer                                       | Jian Xiong; Bingchu Yang; Chenghao Cao et al.                  | 2015 | Organic Electronics                    | 107   | 10.1016/j.orgel.2015.12.010   | 10.1327(2026)                      |
| Boosting mechanical durability under high humidity by bioinspired multisite polymer for high-efficiency flexible perovskite solar cells                       | Zhihao Li; Zhihao Li; Chunmei Jia et al.                       | 2025 | Nature Communications                  | 107   | 10.1038/s41467-025-57102-3    | 10.1327(2026)                      |
| Dual Additive for Simultaneous Improvement of Photovoltaic Performance and Stability of Perovskite Solar Cell                                                 | In Seok Yang; Nam-Gyu Park                                     | 2021 | Advanced Functional Materials          | 105   | 10.1002/adfm.202100376        | 10.1325(2026)                      |
| Solution processed pristine PDPP3T polymer as hole transport layer for efficient perovskite solar cells with slower degradation                               | Ashish Dubey; Nirmal Adhikari; Swaminathan Venkatesan et al.   | 2015 | Solar Energy Materials and Solar Cells | 105   | 10.1016/j.solmat.2015.10.008  | 10.1325(2026)                      |
| Spray reaction prepared FA 1-x Cs x PbI <sub>3</sub> solid solution as a light harvester for perovskite solar cells with improved humidity stability          | Xiang Xia; Wenyi Wu; Hongcui Li et al.                         | 2016 | RSC Advances                           | 104   | 10.1039/c5ra23359c            | 10.1325(2026)                      |
| Amino acid salt-driven planar hybrid perovskite solar cells with enhanced humidity stability                                                                  | Seong-Cheol Yun; Sunihl Ma; Hyeok-Chan Kwon et al.             | 2019 | Nano Energy                            | 103   | 10.1016/j.nanoen.2019.02.064  | 10.1325(2026)                      |
| Degradation and Self-Healing in Perovskite Solar Cells                                                                                                        | Blake P. Finkenauer; Akriti Akriti; Ke Ma et al.               | 2022 | ACS Applied Materials & Interfaces     | 101   | 10.1021/acsami.2c01382        | 10.1325(2026)                      |
| Recent Progress of All-Bromide Inorganic Perovskite Solar Cells                                                                                               | Guoqing Tong; Luis K. Ono; Yabing Qi                           | 2019 | Energy Technology                      | 100   | 10.1002/ente.201900021        | 10.1321(2026)                      |
| Enhancing stability for organic-inorganic perovskite solar cells by atomic layer deposited Al <sub>2</sub> O <sub>3</sub> encapsulation                       | Eun Young Choi; Jincheol Kim; Sean Lim et al.                  | 2018 | Solar Energy Materials and Solar Cells | 99    | 10.1016/j.solmat.2018.08.016  | 10.1321(2026)                      |
| Enhanced Ambient Stability of Efficient Perovskite Solar Cells by Employing a Modified Fullerene Cathode Interlayer                                           | Zonglong Zhu; Chu-Chen Chueh; Francis Lin et al.               | 2016 | Advanced Science                       | 97    | 10.1002/advs.201600027        | 10.1327(2026)                      |



| Title                                                                                                                                                     | Authors                                                                      | Year | Venue                               | Cited | DOI                           | OpenAlex<br>2-yr mean<br>citedness |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|------|-------------------------------------|-------|-------------------------------|------------------------------------|
| Photodecomposition and Morphology Evolution of Organometal Halide Perovskite Solar Cells                                                                  | Fukashi Matsumoto; Sarah M. Vorpahl; Jannel Q. Banks et al.                  | 2015 | The Journal of Physical Chemistry C | 95    | 10.1021/acs.jpcc.5b06269      | 167.12(2025)                       |
| Humidity-Assisted Chlorination with Solid Protection Strategy for Efficient Air-Fabricated Inverted CsPbI <sub>3</sub> Perovskite Solar Cells             | Sheng Fu; Wenxiao Zhang; Xiaodong Li et al.                                  | 2021 | ACS Energy Letters                  | 91    | 10.1021/acsenergy.1c01120     | 167.12(2025)                       |
| Achieving Long-Term Operational Stability of Perovskite Solar Cells with a Stabilized Efficiency Exceeding 20% after 1000 h                               | Tae-Youl Yang; Nam Joong Jeon; Hee-Won Shin et al.                           | 2019 | Advanced Science                    | 91    | 10.1002/advs.201900528        | 167.12(2025)                       |
| Graphdiyne-Based Bulk Heterojunction for Efficient and Moisture-Stable Planar Perovskite Solar Cells                                                      | Hongshi Li; Rui Zhang; Yusheng Li et al.                                     | 2018 | Advanced Energy Materials           | 90    | 10.1002/aenm.201803011        | 167.12(2025)                       |
| Long-term stability of organic-inorganic hybrid perovskite solar cells with high efficiency under high humidity conditions                                | Yue Sun; Yihui Wu; Xiang Fang et al.                                         | 2016 | Journal of Materials Chemistry A    | 89    | 10.1039/c6ta08117g            | 167.12(2025)                       |
| Tetrabutylammonium cations for moisture-resistant and semitransparent perovskite solar cells                                                              | Isabella Poli; Salvador Eslava; Petra J. Cameron                             | 2017 | Journal of Materials Chemistry A    | 87    | 10.1039/c7ta06735f            | 167.12(2025)                       |
| Textured CH <sub>3</sub> NH <sub>3</sub> PbI <sub>3</sub> thin film with enhanced stability for high performance perovskite solar cells                   | Mingzhu Long; Tiankai Zhang; Houyu Zhu et al.                                | 2017 | Nano Energy                         | 86    | 10.1016/j.nanoen.2017.02.002  | 167.12(2025)                       |
| Stability Enhancement in Perovskite Solar Cells with Perovskite/Silver-Graphene Composites in the Active Layer                                            | Tahmineh Mahmoudi; Yousheng Wang; Yoon-Bong Hahn                             | 2018 | ACS Energy Letters                  | 86    | 10.1021/acsenergy.1c01120     | 167.12(2025)                       |
| Reviewing and understanding the stability mechanism of halide perovskite solar cells                                                                      | Caixin Zhang; Tao Shen; Dan Guo et al.                                       | 2020 | InfoMat                             | 84    | 10.1002/inf2.12104            | 167.12(2025)                       |
| Double Barriers for Moisture Degradation: Assembly of Hydrolysable Hydrophobic Molecules for Stable Perovskite Solar Cells with High Open-Circuit Voltage | Pengfei Guo; Qian Ye; Chen Liu et al.                                        | 2020 | Advanced Functional Materials       | 84    | 10.1002/adfm.202004279        | 167.12(2025)                       |
| Spontaneous grain polymerization for efficient and stable perovskite solar cells                                                                          | Xiaodong Li; Wenxiao Zhang; Wenjun Zhang et al.                              | 2019 | Nano Energy                         | 84    | 10.1016/j.nanoen.2019.02.009  | 167.12(2025)                       |
| Glass-to-glass encapsulation with ultraviolet light curable epoxy edge sealing for stable perovskite solar cells                                          | Easwaramoorthi Ramasamy; Vaithinathan Karthikeyan; Kadusu Rameshkumar et al. | 2019 | Materials Letters                   | 83    | 10.1016/j.matlet.2019.04.082  | 167.12(2025)                       |
| Composite Encapsulation Enabled Superior Comprehensive Stability of Perovskite Solar Cells                                                                | Yifan Lv; Hui Zhang; Ruqing Liu et al.                                       | 2020 | ACS Applied Materials & Interfaces  | 82    | 10.1021/acsami.0c05823        | 167.12(2025)                       |
| Degradation mechanism of planar-perovskite solar cells: correlating evolution of iodine distribution and photocurrent hysteresis                          | Riski Titian Ginting; Mi-Kyoung Jeon; Kwang-Jae Lee et al.                   | 2017 | Journal of Materials Chemistry A    | 80    | 10.1039/c6ta09202k            | 167.12(2025)                       |
| Stability Issues of Perovskite Solar Cells: A Critical Review                                                                                             | Shahriyar Safat Dipta; Ashraf Uddin                                          | 2021 | Energy Technology                   | 79    | 10.1002/ente.202100530        | 167.12(2025)                       |
| Advancement in CsPbBr <sub>3</sub> inorganic perovskite solar cells: Fabrication, efficiency and stability                                                | Naveen Kumar; Jyoti Rani; Rajnish Kurchania                                  | 2021 | Solar Energy                        | 79    | 10.1016/j.solener.2021.04.042 | 167.12(2025)                       |
| 3D/2D multidimensional perovskites: Balance of high performance and stability for perovskite solar cells                                                  | Fei Zhang; Dong Hoe Kim; Kai Zhu                                             | 2018 | Current Opinion in Electrochemistry | 79    | 10.1016/j.coelec.2018.04.002  | 167.12(2025)                       |
| Electrospray-Assisted Fabrication of Moisture-Resistant and Highly Stable Perovskite Solar Cells at Ambient Conditions                                    | Shalinee Kavadiya; Dariusz M. Niedzwiedzki; Su Huang et al.                  | 2017 | Advanced Energy Materials           | 76    | 10.1002/aenm.201700421        | 167.12(2025)                       |

| Title                                                                                                                                                    | Authors                                                             | Year | Venue                               | Cited | DOI                           | OpenAlex<br>2-yr mean<br>citedness |
|----------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|------|-------------------------------------|-------|-------------------------------|------------------------------------|
| In Situ Identification of Photo- and Moisture-Dependent Phase Evolution of Perovskite Solar Cells                                                        | Bo-An Chen; Jin-Tai Lin; Nian-Tzu Suen et al.                       | 2017 | ACS Energy Letters                  | 75    | 10.1021/acsenergy.1c00098     | 167.12 (2026)                      |
| Printing High-Efficiency Perovskite Solar Cells in High-Humidity Ambient Environment—An In Situ Guided Investigation                                     | W.K. Fong; Hanlin Hu; Zhiwei Ren et al.                             | 2021 | Advanced Science                    | 75    | 10.1002/advs.202001385        | 133.39 (2026)                      |
| Self-healing perovskite solar cells                                                                                                                      | Yue Yu; Fu Zhang; Hua Yu                                            | 2020 | Solar Energy                        | 75    | 10.1016/j.solener.2020.09.018 |                                    |
| Effect of relative humidity during the preparation of perovskite solar cells: Performance and stability                                                  | Isabel Mesquita; Luísa Andrade; Adélio Mendes                       | 2020 | Solar Energy                        | 72    | 10.1016/j.solener.2020.02.052 |                                    |
| Enhanced Moisture Stability by Butyldimethylsulfonium Cation in Perovskite Solar Cells                                                                   | Bo-Hyung Kim; Maengsuk Kim; Jun Hee Lee et al.                      | 2019 | Advanced Science                    | 71    | 10.1002/advs.201901430        | 149.30 (2026)                      |
| Excellent Moisture Stability and Efficiency of Inverted All-Inorganic CsPbIBr <sub>2</sub> Perovskite Solar Cells through Molecule Interface Engineering | Shuzhang Yang; Liang Wang; Liguao Gao et al.                        | 2020 | ACS Applied Materials & Interfaces  | 70    | 10.1021/acsami.9b23532        | 135.32 (2026)                      |
| Amphiphilic Fullerenes Employed to Improve the Quality of Perovskite Films and the Stability of Perovskite Solar Cells                                   | Qingxia Fu; Shuqin Xiao; Xianglan Tang et al.                       | 2019 | ACS Applied Materials & Interfaces  | 70    | 10.1021/acsami.9b17829        | 134.29 (2026)                      |
| From Exceptional Properties to Stability Challenges of Perovskite Solar Cells                                                                            | Somayeh Gholipour; Michael Saliba                                   | 2018 | Small                               | 69    | 10.1002/sml.201802385         |                                    |
| Aromatic Alkylammonium Spacer Cations for Efficient Two-Dimensional Perovskite Solar Cells with Enhanced Moisture and Thermal Stability                  | Deepak Thrithamarassery Gangadharan; Yujie Han; Ashish Dubey et al. | 2018 | Solar RRL                           | 68    | 10.1002/solr.201700215        |                                    |
| Thermal and Humidity Stability of Mixed Spacer Cations 2D Perovskite Solar Cells                                                                         | Huayang Yu; Yulin Xie; Jia Zhang et al.                             | 2021 | Advanced Science                    | 67    | 10.1002/advs.202001490        | 149.30 (2026)                      |
| Chlorine-terminated MXene quantum dots for improving crystallinity and moisture stability in high-performance perovskite solar cells                     | Xiao Liu; Zhiang Zhang; Jinkun Jiang et al.                         | 2021 | Chemical Engineering Journal        | 64    | 10.1016/j.cej.2021103438      | 133.2 (2026)                       |
| Stability and Performance Enhancement of Perovskite Solar Cells: A Review                                                                                | Maria Khalid; Tapas K. Mallick                                      | 2023 | Energies                            | 62    | 10.3390/en16104041            | 1847 (2026)                        |
| Progress on the stability and encapsulation techniques of perovskite solar cells                                                                         | Ling Xiang; Fangliang Gao; Yunxuan Cao et al.                       | 2022 | Organic Electronics                 | 61    | 10.1016/j.orgel.2022.106515   |                                    |
| Encapsulating perovskite solar cells for long-term stability and prevention of lead toxicity                                                             | Shahriyar Safat Dipta; Md. Arifur Rahim; Ashraf Uddin               | 2024 | Applied Physics Reviews             | 61    | 10.1063/5.0197154             | 11.309 (2026)                      |
| Recent Progress on the Stability of Perovskite Solar Cells in a Humid Environment                                                                        | Mengying Li; Haibo Li; Jing Fu et al.                               | 2020 | The Journal of Physical Chemistry C | 60    | 10.1021/acs.jpcc.0c08019      |                                    |
| Light enhanced moisture degradation of perovskite solar cell material CH <sub>3</sub> NH <sub>3</sub> PbI <sub>3</sub>                                   | Ying-Bo Lu; Wei-Yan Cong; Chengbo Guan et al.                       | 2019 | Journal of Materials Chemistry A    | 60    | 10.1039/c9ta10443g            |                                    |
| Enhanced moisture stability of metal halide perovskite solar cells based on sulfur-oleyamine surface modification                                        | Yu Hou; Zi Ren Zhou; Tian Wen et al.                                | 2018 | Nanoscale Horizons                  | 57    | 10.1039/c8nh00163d            |                                    |
| Homeopathic Perovskite Solar Cells: Effect of Humidity during Fabrication on the Performance and Stability of the Device                                 | Lidia Contreras-Bernal; Clara Aranda; Marta Vallés-Pelarda et al.   | 2018 | The Journal of Physical Chemistry C | 56    | 10.1021/acs.jpcc.8b01558      |                                    |
| Nonionic Sc <sub>3</sub> N@C <sub>80</sub> Dopant for Efficient and Stable Halide Perovskite Photovoltaics                                               | Kai Wang; Xiaoyang Liu; Rong Huang et al.                           | 2019 | ACS Energy Letters                  | 56    | 10.1021/acsenergy.1c00098     | 167.12 (2026)                      |
| Improving the moisture stability of perovskite solar cells by using PMMA/P3HT based hole-transport layers                                                | Soumya Kundu; Timothy L. Kelly                                      | 2017 | Materials Chemistry Frontiers       | 56    | 10.1039/c7qm00396j            |                                    |

| Title                                                                                                                                     | Authors                                                               | Year | Venue                                   | Cited | DOI                           | OpenAlex<br>2-yr mean<br>citedness |
|-------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|------|-----------------------------------------|-------|-------------------------------|------------------------------------|
| Moisture-tolerant super-molecule for the stability enhancement of organic-inorganic perovskite solar cells in ambient air                 | Dong Wei; Hao Huang; Peng Cui et al.                                  | 2018 | Nanoscale                               | 53    | 10.1039/c8nr07638c            |                                    |
| Enhanced moisture stability of MAPbI <sub>3</sub> perovskite solar cells through Barium doping                                            | Jitendra Bahadur; Amir H. Ghahremani; Sunil Gupta et al.              | 2019 | Solar Energy                            | 51    | 10.1016/j.solener.2019.08.033 |                                    |
| Reduced Graphene Oxide Improves Moisture and Thermal Stability of Perovskite Solar Cells                                                  | Hui-Seon Kim; Bowen Yang; Minas M. Stylianakis et al.                 | 2020 | Cell Reports Physical Science           | 50    | 10.1016/j.xcrp.2020.100430    | 2026                               |
| A simple method to evaluate the effectiveness of encapsulation materials for perovskite solar cells                                       | Timothy J. Wilderspin; Francesca De Rossi; Trystan Watson             | 2016 | Solar Energy                            | 48    | 10.1016/j.solener.2016.09.038 |                                    |
| Simultaneously enhanced moisture tolerance and defect passivation of perovskite solar cells with cross-linked grain encapsulation         | Ke Xiao; Qiaolei Han; Yuan Gao et al.                                 | 2020 | Journal of Energy Chemistry             | 46    | 10.1016/j.jechem.2020.08.020  |                                    |
| Protecting Perovskite Solar Cells against Moisture-Induced Degradation with Sputtered Inorganic Barrier Layers                            | Ramez Hosseinian Ahangharnejhad; Zhaoning Song; Tamanna Mariam et al. | 2021 | ACS Applied Energy Materials            | 46    | 10.1021/acsaem.1c06566        | 2026                               |
| Role of interface in stability of perovskite solar cells                                                                                  | Chris Manspeaker; Swaminathan Venkatesan; Alex Zakhidov et al.        | 2016 | Current Opinion in Chemical Engineering | 45    | 10.1016/j.coche.2016.04.008   | 2026                               |
| Hysteresis and Instability Predicted in Moisture Degradation of Perovskite Solar Cells                                                    | Kelvin J. Xu; Ryan Taoran Wang; Fan Xu et al.                         | 2020 | ACS Applied Materials & Interfaces      | 38    | 10.1021/acsaami.3c01382       | 2026                               |
| Morphological Insights into the Degradation of Perovskite Solar Cells under Light and Humidity                                            | Kun Sun; Renjun Guo; Yuxin Liang et al.                               | 2023 | ACS Applied Materials & Interfaces      | 36    | 10.1021/acsaami.3c05382       | 2026                               |
| Moisture stability in nanostructured perovskite solar cells                                                                               | Mahdi Malekshahi Byravanand; Ali Nemati Kharat; Nima Taghavinia       | 2018 | Materials Letters                       | 31    | 10.1016/j.matlet.2018.10.029  |                                    |
| Encapsulation of Perovskite Solar Cells for High Humidity Conditions                                                                      | Qi Dong; Fangzhou Liu; Man Kwong Wong et al.                          | 2016 | ChemSusChem                             | 23    | 10.1002/cssc.201601490        | 2026                               |
| Degradation Kinetics of Inverted Perovskite Solar Cells                                                                                   | Mejd Alsari; Andrew J. Pearson; Jacob Tse-Wei Wang et al.             | 2018 | arXiv preprint                          | 0     | (link)                        |                                    |
| Degradation Dynamics of Perovskite Solar Cells Under Fixed Reverse Current Injection                                                      | Fangyuan Jiang; Haruka Koizumi; Hannah Contreras et al.               | 2026 | arXiv preprint                          | 0     | (link)                        |                                    |
| Effect of Ion Migration Induced Electrode Degradation on the Operational Stability of Perovskite Solar Cells                              | Boris Rivkin; Paul Fassl; Qing Sun et al.                             | 2019 | arXiv preprint                          | 0     | (link)                        |                                    |
| Predictive modeling of ion migration induced degradation in perovskite solar cells                                                        | Vikas Nandal; Pradeep R. Nair                                         | 2017 | arXiv preprint                          | 0     | (link)                        |                                    |
| Spontaneous Radiative Cooling to Enhance the Operational Stability of Perovskite Solar Cells via a Black-body-like Full Carbon Electrode  | Bingcheng Yu; Jiangjian Shi; Yiming Li et al.                         | 2022 | arXiv preprint                          | 0     | (link)                        |                                    |
| Quantifying Perovskite Solar Cell Degradation via Machine Learning from Spatially Resolved Multimodal Luminescence Time Series            | Giulio Barletta; Simon Ternes; Saif Ali et al.                        | 2026 | arXiv preprint                          | 0     | (link)                        |                                    |
| Thermal Degradation Mechanisms and Stability Enhancement Strategies in Perovskite Solar Cells: A Review                                   | Arghya Paul; Kanak Raj; Prince Raj Lawrence Raj et al.                | 2025 | arXiv preprint                          | 0     | (link)                        |                                    |
| Architecture Optimization Dramatically Improves Reverse Bias Stability in Perovskite Solar Cells: A Role of Polymer Hole Transport Layers | Fangyuan Jiang; Yangwei Shi; Tanka R. Rana et al.                     | 2023 | arXiv preprint                          | 0     | (link)                        |                                    |
| Improving efficiency and stability for perovskite solar cell with diethylene glycol dimethacrylate modification                           | Xiaodan Wei; Shuyuan Zhang; Boyang Ma et al.                          | 2025 | arXiv preprint                          | 0     | (link)                        |                                    |

| Title                                                                                                                                                  | Authors                                                                    | Year | Venue                                                                            | Cited | DOI                          | OpenAlex<br>2-yr mean<br>citedness |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|------|----------------------------------------------------------------------------------|-------|------------------------------|------------------------------------|
| Influence of the Cathodes Microstructure on the Stability of Inverted Planar Perovskite Solar Cells                                                    | Svetlana Sirotinskaya; Roland Schmechel; Niels Benson                      | 2019 | arXiv preprint                                                                   | 0     | (link)                       |                                    |
| Boosting Perovskite Solar Cell Stability: Dual Protection with Ultrathin Plasma Polymer Passivation Layers                                             | Mahmoud Nabil; Lidia Contreras-Bernal; Gloria P. Moreno-Martinez et al.    | 2024 | arXiv preprint                                                                   | 0     | (link)                       |                                    |
| Disentangling the origin of degradation in perovskite solar cells via optical imaging and Bayesian inference                                           | Akash Dasgupta; Robert D. J. Oliver; Manuel Kober-Czerny et al.            | 2026 | arXiv preprint                                                                   | 0     | (link)                       |                                    |
| Analysis of degradation in perovskite solar cells through physics-based machine learning                                                               | Kjeld O. Jensen; Gemma Giliberti; Aldo Di Carlo et al.                     | 2026 | arXiv preprint                                                                   | 0     | (link)                       |                                    |
| Critical role of water in defect aggregation and chemical degradation of perovskite solar cells                                                        | Yun-Hyok Kye; Chol-Jun Yu; Un-Gi Jong et al.                               | 2017 | arXiv preprint                                                                   | 0     | (link)                       |                                    |
| Ionic Charge Imbalance in Perovskite Solar Cells                                                                                                       | Dhyana Sivadas; Abhimanyu Singareddy; Chittiboina vinod et al.             | 2023 | arXiv preprint                                                                   | 0     | (link)                       |                                    |
| First-principles studies of oxygen interstitial dopants in RbPbI <sub>3</sub> halide for perovskite solar cells                                        | Chongyao Yang; Wei Wu; Kwang-Leong Choy                                    | 2023 | Modelling and Simulation in Materials Science and Engineering, 32, 015007 (2023) | 0     | 10.1088/1361-651X/ad104f     |                                    |
| Efficient indoor p-i-n hybrid perovskite solar cells using low temperature solution processed NiO as hole extraction layers                            | Lethy Krishnan Jagadamma; Oskar Blaszczyk; Muhammad T. Sajjad et al.       | 2019 | Solar Energy Materials and Solar Cells, 201, 110071, 2019                        | 0     | 10.1016/j.solmat.2019.110071 |                                    |
| Enhancing Intrinsic Stability of Hybrid Perovskite Solar Cell by Strong, yet Balanced, Electronic Coupling                                             | Fedwa El-Mellouhi; El Tayeb Bentría; Sergey N Rashkeev et al.              | 2016 | arXiv preprint                                                                   | 0     | (link)                       |                                    |
| Charge transport layer dependent electronic band bending in perovskite solar cells and its correlation to device degradation                           | Junseop Byeon; Jutae Kim; Ji-Young Kim et al.                              | 2019 | arXiv preprint                                                                   | 0     | (link)                       |                                    |
| Nitrobenzene as Additive to Improve Reproducibility and Degradation Resistance of Highly Efficient Methylammonium-Free Inverted Perovskite Solar Cells | Apostolos Ioakeimidis; Stelios. A. Choulis                                 | 2020 | Materials 2020, 13, 3289                                                         | 0     | 10.3390/ma13153289           |                                    |
| Ultrathin plasma polymer passivation of perovskite solar cells for improved stability and reproducibility                                              | Jose M. Obrero-Perez; Lidia Contreras-Bernal; Fernando Nunez-Galvez et al. | 2022 | arXiv preprint                                                                   | 0     | (link)                       |                                    |
| Material selection method for a perovskite solar cell design based on the genetic algorithm                                                            | Eungkyun Kim; Indranil Bhattacharya                                        | 2020 | arXiv preprint                                                                   | 0     | (link)                       |                                    |
| Low Frequency Carrier Kinetics in Perovskite Solar Cells                                                                                               | Vinod K. Sangwan; Menghua Zhu; Sarah Clark et al.                          | 2019 | arXiv preprint                                                                   | 0     | 10.1021/acsami.9b03884       |                                    |
| Multimodal operando microscopy reveals that interfacial chemistry and nanoscale performance disorder dictate perovskite solar cell stability           | Kyle Frohna; Cullen Chosy; Amran Al-Ashouri et al.                         | 2024 | arXiv preprint                                                                   | 0     | (link)                       |                                    |
| Bayesian parameter estimation for characterising mobile ion vacancies in perovskite solar cells                                                        | Samuel G. McCallum; Oliver Nicholls; Kjeld O. Jensen et al.                | 2023 | arXiv preprint                                                                   | 0     | (link)                       |                                    |
| Enhanced Power Point Tracking for High Hysteresis Perovskite Solar Cells: A Galvanostatic Approach                                                     | Emilio J. Juarez-Perez; Cristina Momblona; Roberto Casas et al.            | 2023 | Cell Reports Physical Science, 2024, 5, 101885                                   | 0     | 10.1016/j.xcrp.2024.101885   |                                    |
| Quantification of mobile ions in perovskite solar cells with thermally activated ion current measurements                                              | Moritz C. Schmidt; Agustin O. Alvarez; Riccardo Pallotta et al.            | 2025 | arXiv preprint                                                                   | 0     | (link)                       |                                    |
| Photo Stabilization of p-i-n Perovskite Solar Cells with Bathocuproine: MXene                                                                          | Anastasia Yusheva; Danila Saranin; Dmitry Muratov et al.                   | 2023 | Small, 2022                                                                      | 0     | (link)                       |                                    |
| Water-resistant hybrid perovskite solar cell – drop triboelectric energy harvester                                                                     | Fernando Nunez-Galvez; Xabier Garcia-Casas; Lidia Contreras Bernal et al.  | 2024 | Nano Energy, vol. 148, 2026, 111678                                              | 0     | 10.1016/j.nanoen.2025.111678 |                                    |
| Deposition of Reduced Graphene Oxide Thin Film by Spray Pyrolysis Method for Perovskite Solar Cell                                                     | Manoj Pandey; Dipendra Hamal; Deepak Subedi et al.                         | 2022 | arXiv preprint                                                                   | 0     | 10.3126/jnphysoc.v7i3.42193  |                                    |

| Title                                                                                                                                                                                       | Authors                                                                     | Year | Venue                                               | Cited | DOI                                | OpenAlex<br>2-yr mean<br>citedness |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|------|-----------------------------------------------------|-------|------------------------------------|------------------------------------|
| Degradation Analysis of Perovskite Solar Cells via Short-Circuit Impedance Spectroscopy: A case study on NiOx passivation                                                                   | Osbel Almora; Pilar López-Varo; Renán Escalante et al.                      | 2024 | arXiv preprint                                      | 0     | 10.1063/5.0216983                  |                                    |
| Simultaneous Optimization of Efficiency and Degradation in Tunable HTL-Free Perovskite Solar Cells with MWCNT-Integrated Back Contact Using a Machine Learning-Derived Polynomial Regressor | Ihtesham Ibn Malek; Hafiz Imtiaz; Samia Subrina                             | 2025 | arXiv preprint                                      | 0     | (link)                             |                                    |
| Influence of Phase Segregation on the Hysteresis of Perovskite Solar Cells                                                                                                                  | Abhimanyu Singareddy; Pradeep R. Nair                                       | 2024 | arXiv preprint                                      | 0     | (link)                             |                                    |
| Pulsatile therapy for perovskite solar cells                                                                                                                                                | Kiwan Jeong; Junseop Byeon; Jihun Jang et al.                               | 2020 | arXiv preprint                                      | 0     | (link)                             |                                    |
| LLM tools in the prediction of the stability of perovskite solar cells                                                                                                                      | S. Frenkel; V. Zakharov; E. A. Katz                                         | 2025 | arXiv preprint                                      | 0     | (link)                             |                                    |
| Efficiency limits of Perovskite Solar Cells with Transition Metal Oxides as Hole Transport Layers                                                                                           | Dhyana Sivadas; Swasti Bhatia; Pradeep Nair                                 | 2021 | arXiv preprint                                      | 0     | 10.1063/5.0059221                  |                                    |
| Enhanced thermal stability of inverted perovskite solar cells by bulky passivation with pyridine-functionalized triphenylamine                                                              | Ekaterina A. Ilicheva; Irina A. Chuyko; Lev O. Luchnikov et al.             | 2025 | arXiv preprint                                      | 0     | (link)                             |                                    |
| Employing surfactant-assisted hydrothermal synthesis to control CuGaO2 nanoparticle formation and improved carrier selectivity of perovskite solar cells                                    | Ioannis T. Papadas; Achilleas Savva; Apostolos Ioakeimidis et al.           | 2018 | Materials Today Energy, Volume 8, Pages 57-64, 2018 | 0     | 10.1016/j.mtener.2018.03.003       |                                    |
| Experimental demonstration of ions induced electric field in perovskite solar cells                                                                                                         | Zeguo Tang; Takashi Minemoto                                                | 2015 | arXiv preprint                                      | 0     | (link)                             |                                    |
| Quantification of Ion Migration in CH3NH3PbI3 Perovskite Solar Cells by Transient Capacitance Measurements                                                                                  | Moritz H. Futscher; Ju Min Lee; Lucie McGovern et al.                       | 2018 | arXiv preprint                                      | 0     | 10.1039/c9mh00445a                 |                                    |
| Enhanced Performance and Stability of Perovskite Solar Cells with Ag-Cu-Zn Alloy Electrodes                                                                                                 | Keshav Kumar Sharma; Ashutosh Ujjwal; Rohit Saini et al.                    | 2025 | Energy Technology, 2025, e202501392                 | 0     | 10.1002/ente.202501392             |                                    |
| The hysteresis-free behavior of perovskite solar cells from the perspective of the measurement conditions                                                                                   | George Alexandru Nemnes; Cristina Besleaga; Andrei Gabriel Tomulescu et al. | 2018 | Journal of Materials Chemistry C 2019               | 0     | 10.1039/C8TC05999C                 |                                    |
| A Design Based on Staircase Band Alignment of Electron Transport Layer for Improving Performance and Stability in Planar Perovskite Solar Cells                                             | Shang-Hsuan Wu; Ming-Yi Lin; Sheng-Hao Chang et al.                         | 2017 | arXiv preprint                                      | 0     | (link)                             |                                    |
| Improved evaluation of deep-level transient spectroscopy on perovskite solar cells reveals ionic defect distribution                                                                        | Sebastian Reichert; Jens Flemming; Qingzhi An et al.                        | 2019 | Phys. Rev. Applied 13, 034018 (2020)                | 0     | 10.1103/PhysRevApplied.13.034018   |                                    |
| Automated and Scalable SEM Image Analysis of Perovskite Solar Cell Materials via a Deep Segmentation Framework                                                                              | Jian Guo Pan; Lin Wang; Xia Cai                                             | 2025 | arXiv preprint                                      | 0     | (link)                             |                                    |
| Double-side Integration of the Fluorinated Self-Assembling Monolayers for Enhanced Stability of Inverted Perovskite Solar Cells                                                             | Ekaterina A. Ilicheva; Polina K. Sukhorukova; Lev O. Luchnikov et al.       | 2024 | arXiv preprint                                      | 0     | (link)                             |                                    |
| Unraveling UV Stability in Metal Halide Perovskites: From Degradation Mechanisms to Molecular Passivation                                                                                   | Xin Wen; Zhiyi Yao; Wenzhuo Li et al.                                       | 2025 | arXiv preprint                                      | 0     | (link)                             |                                    |
| Impact of Doping Spiro-OMeTAD with Li-TFSI, FK209, and tBP on the Performance of Perovskite Solar Cells                                                                                     | Mehran Hosseinzadeh Dizaj                                                   | 2024 | arXiv preprint                                      | 0     | 10.22036/org.chem.2024.494577.1370 |                                    |

| Title                                                                                                                                                          | Authors                                                           | Year | Venue                                             | Cited | DOI                               | OpenAlex<br>2-yr mean<br>citedness |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|------|---------------------------------------------------|-------|-----------------------------------|------------------------------------|
| Touching is believing: interrogating organometal halide perovskite solar cells at the nanoscale via scanning probe microscopy                                  | Jiangyu Li; Boyuan Huang; Ehsan Nasr Esfahani et al.              | 2017 | npj Quantum Materials 2, Article number: 56, 2017 | 0     | 10.1038/s41535-017-0061-4         |                                    |
| Dopant-free molecular hole transport material that mediates a 20% power conversion efficiency in a perovskite solar cell                                       | Yang Cao; Yunlong Li; Thomas Morrissey et al.                     | 2019 | arXiv preprint                                    | 0     | (link)                            |                                    |
| Comparison of Electroluminescence and Photoluminescence Imaging of Mixed-Cation Mixed-Halide Perovskite Solar Cells at Low Temperatures                        | Hurriyet Yuce-Cakir; Hao-ran Chen; Isaac Ogunniran et al.         | 2025 | arXiv preprint                                    | 0     | (link)                            |                                    |
| Operando multidimensional spectroscopy reveals A-site-dependent carrier cooling in perovskite solar cells                                                      | Edoardo Amarotti; Luca Bolzonello; Qi Shi et al.                  | 2026 | arXiv preprint                                    | 0     | (link)                            |                                    |
| La-Doped BaSnO <sub>3</sub> Electron Transport Layer for Perovskite Solar Cells                                                                                | Chang Woo Myung; Geun-sik Lee; Kwang S. Kim                       | 2018 | arXiv preprint                                    | 0     | (link)                            |                                    |
| NiOx passivation in perovskite solar cells: from surface reactivity to device performance                                                                      | John Mohanraj; Bipasa Samanta; Osbel Almora et al.                | 2024 | arXiv preprint                                    | 0     | (link)                            |                                    |
| Phenothiazine-Based Self-Assembled Monolayer with Thiophene Head Groups Minimizes Buried Interface Losses in Tin Perovskite Solar Cells                        | Valerio Stacchini; Madineh Rastgoo; Mantas Marčinskis et al.      | 2025 | arXiv preprint                                    | 0     | 10.1002/aenm.202500841            |                                    |
| Exploring the Way to Approach the Efficiency Limit of Perovskite Solar Cells by Drift-Diffusion Model                                                          | Xingang Ren; Zishuai Wang; Wei E. I. Sha et al.                   | 2017 | ACS Photonics, 2017, 4(4), 934-942                | 0     | 10.1021/acsphotonics.6b01043      |                                    |
| Triphenylamine-based interlayer with carboxyl anchoring group for tuning of charge collection interface in stabilized p-i-n perovskite solar cells and modules | P. K. Sukhorukova; E. A. Ilicheva; P. A. Gostishchev et al.       | 2023 | arXiv preprint                                    | 0     | (link)                            |                                    |
| Ion Induced Passivation of Grain Boundaries in Perovskite Solar Cells                                                                                          | Vikas Nandal; Pradeep R. Nair                                     | 2018 | arXiv preprint                                    | 0     | 10.1063/1.5082967                 |                                    |
| Super-droplet-repellent carbon-based printable perovskite solar cells                                                                                          | Cuc Thi Kim Mai; Janne Halme; Heikki A. Nurmi et al.              | 2024 | arXiv preprint                                    | 0     | (link)                            |                                    |
| An eco-friendly passivation strategy of resveratrol for highly efficient and anti-oxidative perovskite solar cells                                             | Xianhu Wu; Jieyu Bi; Guanglei Cui et al.                          | 2024 | arXiv preprint                                    | 0     | (link)                            |                                    |
| Normal and inverted hysteresis in perovskite solar cells                                                                                                       | George Alexandru Nemnes; Cristina Besleaga; Viorica Stancu et al. | 2017 | arXiv preprint                                    | 0     | (link)                            |                                    |
| Improved Performance and Reliability of p-i-n Perovskite Solar Cells via Doped Metal Oxides                                                                    | Achilleas Savva; Ignasi Burgues-Ceballos; Stelios A. Choulis      | 2016 | Advanced Energy Materials 2016, 1600285           | 0     | 10.1002/aenm.201600285            |                                    |
| Influences of Dielectric Constant and Scan Rate to Hysteresis Effect in Perovskite Solar Cell: Simulation and Experimental Analyses                            | Jun-Yu Huang; You-Wei Yang; Wei-Hsuan Hsu et al.                  | 2021 | Scientific Reports, 12, 7927, 2022                | 0     | 10.1038/s41598-022-11899-x        |                                    |
| Optimization and Performance Evaluation of Cs <sub>2</sub> CuBiCl <sub>6</sub> Double Perovskite Solar Cell for Lead-Free Photovoltaic Applications            | Syeda Anber Urooj Wasti; Sundus Naz; Ammara Sattar et al.         | 2025 | arXiv preprint                                    | 0     | (link)                            |                                    |
| Remarkable performance recovery in highly defective perovskite solar cells by photo-oxidation                                                                  | Katelyn P. Goetz; Fabian T. F. Thome; Qingzhi An et al.           | 2024 | arXiv preprint                                    | 0     | (link)                            |                                    |
| Towards low-temperature processing of efficient Cs-Pb perovskite solar cells                                                                                   | Zongbao Zhang; Ran Ji; Yvonne J. Hofstetter et al.                | 2024 | arXiv preprint                                    | 0     | (link)                            |                                    |
| Polyaniline as a conductive polymer and its role in improving the efficiency and conductivity of perovskite solar cells                                        | Mehran Hosseinzadeh Dizaj                                         | 2026 | arXiv preprint                                    | 0     | 10.11591/ijped.v16.i4.pp2731-2743 |                                    |

| Title                                                                                                                                                                                                 | Authors                                                               | Year | Venue                                       | Cited   | DOI                               | OpenAlex<br>2-yr mean<br>citedness |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|------|---------------------------------------------|---------|-----------------------------------|------------------------------------|
| Gefitinib-Induced Interface Engineering Enhances the Defect Formation Energy for Highly Efficient and Stable Perovskite Solar Cells                                                                   | Xianhu Wu; Guanglei Cui; Jieyu Bi et al.                              | 2025 | arXiv preprint                              | 0       | (link)                            |                                    |
| Imaging real-time amorphization of hybrid perovskite solar cells under electrical biasing                                                                                                             | Min-cheol Kim; Namyoun Ahn; Diyi Cheng et al.                         | 2020 | arXiv preprint                              | 0       | (link)                            |                                    |
| Surface Treatment of Cu:NiOx Hole-Transporting Layer Using $\eta$ -Alanine for Hysteresis-Free and Thermally Stable Inverted Perovskite Solar Cells                                                   | Fedros Galatopoulos; Ioannis T. Papadas; Apostolos Ioakeimidis et al. | 2020 | Nanomaterials 10(10), 1961                  | 2020, 0 | 10.3390/nano10101961              |                                    |
| Exploring Lead Free Mixed Halide Double Perovskites Solar Cell                                                                                                                                        | Md Yekra Rahman; Sharif Mohammad Mominuzzaman                         | 2024 | arXiv preprint                              | 0       | 10.1109/ICECE64886.2024.11024609  |                                    |
| A green solvent system for precursor phase-engineered sequential deposition of stable formamidinium lead triiodide for perovskite solar cells                                                         | Benjamin M. Gallant; Philippe Holzhey; Joel A. Smith et al.           | 2024 | arXiv preprint                              | 0       | (link)                            |                                    |
| Impact of interface defects on the band alignment and performance of $\text{TiO}_2/\text{MAPI}/\text{Cu}_2\text{O}$ perovskite solar cells                                                            | Nicolae Filipoiu; Marina Cuzminschi; Mihaela Cosinschi et al.         | 2024 | Physica Scripta 265906 (2026)               | 101, 0  | 10.1088/1402-4896/ae7cef          |                                    |
| Efficient Passivation of Surface Defects by Lewis Base in Lead-free Tin-based Perovskite Solar Cells                                                                                                  | Hejin Yan; Bowen Wang; Xuefei Yan et al.                              | 2022 | Materials Today Energy (2022)               | 0       | 10.1016/j.mtener.2022.101038      |                                    |
| Capacitive and inductive effects in perovskite solar cells: the different roles of ionic current and ionic charge accumulation                                                                        | Nicolae Filipoiu; Amanda Teodora Preda; Dragos Victor Anghel et al.   | 2022 | Phys. Rev. Applied 18, 064087 (2022)        | 0       | (link)                            |                                    |
| First-Principles Study on Material Properties and Stability of Inorganic Halide Perovskite Solid Solutions $\text{CsPb}(\text{I}_{1-x}\text{Br}_x)_3$ towards High Performance Perovskite Solar Cells | Chol-Jun Yu; Un-Hyok Ko; Suk-Gyong Hwang et al.                       | 2019 | Phys. Rev. Materials 4, 045402 (2020)       | 0       | 10.1103/PhysRevMaterials.4.045402 |                                    |
| An eco-friendly universal strategy via ribavirin to achieve highly efficient and stable perovskite solar cells (111)                                                                                  | Xianhu Wu; Gaojie Xia; Guanglei Cui et al.                            | 2025 | arXiv preprint                              | 0       | (link)                            |                                    |
| Facet-engineered $\text{SnO}_2$ as Electron Transport Layer for Efficient and Stable Triple-Cation Perovskite Solar Cells                                                                             | Keshav Kumar Sharma; Rohit; Sochannao Machinao et al.                 | 2025 | Sustainable Energy Fuels, 2025,9, 3102-3109 | 0       | 10.1039/D5SE00339C                |                                    |
| Transparent Multispectral Photonic Electrode for All-Weather Stable and Efficient Perovskite Solar Cells                                                                                              | George Perrakis; Anna C. Tasolamprou; George Kakavelakis et al.       | 2023 | arXiv preprint                              | 0       | (link)                            |                                    |
| Atomistic mechanism for trapped-charge driven degradation of perovskite solar cells                                                                                                                   | Kwisung Kwak; Eunhak Lim; Namyoun Ahn et al.                          | 2017 | arXiv preprint                              | 0       | (link)                            |                                    |
| A simple all-inorganic hole-only structure for trap density measurement in perovskite solar cells                                                                                                     | Atena mohamadnezhad; Alireza Fathi-Beiraghvandi; Mahmoud Samadpour    | 2023 | arXiv preprint                              | 0       | (link)                            |                                    |
| Deep-level transient spectroscopy of the charged defects in p-i-n perovskite solar cells induced by light-soaking                                                                                     | A. A. Vasilev; D. S. Saranin; P. A. Gostishchev et al.                | 2022 | arXiv preprint                              | 0       | (link)                            |                                    |
| Turning bad into good: a water-splitting-active hole transporting material to preserve the performance of perovskite solar cells in humid environments                                                | Min Kim; Antonio Alfano; Giovanni Perotto et al.                      | 2020 | arXiv preprint                              | 0       | (link)                            |                                    |
| AI-supported Degradation Study of Carbon-based Perovskite Solar Cells: Learning the Device Physics of Perovskite Solar Cells: A Drift-Diffusion Guided Autoencoder Approach                           | Oliver Zbinden; Sharun Parayil Shaji; Wolfgang Tress                  | 2026 | arXiv preprint                              | 0       | (link)                            |                                    |

| Title                                                                                                                                                                                                           | Authors                                                              | Year | Venue                                                          | Cited | DOI                            | OpenAlex<br>2-yr mean<br>citedness |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|------|----------------------------------------------------------------|-------|--------------------------------|------------------------------------|
| Separation of Hoke and Schottky effects for improvement of mixed halide perovskite solar cell stability                                                                                                         | Vladimir Ivanov; Eduard Ageev; Denis Danilov et al.                  | 2025 | arXiv preprint                                                 | 0     | (link)                         |                                    |
| Local Nanoscale Defective Phase Impurities Are the Sites of Degradation in Halide Perovskite Devices                                                                                                            | Stuart Macpherson; Tiaran A. S. Doherty; Andrew J. Winchester et al. | 2021 | arXiv preprint                                                 | 0     | (link)                         |                                    |
| Combined DFT, SCAPS-1D, and wxAMPS frameworks for design optimization of efficient Cs <sub>2</sub> BiAgI <sub>6</sub> -based perovskite solar cells with different charge transport layers                      | M. Khalid Hossain; A. A. Arnab; Ranjit C. Das et al.                 | 2022 | RSC Advance 12 (2022)                                          | 0     | 10.1039/D2RA06734J             |                                    |
| A Comprehensive Computational Photovoltaic Study of Lead-free Inorganic NaSnCl <sub>3</sub> -based Perovskite Solar Cell: Effect of Charge Transport Layers and Material Parameters                             | Md Tashfiq Bin Kashem; Sadia Anjum Esha                              | 2025 | arXiv preprint                                                 | 0     | (link)                         |                                    |
| Ab Initio Thermodynamic Study of PbI <sub>2</sub> and CH <sub>3</sub> NH <sub>3</sub> PbI <sub>3</sub> Surfaces in Reaction with CH <sub>3</sub> NH <sub>2</sub> Gas for Perovskite Solar Cells                 | Yun-Sim Kim; Chol-Hyok Ri; Yun-Hyok Kye et al.                       | 2021 | arXiv preprint                                                 | 0     | (link)                         |                                    |
| Structural Stability and Optoelectronic Properties of Tetragonal MAPbI <sub>3</sub> under Strain                                                                                                                | Lei Guo; Gao Xu; Gang Tang et al.                                    | 2020 | Nanotechnology 31, 225204 (2020)                               | 0     | 10.1088/1361-6528/ab7679       |                                    |
| Stability Engineering of Halide Perovskite via Machine Learning                                                                                                                                                 | Zhenzhu Li; Qichen Xu; Qingde Sun et al.                             | 2018 | arXiv preprint                                                 | 0     | (link)                         |                                    |
| Nanoparticulate Metal Oxide Top Electrode Interface Modification Improves the Thermal Stability of Inverted Perovskite Photovoltaics                                                                            | Ioannis T. Papadas; Fedros Galatopoulos; Gerasimos S. Armatas et al. | 2019 | Nanomaterials 2019, 9(11), 1616                                | 0     | 10.3390/nano9111616            |                                    |
| Long Thermal Stability of Inverted Perovskite Photovoltaics Incorporating Fullerene-based Diffusion Blocking Layer                                                                                              | Fedros Galatopoulos; Ioannis T. Papadas; Gerasimos S. Armatas et al. | 2019 | Adv. Mater. Interfaces 2018, 5, 1800280                        | 0     | 10.1002/admi.201800280         |                                    |
| Revealing the stability and efficiency enhancement in mixed halide perovskites MAPb(I <sub>1-x</sub> Cl <sub>x</sub> ) <sub>3</sub> with ab initio calculations                                                 | Un-Gi Jong; Chol-Jun Yu; Yong-Man Jang et al.                        | 2016 | Journal of Power Sources, Volume 350, 15 May 2017, Pages 65-72 | 0     | 10.1016/j.jpowsour.2017.03.038 |                                    |
| Stabilization of interfaces for double-cation halide perovskites with AVA2FAPb2I7 additives                                                                                                                     | Lev O. Luchnikov; Ekaterina A. Ilicheva; Victor A. Voronov et al.    | 2025 | arXiv preprint                                                 | 0     | (link)                         |                                    |
| Impermeable Inorganic Walls Sandwiching Photoactive Layer toward Inverted Perovskite Solar and Indoor-Photovoltaic Devices                                                                                      | Jie Xu; Jun Xi; Hua Dong et al.                                      | 2020 | arXiv preprint                                                 | 0     | (link)                         |                                    |
| Impact of Grain Boundaries on Efficiency and Stability of Organic-Inorganic Trihalide Perovskites                                                                                                               | Zhaodong Chu; Mengjin Yang; Philip Schulz et al.                     | 2016 | arXiv preprint                                                 | 0     | 10.1038/s41467-017-02331-4     |                                    |
| Computational Prediction of Structural, Electronic, Optical Properties and Phase Stability of Double Perovskites K <sub>2</sub> SnX <sub>6</sub> (X = I, Br, Cl)                                                | Un-Gi Jong; Chol-Jun Yu; Yun-Hyok Kye                                | 2019 | arXiv preprint                                                 | 0     | (link)                         |                                    |
| Charting the thermodynamic stability of hybrid perovskite alloys with machine learning                                                                                                                          | Jarno Laakso; Armi Tiihonen; Patrick Rinke                           | 2026 | arXiv preprint                                                 | 0     | (link)                         |                                    |
| Crystal structure, stability and optoelectronic properties of the organic-inorganic wide bandgap perovskite CH <sub>3</sub> NH <sub>3</sub> BaI <sub>3</sub> : Candidate for transparent conductor applications | Akash Kumar; K. R. Balasubramaniam; Jiban Kangsabanik et al.         | 2016 | Phys. Rev. B 94, 180105 (2016)                                 | 0     | 10.1103/PhysRevB.94.180105     |                                    |



| Title                                                                                                                                                                                                                               | Authors                                                         | Year | Venue                                                                                                                           | Cited | DOI                        | OpenAlex<br>2-yr mean<br>citedness |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|------|---------------------------------------------------------------------------------------------------------------------------------|-------|----------------------------|------------------------------------|
| Eliminating the Electric Field Response in a Perovskite Heterojunction Solar Cell to Improve Operational Stability                                                                                                                  | Jiangjian Shi; Yiming Li; Yusheng Li et al.                     | 2020 | arXiv preprint                                                                                                                  | 0     | (link)                     |                                    |
| Improvement of both performance and stability of photovoltaic devices by in situ formation of a sulfur-based 2D perovskite                                                                                                          | Milon Kundar; Sahil Bhandari; Sein Chung et al.                 | 2022 | arXiv preprint                                                                                                                  | 0     | (link)                     |                                    |
| Enhanced stability of 2D organic-inorganic halide perovskites by doping and heterostructure engineering                                                                                                                             | Rahul Singh; Prashant Singh; Ganesh Balasubramanian             | 2018 | arXiv preprint                                                                                                                  | 0     | (link)                     |                                    |
| Influence of halide composition on the structural, electronic, and optical properties of mixed $\text{CH}_3\text{NH}_3\text{Pb}(\text{I}_{1-x}\text{Br}_x)_3$ perovskites calculated using the virtual crystal approximation method | Un-Gi Jong; Chol-Jun Yu; Nam-Hyok Kim et al.                    | 2016 | Phys. Rev. B 94, 125139 (2016)                                                                                                  | 0     | 10.1103/PhysRevB.94.125139 |                                    |
| Predicting Organic-Inorganic Halide Perovskite Photovoltaic Performance from Optical Properties of Constituent Films through Machine Learning                                                                                       | Ruiqi Zhang; Brandon Motes; Shaun Tan et al.                    | 2024 | arXiv preprint                                                                                                                  | 0     | (link)                     |                                    |
| Protective Coating Interfaces for perovskite Solar Cell Materials: A first Principles Study                                                                                                                                         | Azimatu Fangnon; Marc Dvorak; Ville Havu et al.                 | 2022 | arXiv preprint                                                                                                                  | 0     | 10.1021/acsami.1c21785     |                                    |
| Microscopic Intricacies of Self-Healing in Halide Perovskite-Charge Transport Layer Heterostructures                                                                                                                                | Tejmani Behera; Boris Louis; Lukas Paesen et al.                | 2025 | arXiv preprint                                                                                                                  | 0     | (link)                     |                                    |
| Multi-site mixing and entropy stabilization of $\text{CsPbI}_3$ with potential application in photovoltaics                                                                                                                         | Namitha Anna Koshi; Krishnamohan Thekkepat; Doh-Kwon Lee et al. | 2025 | arXiv preprint                                                                                                                  | 0     | (link)                     |                                    |
| Photo-effect on ion transport in mixed cation and halide perovskites and implications for photo de-mixing                                                                                                                           | Gee Yeong Kim; Alessandro Senocrate; Yaru Wang et al.           | 2019 | arXiv preprint                                                                                                                  | 0     | (link)                     |                                    |
| A microstructural analysis of 2D halide perovskites: Stability and functionality                                                                                                                                                    | Susmita Bhattacharya; Goutam Kumar Chandra; P. Predeep          | 2021 | Frontiers in Nanotechnology 2021                                                                                                | 0     | 10.3389/fnano.2021.657948  |                                    |
| Experimentally observed defect tolerance in the electronic structure of lead bromide perovskites                                                                                                                                    | Gabriel J. Man; Aleksandr Kalinko; Dibya Phuyal et al.          | 2023 | arXiv preprint                                                                                                                  | 0     | (link)                     |                                    |
| What Happens at Surfaces and Grain Boundaries of Halide Perovskites: Insights from Reactive Molecular Dynamics Simulations of $\text{CsPbI}_3$                                                                                      | Mike Pols; Tobias Hilpert; Ivo Filot et al.                     | 2022 | ACS Appl. Mater. Interfaces 2022, 14, 36, 40841-40850                                                                           | 0     | 10.1021/acsami.2c09239     |                                    |
| Influence of interstitial Li on the electronic properties of $\text{Li}_x\text{CsPbI}_3$ for photovoltaic and battery applications                                                                                                  | Wei Wei; Julian Gebhardt; Daniel F. Urban et al.                | 2024 | Physical Review B 112, 245103 (2025)                                                                                            | 0     | 10.1103/nvxh-kwzt          |                                    |
| Transmission electron microscopy of organic-inorganic hybrid perovskites: myths and truths                                                                                                                                          | Shulin Chen; Ying Zhang; Jinjin Zhao et al.                     | 2020 | Science Bulletin. 65, 1643-1649 (2020)                                                                                          | 0     | 10.1016/j.scib.2020.05.020 |                                    |
| Hexagonal $\text{MASnI}_3$ exhibiting strong absorption of ultraviolet photons                                                                                                                                                      | Qiaoqiao Li; Wenhui Wan; Yanfeng Ge et al.                      | 2019 | Appl. Phys. Lett. 114, 101906 (2019);                                                                                           | 0     | 10.1063/1.5087649          |                                    |
| First-principles prediction into robust high-performance photovoltaic double perovskites $\text{A}_2\text{SiI}_6$ (A = K, Rb, Cs)                                                                                                   | Qiaoqiao Li; Liujiang Zhou; Yanfeng Ge et al.                   | 2019 | arXiv preprint                                                                                                                  | 0     | (link)                     |                                    |
| Multiscale Models For Perovskite Optimisation                                                                                                                                                                                       | Philippe Baranek; James P. Connolly; Antoine Gissler et al.     | 2025 | EUPVSEC 2025 42nd European Photovoltaic Solar Energy Conference and Exhibition, WIP Renewable Energies, Sep 2025, Bilbao, Spain | 0     | (link)                     |                                    |

| Title                                                                                                                                                                   | Authors                                                         | Year | Venue                                                              | Cited | DOI                             | OpenAlex<br>2-yr mean<br>citedness |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|------|--------------------------------------------------------------------|-------|---------------------------------|------------------------------------|
| Ionicallly gated perovskite solar cell with tunable carbon nanotube interface at thick fullerene electron transporting layer: comparison to gated OPV                   | D. S. Saranin; D. S. Murov; R. Haroldson et al.                 | 2019 | arXiv preprint                                                     | 0     | (link)                          |                                    |
| Molecular dynamics simulations of nucleation of hexagonal (\$\$) and cubic (\$\$)-FAPbI <sub>3</sub> perovskites from solution                                          | Paramvir Ahlawat                                                | 2023 | arXiv preprint                                                     | 0     | (link)                          |                                    |
| Stable Perovskite Solar Cells via exfoliated graphite as an ion diffusion-blocking layer                                                                                | Abdullah S. Alharbi; Miqad S. Albishi; Temur Maksudov et al.    | 2024 | arXiv preprint                                                     | 0     | (link)                          |                                    |
| Generalised Framework for Controlling and Understanding Ion Dynamics with Passivated Lead Halide Perovskites                                                            | Tomi K. Baikie; Philip Calado; Krzysztof Galkowski et al.       | 2023 | arXiv preprint                                                     | 0     | (link)                          |                                    |
| Stabilizing Solution-Substrate Interaction of Perovskite Ink on PEDOT:PSS for Scalable Blade Coated Narrow Bandgap Perovskite Solar Modules by Gas Quenching            | Severin Siegrist; Johnpaul K. Pious; Huagui Lai et al.          | 2024 | arXiv preprint                                                     | 0     | (link)                          |                                    |
| Design of Lead-Free Inorganic Halide Perovskites for Solar Cells via Cation-Transmutation                                                                               | Xin-Gang Zhao; Ji-Hui Yang; Yuhao Fu et al.                     | 2017 | J. Am. Chem. Soc. 139, 2630 (2017)                                 | 0     | 10.1021/jacs.6b09645            |                                    |
| Unraveling the water degradation mechanism of CH <sub>3</sub> NH <sub>3</sub> PbI <sub>3</sub>                                                                          | Chao Zheng; Oleg Rubel                                          | 2019 | J. Phys. Chem. C 123, 19385-19394 (2019)                           | 0     | 10.1021/acs.jpcc.9b05516        |                                    |
| Physics-based material parameters extraction from perovskite experiments via Bayesian optimization                                                                      | Hualin Zhan; Viqar Ahmad; Azul Mayon et al.                     | 2024 | arXiv preprint                                                     | 0     | 10.1039/D4EE00911H              |                                    |
| Influence of water intercalation and hydration on chemical decomposition and ion transport in methylammonium lead halide perovskites                                    | Un-Gi Jong; Chol-Jun Yu; Gum-Chol Ri et al.                     | 2017 | arXiv preprint                                                     | 0     | (link)                          |                                    |
| Terahertz Time-Domain Spectroscopy as a Universal Defect Fingerprinting Tool for Organic Halide Perovskite Solar Cells                                                  | Inhee Maeng; Young Mi Lee; Jinwoo Park et al.                   | 2026 | arXiv preprint                                                     | 0     | (link)                          |                                    |
| Post-Transition Metal Sn-Based Chalcogenide Perovskites: A Promising Lead-Free and Transition Metal Alternative for Stable, High-Performance Photovoltaics              | Surajit Adhikari; Sankhasuvra Das; Priya Johari                 | 2024 | arXiv preprint                                                     | 0     | 10.1039/D4TC04701J              |                                    |
| Light Intensity Modulated Impedance Spectroscopy (LIMIS) in All-Solid-State Solar Cells at Open Circuit                                                                 | Osbel Almora; Yicheng Zhao; Xiaoyan Du et al.                   | 2019 | arXiv preprint                                                     | 0     | 10.1016/j.nanoen.2020.104982    |                                    |
| Performance Analysis of Double Perovskite-Based Solar Cells Using SCAPS-1D Simulation: A brief review                                                                   | H. Laltlanmawii; Lalrem Kima; Mahabur Rahman et al.             | 2026 | arXiv preprint                                                     | 0     | (link)                          |                                    |
| Intrinsic Instability of the Hybrid Halide Perovskite Semiconductor CH <sub>3</sub> NH <sub>3</sub> PbI <sub>3</sub>                                                    | Yue-Yu Zhang; Shiyu Chen; Peng Xu et al.                        | 2015 | Chin. Phys. Lett. 35, 036104 (2018)                                | 0     | 10.1088/0256-307X/35/3/036104   |                                    |
| Effect of heterostructure engineering on electronic structure and transport properties of two-dimensional halide perovskites                                            | Rahul Singh; Prashant Singh; Ganesh Balasubramania              | 2021 | Computational Materials Science, Volume 200, December 2021, 110823 | 0     | 10.1016/j.commatsci.2021.110823 |                                    |
| Molecular Bond Engineering and Feature Learning for the Design of Hybrid Organic-Inorganic Perovskites Solar Cells with Strong Non-Covalent Halogen-Cation Interactions | Johannes Teunissen; Fabiana da Pieve                            | 2021 | arXiv preprint                                                     | 0     | 10.1021/acs.jpcc.1c07295        |                                    |
| Inverted Perovskite Photovoltaics using Flame Spray Pyrolysis Solution based CuAlO <sub>2</sub> /Cu-O Hole Selective Contact                                            | Achilleas Savva; Ioannis T. Papadas; Dimitris Tsikritzis et al. | 2019 | ACS Applied Energy Materials, 2, 3, 2276-2287, 2019                | 0     | 10.1021/acsaem.9b00070          |                                    |

| Title                                                                                                                                                     | Authors                                                           | Year | Venue                               | Cited | DOI                          | OpenAlex<br>2-yr mean<br>citedness |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|------|-------------------------------------|-------|------------------------------|------------------------------------|
| Unraveling the varied nature and roles of defects in hybrid halide perovskites with time-resolved photoemission electron microscopy                       | Sofia Kosar; Andrew J. Winchester; Tiarnan A. S. Doherty et al.   | 2021 | arXiv preprint                      | 0     | (link)                       |                                    |
| More than just light management – The multiple advantages of nano- and micro-textures in perovskite solar cells                                           | Guillermo Martínez-Denegri; Christiane Becker                     | 2026 | arXiv preprint                      | 0     | (link)                       |                                    |
| Lattice matched heterogeneous nucleation eliminate defective buried interface in halide perovskites                                                       | Paramvir Ahlawat; Cecilia Clementi; Felix Musil et al.            | 2024 | arXiv preprint                      | 0     | (link)                       |                                    |
| Enhanced Electron Extraction in Co-Doped TiO <sub>2</sub> Quantified by Drift-Diffusion Simulation for Stable CsPbI <sub>3</sub> Solar Cells              | Thomas W. Gries; Davide Regaldo; Hans Koebler et al.              | 2024 | Small Sci. 2025, 5, 2400578         | 0     | 10.1002/smsc.202400578       |                                    |
| Resolving the Hydrophobicity of Me-4PACz Hole Transport Layer for High-Efficiency Inverted Perovskite Solar Cells                                         | Kashimul Hossain; Ashish Kulkarni; Urvashi Bothra et al.          | 2023 | arXiv preprint                      | 0     | (link)                       |                                    |
| First-principles study on the chemical decomposition of inorganic perovskites $\text{CsPbI}_3$ and $\text{RbPbI}_3$ at finite temperature and pressure    | Un-Gi Jong; Chol-Jun Yu; Yun-Hyok Kye et al.                      | 2018 | arXiv preprint                      | 0     | (link)                       |                                    |
| Compound Defects in Halide Perovskites: A First-Principles Study of $\text{CsPbI}_3$                                                                      | Haibo Xue; José Manuel Vicent-Luna; Shuxia Tao et al.             | 2022 | arXiv preprint                      | 0     | (link)                       |                                    |
| Quantum optoelectronics in semiconductor solar cell materials and devices                                                                                 | Xi Liu; Wenxi Fang                                                | 2026 | arXiv preprint                      | 0     | (link)                       |                                    |
| Recent advances in $\text{La}_2\text{NiMnO}_6$ Double Perovskites for various applications; Challenges and opportunities                                  | Suresh Chandra Baral; P. Maneesha; E. G. Rini et al.              | 2023 | arXiv preprint                      | 0     | (link)                       |                                    |
| Interface Engineering in Hybrid Iodide $\text{CH}_3\text{NH}_3\text{PbI}_3$ Perovskite Using Lewis Base and Graphene towards High Performance Solar Cells | Chol-Jun Yu; Yun-Hyok Kye; Un-Gi Jong et al.                      | 2019 | arXiv preprint                      | 0     | (link)                       |                                    |
| Ultra-stable 2D/3D hybrid perovskite photovoltaic module                                                                                                  | Giulia Grancini; Cristina Roldán-Carmona; Iwan Zimmermann et al.  | 2016 | arXiv preprint                      | 0     | (link)                       |                                    |
| Comprehensive numerical analysis of doping controlled efficiency in lead free $\text{Cs}(\text{SnGe})\text{I}_3$ perovskites solar cell                   | Nazmul Hasan; M. Husayeen Khan Anik; Mohammed Mehedi Hasan et al. | 2025 | arXiv preprint                      | 0     | 10.1007/s00339-024-08125-y   |                                    |
| Methylamine Vapor Exposure for Improved Morphology and Stability of Cesium-Methylammonium Lead Halide Perovskite Thin-Films                               | Akash Singh; Arun Singh Chouhan; Sushobhan Avasthi                | 2017 | arXiv preprint                      | 0     | 10.1007/978-3-319-97604-4_60 |                                    |
| A Review on Improving PSC Performance through Charge Carrier Management: Where We Stand and What's Next?                                                  | Dilshod Nematov                                                   | 2025 | Next Energy, 2026, 13               | 0     | 10.1016/j.nxener.2026.100763 |                                    |
| Organic-inorganic Copper(II)-based Material: a Low-Toxic, Highly Stable Light Absorber beyond Organolead Perovskites                                      | Xiaolei Li; Xiangli Zhong; Yue Hu et al.                          | 2017 | arXiv preprint                      | 0     | (link)                       |                                    |
| Pressure-induced reversible phase transition and amorphization of $\text{CH}_3\text{NH}_3\text{PbI}_3$                                                    | Kai Wang; Ran Liu; Yuan-cun Qiao et al.                           | 2015 | arXiv preprint                      | 0     | (link)                       |                                    |
| Size dependent solid-solid crystallization of halide perovskites                                                                                          | Paramvir Ahlawat                                                  | 2024 | arXiv preprint                      | 0     | (link)                       |                                    |
| Efficient and ultra-stable perovskite light-emitting diodes                                                                                               | Bingbing Guo; Runchen Lai; Sijie Jiang et al.                     | 2022 | Nature Photonics 16, 637-643 (2022) | 0     | 10.1038/s41566-022-01046-3   |                                    |
| Advances in Modelling and Simulation of Halide Perovskites for Solar Cell Applications                                                                    | Chol-Jun Yu                                                       | 2018 | arXiv preprint                      | 0     | (link)                       |                                    |

| Title                                                                                                                                           | Authors                                                          | Year | Venue                                          | Cited | DOI                               | OpenAlex<br>2-yr mean<br>citedness |
|-------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|------|------------------------------------------------|-------|-----------------------------------|------------------------------------|
| Lessons from Chalcopyrites for Scaling Thin Film Perovskite Photovoltaic Technology                                                             | Mirjana Dimitrievska; Edgardo Saucedo; Stefaan De Wolf et al.    | 2025 | arXiv preprint                                 | 0     | (link)                            |                                    |
| Lead-Free Halide Double Perovskites via Heterovalent Substitution of Noble Metals                                                               | George Volonakis; Marina R. Filip; Amir Abbas Haghighirad et al. | 2016 | arXiv preprint                                 | 0     | (link)                            |                                    |
| Fully printed flexible perovskite solar modules with improved energy alignment by tin oxide surface modification                                | Lirong Dong; Shudi Qiu; Jose Garcia Cerrillo et al.              | 2024 | arXiv preprint                                 | 0     | (link)                            |                                    |
| Perovskite-R1: a domain-specialized large language model for intelligent discovery of precursor additives and experimental design               | Xin-De Wang; Zhi-Rui Chen; Peng-Jie Guo et al.                   | 2025 | Communications Materials                       | 0     | 10.1038/s43246-026-01099-9        |                                    |
| Precise Control of Process Parameters for >23% Efficiency Perovskite Solar Cells in Ambient Air Using an Automated Device Acceleration Platform | Jiyun Zhang; Anastasia Barabash; Tian Du et al.                  | 2024 | arXiv preprint                                 | 0     | (link)                            |                                    |
| Dimethylammonium additives alter both vertical and lateral composition in halide perovskite semiconductors                                      | Sarthak Jariwala; Rishi Kumar; Giles E. Eperon et al.            | 2021 | arXiv preprint                                 | 0     | (link)                            |                                    |
| The effect of B-site alloying on the electronic and opto-electronic properties of RbPbI <sub>3</sub> : A DFT study                              | Anupriya Nyayban; Subhasis Panda; Avijit Chowdhury               | 2022 | arXiv preprint                                 | 0     | 10.1016/j.physb.2022.414384       |                                    |
| Strong Performance Enhancement in Lead-Halide Perovskite Solar Cells through Rapid, Atmospheric Deposition of n-type Buffer Layer Oxides        | Ravi D. Raninga; Robert A. Jagt; Solène Béchu et al.             | 2019 | arXiv preprint                                 | 0     | 10.1016/j.nanoen.2020.104946      |                                    |
| Clever Materials: When Models Identify Good Materials for the Wrong Reasons                                                                     | Kevin Maik Jablonka                                              | 2026 | arXiv preprint                                 | 0     | (link)                            |                                    |
| PeroMAS: A Multi-agent System of Perovskite Material Discovery                                                                                  | Yishu Wang; Wei Liu; Yifan Li et al.                             | 2026 | arXiv preprint                                 | 0     | (link)                            |                                    |
| Interface Modeling of Perovskite Polymer Heterostructures for Enhanced Charge Transfer Efficiency in Hybrid Photovoltaic Materials              | Somayyeh Alidoust; V. Ongun Özçelik                              | 2025 | arXiv preprint                                 | 0     | (link)                            |                                    |
| Antiphase boundary in CH <sub>3</sub> NH <sub>3</sub> PbI <sub>3</sub> repels charge carriers while promotes fast ion migrations                | Shulin Chen; Changwei Wu; Qiuyu Shang et al.                     | 2022 | Acta Materialia, 2022                          | 0     | 10.1016/j.actamat.2022.118010     |                                    |
| Substitution of Lead with Tin Suppresses Ionic Transport in Halide Perovskite Optoelectronics                                                   | Krishanu Dey; Dibyajyoti Ghosh; Matthew Pilot et al.             | 2023 | arXiv preprint                                 | 0     | (link)                            |                                    |
| Understanding and exploiting interfacial interactions between phosphonic acid functional groups and co-evaporated perovskites                   | Thomas Feeney; Julian Petry; Abderrezak Torche et al.            | 2024 | arXiv preprint                                 | 0     | 10.1016/j.matt.2024.02.004        |                                    |
| Illuminating the Future: Nanophotonics for Future Green Technologies, Precision Healthcare, and Optical Computing                               | Osama M. Halawa; Esraa Ahmed; Malk M. Abdelrazek et al.          | 2025 | arXiv preprint                                 | 0     | (link)                            |                                    |
| A Morphotropic Phase Boundary in MA <sub>1-x</sub> FA <sub>x</sub> PbI <sub>3</sub> : Linking Structure, Dynamics, and Electronic Properties    | Tobias Hainer; Erik Fransson; Sangita Dutta et al.               | 2025 | arXiv preprint                                 | 0     | (link)                            |                                    |
| Efficient dataset generation for machine learning perovskite alloys                                                                             | Henrietta Homm; Jarno Laakso; Patrick Rinke                      | 2025 | Physical Review Materials, 9(5), 053802 (2025) | 0     | 10.1103/PhysRevMaterials.9.053802 |                                    |
| Dynamic Nanodomains Dictate Macroscopic Properties in Lead Halide Perovskites                                                                   | Milos Dubajic; James R. Neilson; Johan Klarbring et al.          | 2024 | arXiv preprint                                 | 0     | 10.1038/s41565-025-01917-0        |                                    |

| Title                                                                                                                        | Authors                                                 | Year | Venue          | Cited | DOI                      | OpenAlex<br>2-yr mean<br>citedness |
|------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|------|----------------|-------|--------------------------|------------------------------------|
| Influence of Disorder and Anharmonic Fluctuations on the Dynamical Rashba Effect in Purely Inorganic Lead-Halide Perovskites | Arthur Marronnier; Guido Roma; Marcelo Carignano et al. | 2018 | arXiv preprint | 0     | 10.1021/acs.jpcc.8b11288 |                                    |
| A closed-loop AI framework for hypothesis-driven and interpretable materials design                                          | Kangyu Ji; Tianran Liu; Fang Sheng et al.               | 2025 | arXiv preprint | 0     | (link)                   |                                    |
| Tailoring wetting properties of organic hole-transport interlayers for slot-die coated perovskite solar modules              | T. S. Le; I. A. Chuykoa; L. O. Luchnikova et al.        | 2024 | arXiv preprint | 0     | (link)                   |                                    |

## Block MLP (313 unique)

| Title                                                                                                                                                             | Authors                                                                | Year | Venue                                                             | Cited | DOI                            | OpenAlex<br>2-yr mean<br>citedness |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|------|-------------------------------------------------------------------|-------|--------------------------------|------------------------------------|
| Atom-centered symmetry functions for constructing high-dimensional neural network potentials                                                                      | Jörg Behler                                                            | 2011 | The Journal of Chemical Physics                                   | 1644  | 10.1063/1.3553717              |                                    |
| Realistic Atomistic Structure of Amorphous Silicon from Machine-Learning-Driven Molecular Dynamics                                                                | Volker L. Deringer; Noam Bernstein; Albert P. Bartók et al.            | 2018 | The Journal of Physical Chemistry Letters                         | 269   | 10.1021/acs.jpcclett.8b00902   |                                    |
| Machine-learned interatomic potentials by active learning: amorphous and liquid hafnium dioxide                                                                   | Ganesh Sivaraman; Anand Narayanan Krishnamoorthy; Matthias Baur et al. | 2020 | npj Computational Materials                                       | 185   | 10.1038/s41524-020-00367-7     |                                    |
| Machine-learned interatomic potentials by active learning: amorphous and liquid hafnium dioxide                                                                   | Ganesh Sivaraman; Anand Narayanan Krishnamoorthy; Matthias Baur et al. | 2019 | Cambridge University Engineering Department Publications Database | 184   | 10.17863/cam.554081 (2026)     |                                    |
| Growth Mechanism and Origin of High sp <sup>3</sup> Content in Tetrahedral Amorphous Carbon                                                                       | Miguel A. Caro; Volker L. Deringer; Jari Koskinen et al.               | 2018 | Physical Review Letters                                           | 178   | 10.1103/physrevlett.120.166101 |                                    |
| Constructing first-principles phase diagrams of amorphous Li <sub>x</sub> Si using machine-learning-assisted sampling with an evolutionary algorithm              | Nongnuch Artrith; Alexander Urban; Gerbrand Ceder                      | 2018 | The Journal of Chemical Physics                                   | 155   | 10.1063/1.5017661              |                                    |
| Study of Li atom diffusion in amorphous Li <sub>3</sub> PO <sub>4</sub> with neural network potential                                                             | Wenwen Li; Yasunobu Ando; Emi Minamitani et al.                        | 2017 | The Journal of Chemical Physics                                   | 154   | 10.1063/1.4997242              |                                    |
| Impact of the Local Environment on Li Ion Transport in Inorganic Components of Solid Electrolyte Interphases                                                      | Taiping Hu; Jianxin Tian; Fuzhi Dai et al.                             | 2022 | Journal of the American Chemical Society                          | 107   | 10.1021/jacs.2c11521           |                                    |
| Thermal conductivity modeling using machine learning potentials: application to crystalline and amorphous silicon                                                 | Xin Qian; Shiyu Peng; Xiaobo Li et al.                                 | 2019 | Materials Today Physics                                           | 103   | 10.1016/j.mtphys.2019.100140   |                                    |
| A unified deep neural network potential capable of predicting thermal conductivity of silicon in different phases                                                 | Ruiyang Li; Eungkyu Lee; Tengfei Luo                                   | 2020 | Materials Today Physics                                           | 102   | 10.1016/j.mtphys.2020.100181   |                                    |
| Unraveling the crystallization kinetics of the Ge <sub>2</sub> Sb <sub>2</sub> Te <sub>5</sub> phase change compound with a machine-learned interatomic potential | Omar Abou El Kheir; Luigi Bonati; Michele Parrinello et al.            | 2024 | npj Computational Materials                                       | 62    | 10.1038/s41524-024-01217-6     |                                    |
| Development of a neuroevolution machine learning potential of Pd-Cu-Ni-P alloys                                                                                   | Rui Zhao; Shucheng Wang; Zhuangzhuang Kong et al.                      | 2023 | Materials & Design                                                | 43    | 10.1016/j.matdes.2023.112012   |                                    |
| Training machine-learning potentials for crystal structure prediction using disordered structures                                                                 | Chang-Ho Hong; Jeong Min Choi; Wonseok Jeong et al.                    | 2020 | Physical review. B/Physical review. B                             | 40    | 10.1103/physrevb.102.224104    |                                    |

| Title                                                                                                                                                                                          | Authors                                                           | Year | Venue                                      | Cited | DOI                               | OpenAlex<br>2-yr mean<br>citedness |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|------|--------------------------------------------|-------|-----------------------------------|------------------------------------|
| Crystallization of amorphous GeTe simulated by neural network potential addressing medium-range order                                                                                          | Dong-Heon Lee; Kyeong-pung Lee; Dongsun Yoo et al.                | 2020 | Computational Materials Science            | 40    | 10.1016/j.commatsci.2020.100735   | 2020.100735                        |
| Ultralow and glass-like lattice thermal conductivity in crystalline BaAg <sub>2</sub> Te <sub>2</sub> : Strong fourth-order anharmonicity and crucial diffusive thermal transport              | Zezhu Zeng; Cunzhi Zhang; Hulei Yu et al.                         | 2021 | Materials Today Physics                    | 37    | 10.1016/j.mtphys.2021.100487      |                                    |
| Dependence of a cooling rate on structural and vibrational properties of amorphous silicon: A neural network potential-based molecular dynamics study                                          | Wenwen Li; Yasunobu Ando                                          | 2019 | The Journal of Chemical Physics            | 35    | 10.1063/1.5114652                 |                                    |
| Experiment-Driven Atomistic Materials Modeling: A Case Study Combining X-Ray Photoelectron Spectroscopy and Machine Learning Potentials to Infer the Structure of Oxygen-Rich Amorphous Carbon | Tigany Zarrouk; Rina Ibragimova; Albert P. Bartók et al.          | 2024 | Journal of the American Chemical Society   | 35    | 10.1021/jacs.4c01897              |                                    |
| Accurate and Transferable Machine Learning Potential for Molecular Dynamics Simulation of Sodium Silicate Glasses                                                                              | Marco Bertani; Thibault Charpentier; Francesco Faglioni et al.    | 2024 | Journal of Chemical Theory and Computation | 33    | 10.1021/acs.jctc.3c01115          |                                    |
| Applications of machine-learning interatomic potentials for modeling ceramics, glass, and electrolytes: A review                                                                               | Shingo Urata; Marco Bertani; Alfonso Pedone                       | 2024 | Journal of the American Ceramic Society    | 32    | 10.1111/jace.19934                |                                    |
| Vibrational anharmonicity results in decreased thermal conductivity of amorphous HfO <sub>2</sub> at high temperature                                                                          | Honggang Zhang; Xiaokun Gu; Zheyong Fan et al.                    | 2023 | Physical review. B                         | 31    | 10.1103/physrevb.108.045422       |                                    |
| AENET-LAMMPS and AENET-TINKER: Interfaces for accurate and efficient molecular dynamics simulations with machine learning potentials                                                           | Michael S. Chen; Tobias Morawietz; Hideki Mori et al.             | 2021 | The Journal of Chemical Physics            | 29    | 10.1063/5.0063880                 |                                    |
| The Roadmap of Graphene: From Fundamental Research to Broad Applications                                                                                                                       | Jin Zhang; Hailin Peng; Hua Zhang et al.                          | 2022 | Advanced Functional Materials              | 27    | 10.1002/adfm.202201375            | 1375 (2026)                        |
| Modeling Short-Range and Three-Membered Ring Structures in Lithium Borosilicate Glasses Using a Machine-Learning Potential                                                                     | Shingo Urata                                                      | 2022 | The Journal of Physical Chemistry C        | 26    | 10.1021/acs.jpcc.2c07597          |                                    |
| Atomistic Simulation of HF Etching Process of Amorphous Si <sub>3</sub> N <sub>4</sub> Using Machine Learning Potential                                                                        | Chang-Ho Hong; Sangmin Oh; Hyungmin An et al.                     | 2024 | ACS Applied Materials & Interfaces         | 24    | 10.1021/acsami.4c07448            | 7448 (2026)                        |
| Structure of disordered TiO <sub>2</sub> phases from ab initio based deep neural network simulations                                                                                           | Marcos F. Calegari Andrade; Annabella Selloni                     | 2020 | Physical Review Materials                  | 24    | 10.1103/physrevmaterials.4.113803 |                                    |
| Proton Transport in Perfluorinated Ionomer Simulated by Machine-Learned Interatomic Potential                                                                                                  | Ryosuke Jinnouchi; Saori Minami; Ferenc Karsai et al.             | 2023 | The Journal of Physical Chemistry Letters  | 23    | 10.1021/acs.jpclett.3c00293       |                                    |
| Machine learning interatomic potentials for amorphous zeolitic imidazolate frameworks                                                                                                          | Nicolas Castel; Dune André; Connor Edwards et al.                 | 2024 | Digital Discovery                          | 23    | 10.1039/d3dd00236e                | 236 (2026)                         |
| Simulation of Phase-Change-Memory and Thermoelectric Materials using Machine-Learned Interatomic Potentials: Sb <sub>2</sub> Te <sub>3</sub>                                                   | Konstantinos Konstantinou; Juraj Mavračić; Felix C. Mocanu et al. | 2020 | physica status solidi (b)                  | 21    | 10.1002/pssb.202000416            |                                    |
| Suppression of Rayleigh Scattering in Silica Glass by Codoping Boron and Fluorine: Molecular Dynamics Simulations with Force-Matching and Neural Network Potentials                            | Shingo Urata; Nobuhiro Nakamura; Tomofumi Tada et al.             | 2022 | The Journal of Physical Chemistry C        | 21    | 10.1021/acs.jpcc.1c10300          |                                    |

| Title                                                                                                                                                                                       | Authors                                                   | Year | Venue                                   | Cited | DOI                               | OpenAlex<br>2-yr mean<br>citedness |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|------|-----------------------------------------|-------|-----------------------------------|------------------------------------|
| Nature of the Amorphous–Amorphous Interfaces in Solid-State Batteries Revealed Using Machine-Learned Interatomic Potentials                                                                 | Chuhong Wang; Muratahan Aykol; Tim Mueller                | 2023 | Chemistry of Materials                  | 20    | 10.1021/acs.chemmater.4c0187c     | 6.287 (2025)                       |
| Ab initio molecular dynamics and high-dimensional neural network potential study of VZrNbHfTa melt                                                                                          | I. A. Balyakin; A. A. Yuryev; . . . et al.                | 2020 | Journal of Physics Condensed Matter     | 20    | 10.1088/1361-648x/ab6f87          |                                    |
| Boron coordination and three-membered ring formation in sodium borate glasses: a machine-learning molecular dynamics study                                                                  | Takeyuki Kato; Federica Lodesani; Shingo Urata            | 2023 | Journal of the American Ceramic Society | 20    | 10.1111/jace.19629                |                                    |
| Modeling the aqueous interface of amorphous TiO2 using deep potential molecular dynamics                                                                                                    | Zhutian Ding; Annabella Selloni                           | 2023 | The Journal of Chemical Physics         | 19    | 10.1063/5.0157188                 |                                    |
| Assessing the Reactivity of the Na 3 PS 4 Solid-State Electrolyte with the Sodium Metal Negative Electrode Using Total Trajectory Analysis with Neural-Network Potential Molecular Dynamics | Lieven Bekaert; Suzuno Akatsuka; Naoto Tanibata et al.    | 2023 | The Journal of Physical Chemistry C     | 19    | 10.1021/acs.jpcc.3c02379          |                                    |
| Binding energies and sticking coefficients of H2 on crystalline and amorphous CO ice                                                                                                        | Germán Molpeceres; Viktor Zaverkin; Naoki Watanabe et al. | 2021 | Astronomy and Astrophysics              | 18    | 10.1051/0004-6361/202040023       | 5.028 (2026)                       |
| Density dependence of thermal conductivity in nanoporous and amorphous carbon with machine-learned molecular dynamics                                                                       | Yanzhou Wang; Zheyong Fan; Ping Qian et al.               | 2025 | Physical review. B/Physical review. B   | 17    | 10.1103/physrevb.111.094205       |                                    |
| PotentialMind: Graph Convolutional Machine Learning Potential for Sb–Te Binary Compounds of Multiple Stoichiometries                                                                        | Guanjie Wang; Yuqi Sun; Jian Zhou et al.                  | 2023 | The Journal of Physical Chemistry C     | 17    | 10.1021/acs.jpcc.3c07110          |                                    |
| Crystal-like thermal transport in amorphous carbon                                                                                                                                          | Jaeyun Moon; Zhiting Tian                                 | 2025 | npj Computational Materials             | 17    | 10.1038/s41524-025-01625-2        |                                    |
| Plasma Oxidation of Copper: Molecular Dynamics Study with Neural Network Potentials                                                                                                         | Yantao Xia; Philippe Sautet                               | 2022 | ACS Nano                                | 16    | 10.1021/acs.nano.2c07789          | 17.894 (2026)                      |
| Generalized modeling of carbon film deposition growth via hybrid MD/MC simulations with machine-learning potentials                                                                         | Yutao Liu; Tinghong Gao; Qingquan Xiao et al.             | 2025 | npj Computational Materials             | 16    | 10.1038/s41524-025-01781-5        |                                    |
| The ab initio non-crystalline structure database: empowering machine learning to decode diffusivity                                                                                         | Hui Zheng; Eric Sivonxay; Rasmus Christensen et al.       | 2024 | npj Computational Materials             | 16    | 10.1038/s41524-024-01469-2        |                                    |
| Thermal conductivity of Li 3 PS 4 solid electrolytes with ab initio accuracy                                                                                                                | Davide Tisi; Federico Grasselli; Lorenzo Gigli et al.     | 2024 | Physical Review Materials               | 15    | 10.1103/physrevmaterials.8.065403 |                                    |
| Structural Evolution of C 60 Molecular Crystal Predicted by Neural Network Potential                                                                                                        | Kun Ni; Fei Pan; Yanwu Zhu                                | 2022 | Advanced Functional Materials           | 14    | 10.1002/adfm.202203497            | 14.975 (2026)                      |
| Role of hydrogen-doping for compensating oxygen-defect in non-stoichiometric amorphous In2O3–x: Modeling with a machine-learning potential                                                  | Shingo Urata; Nobuhiro Nakamura; Junghwan Kim et al.      | 2023 | Journal of Applied Physics              | 13    | 10.1063/5.0149199                 |                                    |
| Nuclear quantum effects in the acetylene:ammonia plastic co-crystal                                                                                                                         | Atul C. Thakur; Richard C. Remsing                        | 2024 | The Journal of Chemical Physics         | 13    | 10.1063/5.0179161                 |                                    |
| Development and Validation of Neural Network Potentials for Multicomponent Oxide Glasses                                                                                                    | Ryuki Kayano; Yaohiro Inagaki; Ryuta Matsubara et al.     | 2024 | The Journal of Physical Chemistry C     | 12    | 10.1021/acs.jpcc.4c04604          |                                    |
| Atomic Structure of Na 4 P 2 S 7 Glass Solid Electrolyte: Fine-Tuning Machine Learning Potentials for Enhanced Accuracy                                                                     | Marco Bertani; Alfonso Pedone                             | 2025 | The Journal of Physical Chemistry C     | 12    | 10.1021/acs.jpcc.5c01857          |                                    |

| Title                                                                                                                                                           | Authors                                                              | Year | Venue                                                | Cited | DOI                                      | OpenAlex<br>2-yr mean<br>citedness |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|------|------------------------------------------------------|-------|------------------------------------------|------------------------------------|
| Enhanced heat transport in amorphous silicon via microstructure modulation                                                                                      | Youtian Li; Yangyu Guo; Shiyun Xiong et al.                          | 2024 | International Journal of Heat and Mass Transfer      | 12    | 10.1016/j.ijheatmasstransfer.2023.125167 | 2024 (2024)                        |
| Elaboration of a neural-network interatomic potential for silica glass and melt                                                                                 | Salomé Trillot; Julien Lam; Simona Ispas et al.                      | 2024 | Computational Materials Science                      | 12    | 10.1016/j.commatsci.2024.113648          | 2024 (2024)                        |
| Machine Learning Interatomic Potential for Modeling the Mechanical and Thermal Properties of Naphthyl-Based Nanotubes                                           | Hugo X. Rodrigues; Hudson R. Armando; Daniel A. da Silva et al.      | 2025 | Journal of Chemical Theory and Computation           | 12    | 10.1021/acs.jctc.4c01578                 | 2025 (2025)                        |
| Neural Network Force Fields for Metal Growth Based on Energy Decompositions                                                                                     | Qin Hu; Mouyi Weng; Xin Chen et al.                                  | 2020 | The Journal of Physical Chemistry Letters            | 12    | 10.1021/acs.jpcllett.9b03780             | 2020 (2020)                        |
| Dynamic mesophase transition induces anomalous suppressed and anisotropic phonon thermal transport                                                              | Linfeng Yu; Kexin Dong; Qi Yang et al.                               | 2024 | npj Computational Materials                          | 12    | 10.1038/s41524-024-01442-z               | 2024 (2024)                        |
| Accurate prediction of structural and mechanical properties on amorphous materials enabled through machine-learning potentials: A case study of silicon nitride | Ganesh Kumar Nayak; Prashanth Srinivasan; Juraj Todt et al.          | 2025 | Computational Materials Science                      | 11    | 10.1016/j.commatsci.2025.113649          | 2025 (2025)                        |
| Efficient Training of Machine Learning Potentials by a Randomized Atomic-System Generator                                                                       | Young-Jae Choi; Seung-Hoon Jhi                                       | 2020 | The Journal of Physical Chemistry B                  | 11    | 10.1021/acs.jpcc.0c05075                 | 2020 (2020)                        |
| Structural and mechanical properties of monolayer amorphous carbon and boron nitride                                                                            | Xi Zhang; Yutian Zhang; Yun-Peng Wang et al.                         | 2024 | Physical review. B/Physical review. B                | 11    | 10.1103/physrevb.109.174106              | 2024 (2024)                        |
| Structure and Crystallization Kinetics of As-Deposited Films of the GeTe Phase Change Compound from Atomistic Simulations                                       | Simone Perego; Daniele Dragoni; Silvia Gabardi et al.                | 2022 | physica status solidi (RRL) - Rapid Research Letters | 11    | 10.1002/pssr.202200433                   | 2022 (2022)                        |
| High-Accuracy Neural Network Interatomic Potential for Silicon Nitride                                                                                          | Hui Xu; Zeyuan Li; Zhaofu Zhang et al.                               | 2023 | Nanomaterials                                        | 11    | 10.3390/nano13081352                     | 2023 (2023)                        |
| Crystallization kinetics of nanoconfined GeTe slabs in GeTe/TiTe <sub>2</sub> -like superlattices for phase change memories                                     | Debdipto Acharya; Omar Abou El Kheir; Davide Campi et al.            | 2024 | Scientific Reports                                   | 11    | 10.1038/s41598-024-53192-z               | 2024 (2024)                        |
| Graph-EAM: An Interpretable and Efficient Graph Neural Network Potential Framework                                                                              | Jun Yang; Zhitao Chen; Hong Sun et al.                               | 2023 | Journal of Chemical Theory and Computation           | 9     | 10.1021/acs.jctc.3c00344                 | 2023 (2023)                        |
| Exploration of Amorphous V <sub>2</sub> O <sub>5</sub> as Cathode for Magnesium Batteries                                                                       | Vijay Choyal; Debsundar Dey; Gopalakrishnan Sai Gautam               | 2025 | Small                                                | 9     | 10.1002/sml.202505851                    | 2025 (2025)                        |
| Atomistic Study of the Configurational Entropy and the Fragility of Supercooled Liquid GeTe                                                                     | Yuhan Chen; Davide Campi; Marco Bernasconi et al.                    | 2024 | Advanced Functional Materials                        | 9     | 10.1002/adfm.202314267                   | 2024 (2024)                        |
| Neural network potential for Zr-Rh system by machine learning                                                                                                   | Kun Xie; Chong Qiao; Hong Shen et al.                                | 2021 | Journal of Physics Condensed Matter                  | 8     | 10.1088/1361-648x/ac37dc                 | 2021 (2021)                        |
| Investigating Ionic Diffusivity in Amorphous LiPON using Machine-Learned Interatomic Potentials                                                                 | Aqshat Seth; Rutvij Pankaj Kulkarni; Gopalakrishnan Sai Gautam       | 2025 | ACS Materials Au                                     | 7     | 10.1021/acsmaterialsau.1c00164           | 2025 (2025)                        |
| Machine-Learning Molecular Dynamics Study on the Structure and Glass Transition of Calcium Aluminosilicate Glasses                                              | Takeyuki Kato; Ryuki Kayano; Takahiro Ohkubo                         | 2025 | The Journal of Physical Chemistry B                  | 7     | 10.1021/acs.jpcc.5c03404                 | 2025 (2025)                        |
| Umbrella sampling with machine learning potentials applied for solid phase transition of GeSbTe                                                                 | Yanliang Zhao; Jikai Sun; Li Yang et al.                             | 2022 | Chemical Physics Letters                             | 7     | 10.1016/j.cplett.2022.339781             | 2022 (2022)                        |
| Realistic Atomistic Structure of Amorphous Silicon from Machine-Learning-Driven Molecular Dynamics.                                                             | Volker L. Deringer; Noam Bernstein; Albert P. Bartók et al.          | 2018 | arXiv (Cornell University)                           | 6     | 10.17863/cam.28044                       | 2018 (2018)                        |
| Basic Stability Tests of Machine Learning Potentials for Molecular Simulations in Computational Drug Discovery                                                  | Kavindri Ranasinghe; Adam L. Baskerville; Geoffrey P. F. Wood et al. | 2025 | Journal of Chemical Information and Modeling         | 6     | 10.1021/acs.jcim.5c01150                 | 2025 (2025)                        |



| Title                                                                                                                                                            | Authors                                                      | Year | Venue                                           | Cited | DOI                                      | OpenAlex<br>2-yr mean<br>citedness |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|------|-------------------------------------------------|-------|------------------------------------------|------------------------------------|
| Sampling rare events using nanostructures for universal Pt neural network potential                                                                              | Joonhee Kang; Byung-Hyun Kim; Min Ho Seo et al.              | 2024 | Current Applied Physics                         | 6     | 10.1016/j.cap.2024.07.005                | 0.305 (2026)                       |
| Study of vacancy ordering and the boson peak in metastable cubic Ge-Sb-Te using machine learning potentials                                                      | Young-Jae Choi; Minjae Ghim; Seung-Hoon Jhi                  | 2024 | Physical Review Materials                       | 6     | 10.1103/physrevmaterials.8.013606        |                                    |
| Atomistic Simulations of the Crystallization of Amorphous GeTe Nanoparticles                                                                                     | Debdipto Acharya; Omar Abou El Kheir; Simone Perego et al.   | 2024 | The Journal of Physical Chemistry C             | 6     | 10.1021/acs.jpcc.4c05157                 |                                    |
| Multikernel similarity-based clustering of amorphous systems and machine-learned interatomic potentials by active learning                                       | Firas Shuaib; Guido Ori; Philippe Thomas et al.              | 2024 | Journal of the American Ceramic Society         | 5     | 10.1111/jace.20128                       |                                    |
| Crystallization kinetics in Ge-rich Ge x Te alloys from large scale simulations with a machine-learned interatomic potential                                     | Dario Baratella; Omar Abou El Kheir; Marco Bernasconi        | 2024 | Acta Materialia                                 | 5     | 10.1016/j.actamat.2024.11720             | 0.202 (2026)                       |
| Amorphous Zirconia-doped Tantalum modeling and simulations using explicit multi-element spectral neighbor analysis machine learning potentials (EME-SNAP)        | Jun Jiang; Xiangguo Li; A. Mishkin et al.                    | 2023 | Physical Review Materials                       | 5     | 10.1103/physrevmaterials.7.045602        |                                    |
| Temperature-density dependent hidden order of amorphous carbon                                                                                                   | Yutao Liu; Tinghong Gao; Qingquan Xiao et al.                | 2025 | Communications Physics                          | 5     | 10.1038/s42005-025-02320-w               | 5.555 (2026)                       |
| The ab initio amorphous materials database: Empowering machine learning to decode diffusivity                                                                    | Hui Zheng; Eric Sivonxay; Max C. Gallant et al.              | 2024 | arXiv (Cornell University)                      | 5     | 10.48550/arxiv.2402.00177                |                                    |
| Mechanisms of nanodiamond and amorphous diamond-like carbon formation following room temperature compression of C60                                              | Hendrik Heimes; Ethan P. Turner; Alan Salek et al.           | 2025 | Diamond and Related Materials                   | 5     | 10.1016/j.diamond.2025.108221            | 0.123 (2026)                       |
| Non-Monotonic Ion Conductivity in Lithium-Aluminum-Chloride Glass Solid-State Electrolytes Explained by Cascading Hopping                                        | Beomgyu Kang; Jinqui Yu; Shinji Saito et al.                 | 2025 | Advanced Science                                | 5     | 10.1002/advs.202509206                   | 0.203 (2026)                       |
| A high-efficiency neuroevolution potential for tobermorite and calcium silicate hydrate systems with ab initio accuracy                                          | Xiao Xu; Shijie Wang; Haifeng Qin et al.                     | 2025 | Cement and Concrete Research                    | 5     | 10.1016/j.cemconres.2025.120291          | 0.123 (2026)                       |
| Softening of Vibrational Modes and Anharmonicity Induced Thermal Conductivity Reduction in a-Si:H at High Temperatures                                           | Zhuo Chen; Yuejin Yuan; Yanzhou Wang et al.                  | 2025 | Advanced Electronic Materials                   | 5     | 10.1002/aelm.202504197                   | 0.107 (2026)                       |
| Machine learning potential for predicting thermal conductivity of -phase and amorphous tantalum nitride                                                          | Zhicheng Zong; Yangjun Qin; Jiahong Zhan et al.              | 2025 | International Journal of Heat and Mass Transfer | 4     | 10.1016/j.ijheatmasstransfer.2025.128155 |                                    |
| Molecular Dynamics Study of Silicon Carbide Using an Ab Initio-Based Neural Network Potential: Effect of Composition and Temperature on Crystallization Behavior | Jinyoung Lim; Youngseon Shim; Jaehong Park et al.            | 2023 | The Journal of Physical Chemistry C             | 4     | 10.1021/acs.jpcc.3c04224                 |                                    |
| Slowly quenched, high pressure glassy B2O3 at DFT accuracy                                                                                                       | Debendra Meher; Nikhil V. S. Avula; Sundaram Balasubramanian | 2025 | The Journal of Chemical Physics                 | 4     | 10.1063/5.0240030                        |                                    |
| Nature of the amorphous-amorphous interfaces in solid-state batteries revealed using machine-learned interatomic potentials                                      | Chuhong Wang; Murat-Ahan Aykol; Tim Mueller                  | 2023 | ChemRxiv                                        | 3     | 10.26434/chemrxiv-0.467 2023-frr79-v2    | 0.467 (2026)                       |

| Title                                                                                                                                                                                                | Authors                                                              | Year | Venue                               | Cited | DOI                               | OpenAlex<br>2-yr mean<br>citedness |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|------|-------------------------------------|-------|-----------------------------------|------------------------------------|
| Viscosity, breakdown of Stokes-Einstein relation, and dynamical heterogeneity in supercooled liquid Ge <sub>2</sub> Sb <sub>2</sub> Te <sub>5</sub> from simulations with a neural network potential | Simone Marcorini; Rocco Pomodoro; Omar Abou El Kheir et al.          | 2025 | The Journal of Chemical Physics     | 3     | 10.1063/5.0282855                 |                                    |
| Anisotropic response under shock compression in refractory high-entropy alloy via machine-learning potential                                                                                         | Hongyang Liu; Rong Chen; Bo Chen et al.                              | 2025 | Materials & Design                  | 3     | 10.1016/j.matdes.2025.115222      |                                    |
| Comparison of intermediate-range order in GeO <sub>2</sub> glass: Molecular dynamics using machine-learning interatomic potential vs reverse Monte Carlo fitting to experimental data                | Kenta Matsutani; Shusuke Kasamatsu; Takeshi Usuki                    | 2024 | The Journal of Chemical Physics     | 3     | 10.1063/5.0240087                 |                                    |
| A Generalized Bond Switching Monte Carlo method for amorphous structure generation                                                                                                                   | Zhengneng Zheng; Fan Zheng; Yuning Wu et al.                         | 2025 | Computational Materials Today       | 3     | 10.1016/j.commt.2025.100000       | 251830(2026)                       |
| Atomic-scale factors that control the rate capability of nanostructured amorphous Si for high-energy-density batteries                                                                               | Nongnuch Artrith; Alexander S. Urban; Yan Wang et al.                | 2019 | arXiv (Cornell University)          | 3     | 10.48550/arxiv.1901.09272         |                                    |
| Machine learning metallic glass critical cooling rates through elemental and molecular simulation based featurization                                                                                | Lane E. Schultz; Benjamin Afflerbach; Paul M. Voyles et al.          | 2024 | Journal of Materiomics              | 3     | 10.1016/j.jmat.2024.100964        |                                    |
| When more data hurts: Optimizing data coverage while mitigating diversity-induced underfitting in an ultrafast machine-learned potential                                                             | Jason Gibson; Tesia D. Janicki; Ajinkya C. Hire et al.               | 2025 | Physical Review Materials           | 3     | 10.1103/physrevmaterials.9.043802 |                                    |
| Insights into radiation damage in YBa <sub>2</sub> Cu <sub>3</sub> O <sub>7</sub> —from machine learned interatomic potentials                                                                       | Ashley Dickson; Niccolò Di Eugenio; Federico Ledda et al.            | 2026 | Superconductivity                   | 2     | 10.1016/j.supcon.2026.100264      |                                    |
| Critical Composition Window for Co-P Bulk Metallic Glasses Revealed by Machine-Learned Interatomic Potentials                                                                                        | Md Sharif Khan; Bourgeois Biova Irénée Gadjaboui; Oliviero Andreussi | 2025 | The Journal of Physical Chemistry C | 2     | 10.1021/acs.jpcc.5c05928          |                                    |
| The structural evolution of SiBCNZr amorphous ceramics analyzed by machine learning potential                                                                                                        | Meng Zhang; Siyan Deng; Jianghong Zhang et al.                       | 2024 | Ceramics International              | 2     | 10.1016/j.ceramint.2024.129396    | 2024932026                         |
| Unraveling the Crystallization Kinetics of the Ge <sub>2</sub> Sb <sub>2</sub> Te <sub>5</sub> Phase Change Compound with a Machine-Learned Interatomic Potential                                    | Omar Abou El Kheir; Luigi Bonati; Michele Parrinello et al.          | 2023 | arXiv (Cornell University)          | 2     | 10.48550/arxiv.2304.03109         |                                    |
| Thermal Conductivity Modeling using Machine Learning Potentials: Application to Crystalline and Amorphous Silicon                                                                                    | Xin Qian; Shenyong Peng; Xiaobo Li et al.                            | 2019 | arXiv (Cornell University)          | 2     | 10.48550/arxiv.1907.09088         |                                    |
| Density dependence of thermal conductivity in nanoporous and amorphous carbon with machine-learned molecular dynamics                                                                                | Yanzhou Wang; Zheyong Fan; Ping Qian et al.                          | 2024 | Aaltodoc (Aalto University)         | 2     | 10.48550/arxiv.2408010390         | 10390(2026)                        |
| Strong lattice anharmonicity and glass-like lattice thermal conductivity in nitrohalide double antiperovskites: A case study based on machine-learning potentials                                    | Yuan Li; Weidong Zheng; Jin Huan Pu et al.                           | 2025 | Thermo-X                            | 2     | 10.70401/tx.2025.0001             |                                    |
| Crystal-like thermal transport in amorphous carbon                                                                                                                                                   | Jaeyun Moon; Zhiting Tian                                            | 2024 | arXiv (Cornell University)          | 2     | 10.48550/arxiv.2405.07298         |                                    |
| Universal Interatomic Potentials with DFT for Understanding Orbital Localization in Polydimethylsiloxane-Amorphous Silica Nanocomposites                                                             | Carson Farmer; Hector Medina                                         | 2025 | ACS Omega                           | 2     | 10.1021/acsomega.5c06973          | 60973(2026)                        |

| Title                                                                                                                                                              | Authors                                                            | Year | Venue                                             | Cited | DOI                             | OpenAlex<br>2-yr mean<br>citedness |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|------|---------------------------------------------------|-------|---------------------------------|------------------------------------|
| Ballistic transport from propagating vibrational modes in amorphous silicon dioxide: Thermal experiments and atomistic-machine learning modeling                   | Man Li; Lingyun Dai; Huan Wu et al.                                | 2025 | Materials Today Physics                           | 2     | 10.1016/j.mtphys.2025.101659    |                                    |
| Advanced active learning techniques for precision-driven machine-learned potentials in Sb <sub>2</sub> S <sub>3</sub>                                              | V Priyadarshini; Pengfei Zou; Debashish Dash et al.                | 2025 | Results in Engineering                            | 2     | 10.1016/j.rineng.2025.106626    |                                    |
| Atomic armor for thermal stability in nanoporous structures                                                                                                        | Rui Yang; Qiaoling Si; Qiang Sheng et al.                          | 2025 | Proceedings of the National Academy of Sciences   | 2     | 10.1073/pnas.2510746122         |                                    |
| Physics-informed machine learning exploration of Na storage mechanisms in disordered carbon                                                                        | Nikhil Rampal; Stephen E. Weitzner; Fredrick Omenya et al.         | 2026 | Energy storage materials                          | 2     | 10.1016/j.ensm.2026.100498      | (2026)                             |
| Thermal conductivity of Li <sub>3</sub> PS <sub>4</sub> solid electrolytes with ab initio accuracy                                                                 | Davide Tisi; Federico Grasselli; Lorenzo Gigli et al.              | 2024 | arXiv (Cornell University)                        | 2     | 10.48550/arxiv.2401.12936       |                                    |
| Unveiling the crystallization kinetics in Ge-rich Ge <sub>x</sub> Te alloys by large scale simulations with a machine-learned interatomic potential                | Dario Baratella; Omar Abou El Kheir; Marco Bernasconi              | 2024 | arXiv (Cornell University)                        | 1     | 10.48550/arxiv.2404.15128       |                                    |
| On the boroxol ring fraction in melt-quenched B <sub>2</sub> O <sub>3</sub> glass: Insights from machine learning potentials                                       | Debendra Meher; Nikhil V. S. Avula; Sundaram Balasubramanian       | 2026 | The Journal of Chemical Physics                   | 1     | 10.1063/5.0321441               |                                    |
| Accurate prediction of structural and mechanical properties on amorphous materials enabled through machine-learning potentials: a case study of silicon nitride    | Ganesh Kumar Nayak; Prashanth Srinivasan; Juraj Todt et al.        | 2024 | DESY Database (Deutsches Elektronen-Synchrotron)  | 1     | 10.48550/arxiv.2408.06782       | (2026)                             |
| Developing a Neural Network potential to investigate interface phenomena in solid-phase epitaxy                                                                    | Ruggero Lot; Layla Martin-Samos; Stefano de Gironcoli et al.       | 2021 |                                                   | 1     | 10.1109/nmdc50713.2021.9677541  |                                    |
| A Unified Deep Neural Network Potential Capable of Predicting Thermal Conductivity of Silicon in Different Phases                                                  | Ruiyang Li; Eungkyu Lee; Tengfei Luo                               | 2019 | arXiv (Cornell University)                        | 1     | 10.48550/arxiv.1912.05044       |                                    |
| Simulation of structure and dynamics of phosphate glasses with fluoride additives: A hybrid approach combining universal and specialized neural network potentials | I. A. Balyakin; K. A. Arslanov; Maxim I. Vlasov                    | 2025 | Computational Materials Science                   | 1     | 10.1016/j.commatsci.2025.120405 | (2026)                             |
| Nature of adsorption of amino acids and precursors on interstellar amorphous solid water                                                                           | Natsuki Watanabe; Yuta Hori; Kazuki Okazawa et al.                 | 2025 | Monthly Notices of the Royal Astronomical Society | 1     | 10.1093/mnras/staf1913          |                                    |
| AI-Aided Mapping of the Structure-Composition-Conductivity Relationships of Glass-Ceramic Lithium Thiophosphate Electrolytes                                       | Haoyue Guo; Qian Wang; Alexander E. Urban et al.                   | 2022 | arXiv (Cornell University)                        | 1     | 10.48550/arxiv.2201.11203       |                                    |
| Liquid anomalies and fragility of supercooled antimony                                                                                                             | Flavio Giuliani; Francesco Guidarelli Mattioli; Yuhang Chen et al. | 2026 | Proceedings of the National Academy of Sciences   | 1     | 10.1073/pnas.2531605123         |                                    |
| Origin of Doping-Induced Structural Amorphization and Improved Cycling Performance in Aluminum Anodes for Lithium-Ion Batteries                                    | Jiwon Yu; Gyeong S. Hwang                                          | 2025 | The Journal of Physical Chemistry C               | 1     | 10.1021/acs.jpcc.5c06267        |                                    |
| Softening of Vibrational Modes and Anharmonicity Induced Thermal Conductivity Reduction in a-Si:H at High Temperatures                                             | Zhuo Chen; Yuejin Yuan; Yanzhou Wang et al.                        | 2025 | arXiv (Cornell University)                        | 1     | 10.48550/arxiv.2502.05584       |                                    |
| Atomistic Insights into Stress-Driven Lithiation at Silicon Anode Crack Tips                                                                                       | Bowen Zhang; Peiyao Zhang; Changguo Wang et al.                    | 2025 | ACS Applied Materials & Interfaces                | 1     | 10.1021/acsami.5c11238          | (2026)                             |
| Form and Function of Disorder                                                                                                                                      | Parthapratiim Biswas; Gang Chen; Serge Nakhmanson et al.           | 2021 | physica status solidi (b)                         | 1     | 10.1002/pssb.202100366          |                                    |

| Title                                                                                                                                                                  | Authors                                                                          | Year | Venue                                                                    | Cited | DOI                              | OpenAlex<br>2-yr mean<br>citedness |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|------|--------------------------------------------------------------------------|-------|----------------------------------|------------------------------------|
| Nature of the amorphous-amorphous interfaces in solid-state batteries revealed using machine-learned interatomic potentials                                            | Chuhong Wang; Muratahan Aykol; Tim Mueller                                       | 2023 | ChemRxiv                                                                 | 0     | 10.26434/chemrxiv-2023-frr79     | 0.467 (2026)                       |
| Investigating Ionic Diffusivity in Amorphous Solid Electrolytes using Machine Learned Interatomic Potentials                                                           | Aqshat Seth; Rutvij Pankaj Kulkarni; Gopalakrishnan Sai Gautam                   | 2024 | arXiv (Cornell University)                                               | 0     | 10.48550/arxiv.2409.06242        |                                    |
| Machine-learned interatomic potentials for modelling nanoscale fracturing in silica and basalt                                                                         | Marthe Grønlie Guren; Henrik Anderson Sveinsson; Anders Malthes-Sørenssen et al. | 2023 |                                                                          | 0     | 10.5194/egusphere-egu23-6256     |                                    |
| Dynamic Metrics to Validate the Use of Machine-Learned Interatomic Potential for Dynamic Sampling                                                                      | Bourgeois Biova Irénée Gadjaboui; Md Sharif Khan; Oliviero Andreussi             | 2026 |                                                                          | 0     | 10.1145/3773966.3785507          |                                    |
| Dependence of a cooling rate on structural and vibrational properties of amorphous silicon: A neural network potential-based molecular dynamics study                  | ; Yasunobu Ando                                                                  | 2019 | Institutional Repositories DataBase (IRDB)                               | 0     | (link)                           |                                    |
| Critical Composition Window for Co-P Bulk Metallic Glasses Revealed by Machine-Learned Interatomic Potentials                                                          | Md Sharif Khan; Bourgeois Biova Irénée Gadjaboui; Oliviero Andreussi             | 2025 | ChemRxiv                                                                 | 0     | 10.26434/chemrxiv-2025-vl9mj     | 0.467 (2026)                       |
| Suppression of Rayleigh Scattering in Silica Glass by Codoping Boron and Fluorine: Molecular Dynamics Simulations with ForceMatching and Neural Network Potentials     | ; Nobuhiro Nakamura; et al.                                                      | 2022 | Institutional Repositories DataBase (IRDB)                               | 0     | (link)                           |                                    |
| Insights Into Radiation Damage in YBa <sub>2</sub> Cu <sub>3</sub> O <sub>7-x</sub> From Machine-Learned Interatomic Potentials                                        | Ashley Dicksona; Niccolò Di Eugenio; Federico Ledda et al.                       | 2025 | arXiv (Cornell University)                                               | 0     | (link)                           |                                    |
| Modeling Carbon Nanodomain Formation in Amorphous SiOC Using MACE Neural Network Potentials                                                                            | Tuuli Tettenborn                                                                 | 2026 | Digital Library of the University of Innsbruck (University of Innsbruck) | 0     | (link)                           | 0.006 (2026)                       |
| Insights Into Radiation Damage in YBa <sub>2</sub> Cu <sub>3</sub> O <sub>7-x</sub> From Machine-Learned Interatomic Potentials                                        | Ashley Dicksona; Niccolò Di Eugenio; Federico Ledda et al.                       | 2025 | arXiv (Cornell University)                                               | 0     | 10.48550/arxiv.2512.24430        |                                    |
| Ion Dynamics in Amorphous Solid Electrolytes Studied Using Neural Network Potentials                                                                                   | Kōji Shimizu                                                                     | 2024 | The Materials Research Society series                                    | 0     | 10.1007/978-981-97-6039-8_35     |                                    |
| On the Boroxol Ring Fraction in Melt-Quenched B <sub>2</sub> O <sub>3</sub> Glass: Insights from Machine Learning Potentials                                           | Nikhil Avula; Sundaram Balasubramanian; Debendra Meher                           | 2026 | AIP Publishing                                                           | 0     | 10.60893/figshare.jcp.4321831.v1 | 0.4321831 (2026)                   |
| A neural network potential for the phase change material gete: large scale molecular dynamics simulations with close to ab initio accuracy                             | Gabriele C. Sossio                                                               | 2013 | BOA (University of Milano-Bicocca)                                       | 0     | (link)                           | 0.014 (2026)                       |
| On the Boroxol Ring Fraction in Melt-Quenched B <sub>2</sub> O <sub>3</sub> Glass: Insights from Machine Learning Potentials                                           | Nikhil Avula; Sundaram Balasubramanian; Debendra Meher                           | 2026 | AIP Publishing                                                           | 0     | 10.60893/figshare.jcp.4321831    | 0.4321831 (2026)                   |
| Revealing Phonon Bridge Effect for Amorphous vs Crystalline Metal-Silicide Layers at Si/Ti Interfaces by a Machine Learning Potential                                  | Mayur Pratap Singh; Lokanath Patra; Cheng Zhang et al.                           | 2026 | ACS Applied Electronic Materials                                         | 0     | 10.1021/acsaelm.5c02582          | 0.2582 (2026)                      |
| Investigation of medium-range order in amorphous GeTe and Ge <sub>2</sub> Sb <sub>2</sub> Te <sub>5</sub> using density functional theory and neural network potential |                                                                                  | 2020 | Seoul National University Open Repository (Seoul National University)    | 0     | (link)                           |                                    |

| Title                                                                                                                                               | Authors                                                         | Year | Venue                                      | Cited | DOI                                          | OpenAlex<br>2-yr mean<br>citedness |
|-----------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|------|--------------------------------------------|-------|----------------------------------------------|------------------------------------|
| Charge integrated graph neural network-based machine learning potential for amorphous and non-stoichiometric hafnium oxide                          | Hyo Gyeong Shin; Seong Hun Kim; Eun Ho Kim et al.               | 2025 | npj Computational Materials                | 0     | 10.1038/s41524-025-01864-3                   |                                    |
| Application of Neural Network Potentials to Materials Science Analysis of Partial Crystallization in Glass Structures and Ion Conduction Mechanisms | Kōji Shimizu                                                    | 2025 | Vacuum and Surface Science                 | 0     | 10.1380/vss.68.350                           |                                    |
| Desorption dynamics of interstellar molecule on amorphous solid water investigated by machine learning potential-based PaCS-MD simulation           | Watanabe Natsuki; Kästner Johannes; Yuta Hori et al.            | 2026 | Institutional Repositories DataBase (IRDB) | 0     | (link)                                       |                                    |
| Desorption Dynamics of Interstellar Molecule on Amorphous Solid Water Investigated by Machine Learning Potential-based PaCS-MD Simulation           | Yuta Hori; Natsuki Watanabe; Johannes Kästner et al.            | 2026 | AIP Publishing                             | 0     | 10.60893/figshare.jcp.42064920               |                                    |
| Desorption dynamics of interstellar molecule on amorphous solid water investigated by machine learning potential-based PaCS-MD simulation           | Natsuki Watanabe; Johannes Kästner; Yuta Hori et al.            | 2026 | The Journal of Chemical Physics            | 0     | 10.1063/5.0325531                            |                                    |
| Development and Validation of Neural Network Potentials for Multicomponent Oxide Glasses                                                            | Ryuki Kayano; Yaohiro Inagaki; Ryuta Matsubara et al.           | 2024 | ChemRxiv                                   | 0     | 10.26434/chemrxiv-0.467 (2026) 2024-hb50k-v2 |                                    |
| Development and Validation of Neural Network Potentials for Multicomponent Oxide Glasses                                                            | Ryuki Kayano; Yaohiro Inagaki; Ryuta Matsubara et al.           | 2024 | ChemRxiv                                   | 0     | 10.26434/chemrxiv-0.467 (2026) 2024-hb50k    |                                    |
| Neutron and X-ray Diffraction Reveal the Limits of Long-Range Machine Learning Potentials for Medium-Range Order in Silica Glass                    | Sai Harshit Balantrapu; Atul C. Thakur; Chris J. Benmore et al. | 2026 | arXiv (Cornell University)                 | 0     | (link)                                       |                                    |
| Neutron and X-ray Diffraction Reveal the Limits of Long-Range Machine Learning Potentials for Medium-Range Order in Silica Glass                    | Sai Harshit Balantrapu; Atul C. Thakur; Chris J. Benmore et al. | 2026 | arXiv (Cornell University)                 | 0     | 10.48550/arxiv.2604.21222                    |                                    |
| Neutron and x-ray diffraction reveal the limits of long-range machine learning potentials for medium-range order in silica glass                    | Sai Harshit; Atul C. Thakur; Chris J. Benmore et al.            | 2026 | Journal of Physics Materials               | 0     | 10.1088/2515-7639/ae8643                     |                                    |
| A systematic approach to generating accurate neural network potentials: the case of carbon                                                          | Yusuf Shaidu; Emine Küçükbenli; Ruggero Lot et al.              | 2021 | npj Computational Materials                | 0     | 10.1038/s41524-021-00508-6                   |                                    |
| High-dimensional neural-network potentials for phase change materials for data storage                                                              | Marco Bernasconi                                                | 2012 | BOA (University of Milano-Bicocca)         | 0     | (link)                                       | 0.014 (2026)                       |
| ALDo 2.0: Scalable thermal transport from first principles and machine learning potentials                                                          | Giuseppe Barbalinardo; Zekun Chen; Dylan A. Folkner et al.      | 2026 | Mendeley Data                              | 0     | 10.17632/t3f42nv4fw.1                        |                                    |
| ALDo 2.0: Scalable thermal transport from first principles and machine learning potentials                                                          | Giuseppe Barbalinardo; Zekun Chen; Dylan A. Folkner et al.      | 2026 | Mendeley Data                              | 0     | 10.17632/t3f42nv4fw                          |                                    |
| Dynamic heterogeneity in sodium silicate melts via machine-learning potential                                                                       | Kumpei Shiraishi; Rikuta Nozawa; Emi Minamitani                 | 2026 | arXiv (Cornell University)                 | 0     | (link)                                       |                                    |
| Dynamic heterogeneity in sodium silicate melts via machine-learning potential                                                                       | Kumpei Shiraishi; Rikuta Nozawa; Emi Minamitani                 | 2026 | arXiv (Cornell University)                 | 0     | 10.48550/arxiv.2606.26646                    |                                    |
| A Systematic Approach to Generating Accurate Neural Network Potentials: the Case of Carbon                                                          | Yusuf Shaidu; Emine Küçükbenli; Ruggero Lot et al.              | 2020 | arXiv (Cornell University)                 | 0     | 10.48550/arxiv.2011.04604                    |                                    |
| ALDo 2.0: Scalable thermal transport from first principles and machine learning potentials                                                          | Giuseppe Barbalinardo; Zekun Chen; Dylan A. Folkner et al.      | 2026 | Computer Physics Communications            | 0     | 10.1016/j.cpc.2026.110782                    | 0.022 (2026)                       |

| Title                                                                                                                                                                                                  | Authors                                                                | Year | Venue                                                       | Cited | DOI                           | OpenAlex<br>2-yr mean<br>citedness |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|------|-------------------------------------------------------------|-------|-------------------------------|------------------------------------|
| Deformation mechanisms and compressive response of NbTaTiZr alloy via machine learning potentials                                                                                                      | LIU Hongyang; Bo Chen; Rong Chen et al.                                | 2025 | Acta Physica Sinica                                         | 0     | 10.7498/aps.74.20250738       | 0.000 (2026)                       |
| Density functional and neural-network potential simulations of Ag migration in disordered GeS <sub>2</sub> electrolytes                                                                                | Jaakko Akola; Sondre Dahl; A Götz et al.                               | 2026 | Physical Review Materials                                   | 0     | 10.1103/75rr-pqqp             |                                    |
| Development of a TaN-Ce Machine Learning Potential and Its Application to Solid-Liquid Interface Simulations                                                                                           | Yunhan Zhang; Jianfeng Cai; Hongjian Chen et al.                       | 2025 | Metals                                                      | 0     | 10.3390/met15090972           |                                    |
| Modeling Short-Range and Three-Membered Ring Structures in Lithium Borosilicate Glasses using Machine Learning Potential                                                                               | Shingo Urata                                                           | 2022 | arXiv (Cornell University)                                  | 0     | 10.48550/arxiv.2209.00316     |                                    |
| Neural Network Potentials with Effective Charge Separation for Non-Equilibrium Dynamics of Ionic Solids: A ZnO Case Study                                                                              | Gang Seob Jung; Lei Cheng                                              | 2025 | ChemRxiv                                                    | 0     | 10.26434/chemrxiv-2025-fm5t6  | 0.467 (2026)                       |
| Study of vacancy ordering and the boson peak in metastable cubic Ge-Sb-Te using machine learning potentials                                                                                            | Young-Jae Choi; Minjae Ghim; Seung-Hoon Jhi                            | 2023 | arXiv (Cornell University)                                  | 0     | 10.48550/arxiv.2309.01089     |                                    |
| Dynamic mesophase transition induces anomalous suppressed and anisotropic phonon transport revealed by unified machine learning potential                                                              | ; Linfeng Yu; et al.                                                   | 2024 | Research Square                                             | 0     | 10.21203/rs.3.rs-4082274/v1   |                                    |
| Viscosity, breakdown of Stokes-Einstein relation and dynamical heterogeneity in supercooled liquid Ge <sub>25</sub> Sb <sub>25</sub> Te <sub>50</sub> from simulations with a neural network potential | Simone Marcorini; Rocco Pomodoro; Omar Abou El Kheir et al.            | 2025 | arXiv (Cornell University)                                  | 0     | 10.48550/arxiv.2506.13668     |                                    |
| Comparison of intermediate-range order in GeO <sub>2</sub> glass: molecular dynamics using machine-learning interatomic potential vs. reverse Monte Carlo fitting to experimental data                 | Kenta Matsutani; Shusuke Kasamatsu; Takeshi Usuki                      | 2024 | arXiv (Cornell University)                                  | 0     | 10.48550/arxiv.2409.06982     |                                    |
| Study of the thermodynamics and kinetic anomalies of Antimony-based phase-change materials by machine-learned molecular dynamics                                                                       | FLAVIO GIULIANI                                                        | 2026 | IRIS Research product catalog (Sapienza University of Rome) | 0     | (link)                        |                                    |
| Electronic structure and reactive molecular dynamics simulations for modelling solid-state batteries                                                                                                   | ;                                                                      | 2025 | The HKU Scholars Hub (University of Hong Kong)              | 0     | (link)                        |                                    |
| Elucidating Lithium Transport Mechanisms in Disordered LiF from Machine-Learning Molecular Dynamics Simulations                                                                                        | ; Nikhil Rampal; Nicole Adelstein et al.                               | 2026 | ACS Materials Letters                                       | 0     | 10.1021/acsmaterials.1c0326   | 0.000 (2026)                       |
| Machine Learning Guided Exploration of Amorphous Electrodes and Electrolytes                                                                                                                           | Sai Gautam Gopalakrishnan; Aqshat Seth; Debsundar Dey et al.           | 2025 | ECS Meeting Abstracts                                       | 0     | 10.1149/ma2025-0271034mtgabs  | 0.091 (2026)                       |
| Large-scale first-principle simulations of amorphous indium oxide                                                                                                                                      | Matthew Bousquet; Francois Gygi; Giulia Galli                          | 2026 | arXiv (Cornell University)                                  | 0     | (link)                        |                                    |
| Large-scale first-principle simulations of amorphous indium oxide                                                                                                                                      | Matthew Bousquet; Francois Gygi; Giulia Galli                          | 2026 | arXiv (Cornell University)                                  | 0     | 10.48550/arxiv.2607.08617     |                                    |
| Amorphous boron nitride as an ultrathin copper diffusion barrier for advanced interconnects                                                                                                            | Onurcan Kaya; Hyeonjoon Kim; Byeongkyu Kim et al.                      | 2025 | 2D Materials                                                | 0     | 10.1088/2053-1583/ae2521      | 4.053 (2026)                       |
| Large scale molecular modelling of basalt surfaces and fracturing in basaltic glass                                                                                                                    | Marthe Grønlie Guren; Henrik Andersen Sveinsson; Razvan Caracas et al. | 2024 |                                                             | 0     | 10.5194/egusphere-egu24-11861 |                                    |

| Title                                                                                                                                                                              | Authors                                                            | Year | Venue                                                                               | Cited | DOI                             | OpenAlex<br>2-yr mean<br>citedness |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|------|-------------------------------------------------------------------------------------|-------|---------------------------------|------------------------------------|
| Structure and mechanical properties of monolayer amorphous carbon and boron nitride                                                                                                | Xi Zhang; Yutian Zhang; Yunpeng Wang et al.                        | 2023 | arXiv (Cornell University)                                                          | 0     | 10.48550/arxiv.2309.15352       |                                    |
| Large scale simulations of amorphous phase change materials for data storage                                                                                                       | Marco Bernasconi                                                   | 2012 | BOA (University of Milano-Bicocca)                                                  | 0     | (link)                          | 0.014 (2026)                       |
| The ab initio amorphous materials database: Empowering machine learning to decode diffusivity                                                                                      | Hui Zheng; Patrick Huck                                            | 2024 | OSTI OAI (U.S. Department of Energy Office of Scientific and Technical Information) | 0     | 10.17188/mpcontribs/2281590     |                                    |
| Energy dissipation at the atomic scale explains how fracture energy depends on crack velocity in silica glass                                                                      | Marthe Grønlie Guren; Sigbjørn Løland Bore; François Renard et al. | 2026 | arXiv (Cornell University)                                                          | 0     | 10.48550/arxiv.2605.03457       |                                    |
| Energy dissipation at the atomic scale explains how fracture energy depends on crack velocity in silica glass                                                                      | Marthe Grønlie Guren; Sigbjørn Løland Bore; François Renard et al. | 2026 | arXiv (Cornell University)                                                          | 0     | (link)                          |                                    |
| Role of Characteristic Length Scale in Interface Graphitization-Induced Wear Resistance of Diamond and Amorphous Carbon                                                            | Wenliang Shi; YanYu Zhang; WenZheng Liao et al.                    | 2026 | arXiv (Cornell University)                                                          | 0     | (link)                          |                                    |
| Role of Characteristic Length Scale in Interface Graphitization-Induced Wear Resistance of Diamond and Amorphous Carbon                                                            | Wenliang Shi; YanYu Zhang; WenZheng Liao et al.                    | 2026 | arXiv (Cornell University)                                                          | 0     | 10.48550/arxiv.2606.03325       |                                    |
| Neural Network-Based Simulation Method to Examine Ion Behaviors under Electric Fields: Application to Ion Migration in Amorphous Li3PO4                                            | Kōji Shimizu; Ryuji Otsuka; Satoshi Watanabe                       | 2023 | Journal of the Japan Society of Powder and Powder Metallurgy                        | 0     | 10.2497/jjspm.23-00043          |                                    |
| Machine-Learned Potential-Driven Insights into Li S-P S Glass Electrolytes and Li PS Cl Interfacial Chemistry for Solid-State Batteries                                            | Byungju Lee; C S Kim; Sojeong Yang                                 | 2026 | ECS Meeting Abstracts                                                               | 0     | 10.1149/ma2026-016700mtgabs     | 0.091 (2026)                       |
| Suppressed glass transition behavior through high configurational entropy in high-entropy metallic glasses                                                                         | Jinyue Wang; Fuzhi Dai; Yihuan Cao et al.                          | 2026 | Acta Materialia                                                                     | 0     | 10.1016/j.actamat.2026.122302   | 0.061 (2026)                       |
| Visualizing a Li-depleted amorphous cathode-electrolyte interphase in sulfide solid-state batteries using in situ cryogenic electron microscopy                                    | Yuki Nomura; Ryoma Sasaki; Huu Duc Luong et al.                    | 2026 | ChemRxiv                                                                            | 0     | 10.26434/chemrxiv.15463253026   | 0.003 (2026)                       |
| Quantitative assessment of the structure of (Na2O)0.1-(V2O5)0.6-(TeO2)0.3 glass: An ab initio and machine learning interatomic potential study of oxide systems with mixed-valence | Firas Shuaib; Rémi Piotrowski; Gaëlle Delaizir et al.              | 2026 | Artificial Intelligence Chemistry                                                   | 0     | 10.1016/j.aichem.2026.034012326 | 0.004 (2026)                       |
| Thermochemical properties and glass transition of Na 2 O-Al 2 O 3 -P 2 O 5 system: a deep learning potential approach and its experimental validation                              | Dmitry Zakiryanov; Maxim I. Vlasov; Victor Bykov                   | 2026 | Modelling and Simulation in Materials Science and Engineering                       | 0     | 10.1088/1361-651x/ae39f7        |                                    |
| Study on Cu 50 Zr 50 Metallic Glass Based on Machine Learning Force Field: Correlation between Virial and Equipartition Theorem Violations and Atomic-Scale Dynamic Behaviors      | Wenliang Shi; YanYu Zhang; HongYu Wu et al.                        | 2026 | Chinese Physics B                                                                   | 0     | 10.1088/1674-1056/ae42ba        | 1.149 (2026)                       |
| Zero-Dimensional Ice Ring Topological Proton Glass: Gapless Texture Soft Modes, Anharmonic Wandering, and Three Conserved Invariants                                               | Zhongzheng Miao; Miao Zhongzheng                                   | 2026 | ChemRxiv                                                                            | 0     | 10.26434/chemrxiv.15463253026   | 0.003 (2026)                       |

| Title                                                                                                                                                                              | Authors                                                                    | Year | Venue                                                      | Cited | DOI                           | OpenAlex<br>2-yr mean<br>citedness |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|------|------------------------------------------------------------|-------|-------------------------------|------------------------------------|
| Nanoparticules d'argent enrobées dans une matrice de silice pour des dispositifs antibactériens : synthèse en phase gazeuse et modélisation assistée par apprentissage automatique | Marie-Salomé Trillot                                                       | 2025 | HAL (Le Centre pour la Communication Scientifique Directe) | 0     | (link)                        | 0.03 (2026)                        |
| Influence of crystal orientation and polymorphism on the shock response of boron carbide                                                                                           | Kimia Ghaffari; Salil Bavdekar; Douglas E. Spearot et al.                  | 2025 | Acta Materialia                                            | 0     | 10.1016/j.actamat.2025.172116 | 0.03 (2026)                        |
| Phonon Transport in Disordered Two-Dimensional Graphene                                                                                                                            | Hassan Shoaib                                                              | 2025 | McGill-DEV                                                 | 0     | 10.82308/27920                |                                    |
| Atomic structure at the surface of warm basaltic glasses                                                                                                                           | Marthe Grønlie Guren; Henrik Anderson Sveinson; Razvan Caracas et al.      | 2025 |                                                            | 0     | 10.5194/egusphere-egu25-12944 |                                    |
| Through Rainbow-Tinted Glasses: Machine Learning-Driven Modeling of Chromophores                                                                                                   | Eric Lindgren                                                              | 2026 |                                                            | 0     | 10.63959/chalmers.dt/5890     |                                    |
| When More Data Hurts: Optimizing Data Coverage While Mitigating Diversity Induced Underfitting in an Ultra-Fast Machine-Learned Potential                                          | Jason B. Gibson; Tesia Janicki; Ajinkya C. Hire et al.                     | 2024 | arXiv (Cornell University)                                 | 0     | 10.48550/arxiv.2409.07610     |                                    |
| Dynamic Anion Sublattices, Cooperative Ion Migration and Metastable Pathway Engineering for Next-Generation Solid Electrolytes                                                     | Rajab Abbas; Rabia Shahzad; Aasma Bibi et al.                              | 2026 | Saudi Journal of Engineering and Technology                | 0     | 10.36348/sjet.2026.v11i08.003 |                                    |
| Structural origin of disorder-induced ion conduction in NaFePO <sub>4</sub> cathode materials                                                                                      | Rasmus Christensen; Kristin A. Persson; Morten M. Smedskjær                | 2025 | ChemRxiv                                                   | 0     | 10.26434/chemrxiv-2025-9sf4n  | 0.467 (2026)                       |
| MOLECULAR SIMULATION METHODS FOR MODELING                                                                                                                                          | Dylan David Fortney                                                        | 2026 | Purdue                                                     | 0     | 10.25394/pgs.32048331.v1      |                                    |
| MOLECULAR SIMULATION METHODS FOR MODELING                                                                                                                                          | Dylan David Fortney                                                        | 2026 | Purdue                                                     | 0     | 10.25394/pgs.32048331         |                                    |
| Nuclear Quantum Effects in the Acetylene:Ammonia Plastic Co-crystal                                                                                                                | Atul C. Thakur; Richard C. Remsing                                         | 2023 | arXiv (Cornell University)                                 | 0     | 10.48550/arxiv.2310.00480     |                                    |
| Beyond Boltzmann transport: Green-Kubo prediction of lattice thermal conductivity with machine-learned potentials                                                                  | Xiaoying Zhang; Ning Liu; Qian Guo et al.                                  | 2026 | The Journal of Chemical Physics                            | 0     | 10.1063/5.0297462             |                                    |
| Revisit Many-body Interaction Heat Current and Thermal Conductivity Calculation in Moment Tensor Potential/LAMMPS Interface                                                        | Tai, Siu Ting; Chen Wang; Ruihuan Cheng et al.                             | 2024 | arXiv (Cornell University)                                 | 0     | 10.48550/arxiv.2411.01255     |                                    |
| Probing the Formation Mechanisms and Vibrational Spectra of Silicate Cloud Particles on WASP-17b using Molecular Simulation                                                        | Stephen Ingram; Yifei Chen; Hanna Vehkamäki                                | 2026 |                                                            | 0     | 10.5194/egusphere-egu26-9272  |                                    |
| Competitive Hydrogen-Bond Partitioning in Deep Eutectic Solvents: From Cooperative Charge Spreading to Structure-Property Design Rules                                             | Sergio de-la-Huerta-Sainz; Valentín Díez-Cabanes; Alberto Gutiérrez et al. | 2026 | ACS Omega                                                  | 0     | 10.1021/acsomega.6c00376      | 0.0376 (2026)                      |
| Thermal Transport Mechanisms of Phonons, Diffusions, Electrons, and Ions in Functional Materials                                                                                   | Yatian Zhang                                                               | 2026 | Media (https://www.suub.uni-bremen.de/)                    | 0     | 10.26092/elib/5520            |                                    |
| Accelerating Amorphous Alloy Discovery: Data-Driven Property Prediction via General-Purpose Machine Learning Interatomic Potential                                                 | Xuhe Gong; Hengbo Zhao; Xiao Fu et al.                                     | 2025 | arXiv preprint                                             | 0     | (link)                        |                                    |
| Unveiling defect motifs in amorphous GeSe using machine learning interatomic potentials                                                                                            | Minseok Moon; Seungwoo Hwang; Jaesun Kim et al.                            | 2025 | arXiv preprint                                             | 0     | (link)                        |                                    |



| Title                                                                                                                                                                                  | Authors                                                   | Year | Venue          | Cited | DOI                             | OpenAlex<br>2-yr mean<br>citedness |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|------|----------------|-------|---------------------------------|------------------------------------|
| Li-P-S Electrolyte Materials as a Benchmark for Machine-Learned Interatomic Potentials                                                                                                 | Nataschia L. Fragapane; Volker L. Deringer                | 2025 | arXiv preprint | 0     | (link)                          |                                    |
| Persistent homology-based descriptor for machine-learning potential of amorphous structures                                                                                            | Emi Minamitani; Ippei Obayashi; Koji Shimizu et al.       | 2022 | arXiv preprint | 0     | (link)                          |                                    |
| Unveiling the crystallization kinetics in Ge-rich Ge <sub>x</sub> Te alloys by large scale simulations with a machine-learned interatomic potential                                    | Dario Baratella; Omar Abou El Kheir; Marco Bernasconi     | 2024 | arXiv preprint | 0     | (link)                          |                                    |
| Melt-Quench Failures and Practical Solutions for Universal Machine-Learning Interatomic Potentials in Amorphous Structure Generation                                                   | Shuwei Li; Yuqi An; Xingyu Guo et al.                     | 2026 | arXiv preprint | 0     | (link)                          |                                    |
| Comparison of intermediate-range order in GeO <sub>2</sub> glass: molecular dynamics using machine-learning interatomic potential vs. reverse Monte Carlo fitting to experimental data | Kenta Matsutani; Shusuke Kasamatsu; Takeshi Usuki         | 2024 | arXiv preprint | 0     | (link)                          |                                    |
| Machine-learned potential for amorphous Indium-Tin-Oxide alloys                                                                                                                        | Shuaiyang Guo; Yuan Wang; Wei Zhang                       | 2026 | arXiv preprint | 0     | (link)                          |                                    |
| Modeling Short-Range and Three-Membered Ring Structures in Lithium Borosilicate Glasses using Machine Learning Potential                                                               | Shingo Urata                                              | 2022 | arXiv preprint | 0     | (link)                          |                                    |
| Large-scale atomistic study of plasticity in amorphous gallium oxide with a machine-learning potential                                                                                 | Jiahui Zhang; Junlei Zhao; Jesper Byggmästar et al.       | 2024 | arXiv preprint | 0     | (link)                          |                                    |
| Validation Workflow for Machine Learning Interatomic Potentials for Complex Ceramics                                                                                                   | Kimia Ghaffari; Salil Bavdekar; Douglas E. Spearot et al. | 2024 | arXiv preprint | 0     | 10.1016/j.commatsci.2024.112983 |                                    |
| Atomistic Insights into Cu/amorphous-Ta <sub>x</sub> N                                                                                                                                 | Jeong Min Choi; Jaehoon Kim; Ji-Hwan Lee et al.           | 2025 | arXiv preprint | 0     | (link)                          |                                    |
| Interfacial Adhesion via Machine Learning Interatomic Potentials: Effects of Stoichiometry and Interface Construction                                                                  | Zixiong Wei; Nongnuch Artrith                             | 2024 | arXiv preprint | 0     | (link)                          |                                    |
| Machine Learning Potential Powered Insights into the Mechanical Stability of Amorphous Li-Si Alloys                                                                                    | Ata Madanchi; Emna Azek; Karim Zongo et al.               | 2024 | arXiv preprint | 0     | (link)                          |                                    |
| Is the Future of Materials Amorphous? Challenges and Opportunities in Simulations of Amorphous Materials                                                                               | Laura-Bianca Paşca; Yuanbin Liu; Andy S. Anker et al.     | 2025 | arXiv preprint | 0     | (link)                          |                                    |
| Machine-learning-driven modelling of amorphous and polycrystalline BaZrS <sub>3</sub>                                                                                                  | Fraser Birks; Ibrahim Ghanem; Lars Pastewka et al.        | 2026 | arXiv preprint | 0     | 10.1103/6n5m-rxc1               |                                    |
| Resolving Structural Avalanches in Amorphous Carbon with Arclength Continuation                                                                                                        | Omar Abou El Kheir; Dario Baratella; Marco Bernasconi     | 2026 | arXiv preprint | 0     | (link)                          |                                    |
| On phase separation and crystallization of Ge-rich GeSbTe alloys from atomistic simulations with a machine learning interatomic potential                                              | Rustam Arabov; Jiaxuan Li; Xiaotong Chen et al.           | 2026 | arXiv preprint | 0     | (link)                          |                                    |
| Thermal Conductivity and Temperature-Induced Band Gap Renormalization in Crystalline and Amorphous Ga <sub>2</sub> O <sub>3</sub>                                                      | Mirko Fischer; Andreas Heuer                              | 2026 | arXiv preprint | 0     | (link)                          |                                    |
| From oligomers to entangled polymers: How to train a transferable machine learning interatomic potential                                                                               |                                                           |      |                |       |                                 |                                    |

| Title                                                                                                                                                              | Authors                                                               | Year | Venue                                               | Cited | DOI                          | OpenAlex<br>2-yr mean<br>citedness |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|------|-----------------------------------------------------|-------|------------------------------|------------------------------------|
| ATLAS: A Foundation Neural Sampler for Amorphous Materials                                                                                                         | Mouyang Cheng; Denis Blessing; Botao Yu et al.                        | 2026 | arXiv preprint                                      | 0     | (link)                       |                                    |
| Polymorphic crystallites model for monolayer amorphous materials                                                                                                   | Le-Ye Zhu; Xi Zhang; Yun-Peng Wang et al.                             | 2026 | arXiv preprint                                      | 0     | (link)                       |                                    |
| A Unified Deep Neural Network Potential Capable of Predicting Thermal Conductivity of Silicon in Different Phases                                                  | Ruiyang Li; Eungkyu Lee; Tengfei Luo                                  | 2019 | arXiv preprint                                      | 0     | (link)                       |                                    |
| Atomistic insights into hydrogen migration in IGZO from machine-learning interatomic potential: linking atomic diffusion to device performance                     | Hyunsung Cho; Minseok Moon; Jaehoon Kim et al.                        | 2025 | arXiv preprint                                      | 0     | (link)                       |                                    |
| Thermal Conductivity Modeling using Machine Learning Potentials: Application to Crystalline and Amorphous Silicon                                                  | Xin Qian; Shenyong Peng; Xiaobo Li et al.                             | 2019 | arXiv preprint                                      | 0     | (link)                       |                                    |
| Amorphous materials as a frontier challenge for universal interatomic potentials                                                                                   | Nataascia L. Fragapane; Volker L. Deringer                            | 2026 | arXiv preprint                                      | 0     | (link)                       |                                    |
| Efficient training of machine learning potentials for metallic glasses: CuZrAl validation                                                                          | Antoni Wadowski; Anshul D. S. Parmar; Filip Kaškosz et al.            | 2024 | arXiv preprint                                      | 0     | (link)                       |                                    |
| Constitutive relations for plasticity of amorphous carbon                                                                                                          | Richard Jana; Julian von Lautz; S. Mostafa Khosrownejad et al.        | 2020 | J. Phys. Mater. 3, 035005 (2020)                    | 0     | 10.1088/2515-7639/ab953c     |                                    |
| Crystal-like thermal transport in amorphous carbon                                                                                                                 | Jaeyun Moon; Zhiting Tian                                             | 2024 | arXiv preprint                                      | 0     | (link)                       |                                    |
| Structural and elastic properties of amorphous carbon from simulated quenching at low rates                                                                        | Richard Jana; Daniele Savio; Volker L. Deringer et al.                | 2019 | Modelling Simul. Mater. Sci. Eng. 27, 085009 (2019) | 0     | 10.1088/1361-651X/ab45da     |                                    |
| A Gaussian Approximation Potential for Amorphous Si:H                                                                                                              | Davis Unruh; Reza Vatan Meidanshahi; Stephen M. Goodnick et al.       | 2021 | arXiv preprint                                      | 0     | (link)                       |                                    |
| On the origin of in-gap states in amorphous Ge <sub>2</sub> Sb <sub>2</sub> Te <sub>5</sub>                                                                        | Omar Abou El Kheir; Marco Bernasconi                                  | 2026 | arXiv preprint                                      | 0     | (link)                       |                                    |
| A Systematic Approach to Generating Accurate Neural Network Potentials: the Case of Carbon                                                                         | Yusuf Shaidu; Emine Kucukbenli; Ruggero Lot et al.                    | 2020 | arXiv preprint                                      | 0     | (link)                       |                                    |
| Modulation of structural short-range order due to chemical patterning in multi-component amorphous interfacial complexions                                         | Esther C. Hessong; Zhengyu Zhang; Tianjiao Lei et al.                 | 2025 | arXiv preprint                                      | 0     | (link)                       |                                    |
| Medium-range structural order in amorphous arsenic                                                                                                                 | Yuanbin Liu; Yuxing Zhou; Richard Ademuwagun et al.                   | 2025 | arXiv preprint                                      | 0     | (link)                       |                                    |
| From Bulk to Surface: Structure and Dynamics of Amorphous Alumina from Deep Potential Molecular Dynamics                                                           | Zheng Yu; Jiayan Xu; Abhirup Patra et al.                             | 2026 | arXiv preprint                                      | 0     | (link)                       |                                    |
| Recent Advances in Metallic Glasses                                                                                                                                | Silvia Bonfanti; Ralf Busch; Jesper Byggmästar et al.                 | 2025 | arXiv preprint                                      | 0     | (link)                       |                                    |
| Bridging classical and quantum interpretation of chemical state analysis by XPS/HAXPES to resolve short-range order in amorphous alumina films                     | Simon Gramatte; Xing Wang; Michael Alejandro Hernández Bertrán et al. | 2024 | arXiv preprint                                      | 0     | (link)                       |                                    |
| Machine-Learning-Guided Insights into Solid-Electrolyte Interphase Conductivity: Are Amorphous Lithium Fluorophosphates the Key?                                   | Peichen Zhong; Kristin A. Persson                                     | 2025 | arXiv preprint                                      | 0     | 10.1021/acsenerylett.5c03526 |                                    |
| Atomic insight into Li <sup>+</sup> ion transport in amorphous electrolytes Li <sub>x</sub> AlO <sub>y</sub> Cl <sub>3+x-2y</sub> (0.5 ≤ x ≤ 1.5, 0.25 ≤ y ≤ 0.75) | Yang Qifan; Xu Jing; Fu Xiao et al.                                   | 2024 | arXiv preprint                                      | 0     | (link)                       |                                    |

| Title                                                                                                                                                    | Authors                                                     | Year | Venue                        | Cited  | DOI                         | OpenAlex<br>2-yr mean<br>citedness |
|----------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|------|------------------------------|--------|-----------------------------|------------------------------------|
| Turing Pattern Engineering Enables Kinetically Ultrastable yet Ductile Metallic Glasses                                                                  | Huanrong Liu; Qingan Li; Shan Zhang et al.                  | 2025 | arXiv preprint               | 0      | (link)                      |                                    |
| Realistic atomistic structure of amorphous silicon from machine-learning-driven molecular dynamics                                                       | Volker L. Deringer; Noam Bernstein; Albert P. Bartók et al. | 2018 | arXiv preprint               | 0      | (link)                      |                                    |
| Density dependence of thermal conductivity in nanoporous and amorphous carbon with machine-learned molecular dynamics                                    | Yanzhou Wang; Zheyong Fan; Ping Qian et al.                 | 2024 | arXiv preprint               | 0      | (link)                      |                                    |
| The ab initio amorphous materials database: Empowering machine learning to decode diffusivity                                                            | Hui Zheng; Eric Sivonxay; Max Gallant et al.                | 2024 | arXiv preprint               | 0      | (link)                      |                                    |
| Structural properties of amorphous Na <sub>3</sub> SOCl electrolyte by first-principles and machine learning molecular dynamics                          | T. -L. Pham; M. Guerboub; S. D. Wansi Wendj et al.          | 2024 | arXiv preprint               | 0      | (link)                      |                                    |
| Enhanced ionic conductivity through crystallization of glass-Li <sub>3</sub> PS <sub>4</sub> by machine learning molecular dynamics simulations          | Koji Shimizu; Parth Bahuguna; Shigeo Mori et al.            | 2023 | arXiv preprint               | 0      | (link)                      |                                    |
| Machine learning potential for predicting thermal conductivity of $\alpha$ -phase and amorphous Tantalum Nitride                                         | Zhicheng Zong; Yangjun Qin; Jiahong Zhan et al.             | 2025 | arXiv preprint               | 0      | (link)                      |                                    |
| Optimization and Validation of a Deep Learning CuZr Atomistic Potential: Robust Applications for Crystalline and Amorphous Phases with near-DFT Accuracy | Christopher M. Andolina; Philip Williamson; Wisam A. Saidi  | 2020 | J. Chem. Phys. 154701 (2020) | 152, 0 | 10.1063/5.0005347           |                                    |
| Mechanisms of alkali ionic transport in amorphous oxyhalides solid state conductors                                                                      | Luca Binci; KyuJung Jun; Bowen Deng et al.                  | 2026 | arXiv preprint               | 0      | (link)                      |                                    |
| Neural-Network Assisted Study of Nitrogen Atom Dynamics on Amorphous Solid Water – II. Diffusion                                                         | Viktor Zaverkin; Germán Molpeceres; Johannes Kästner        | 2021 | arXiv preprint               | 0      | 10.1093/mnras/stab3631      |                                    |
| Machine Learning Assisted Modeling of Amorphous TiO <sub>2</sub> -Doped GeO <sub>2</sub> for Advanced LIGO Mirror Coatings                               | Jun Jiang; Rui Zhang; Kiran Prasai et al.                   | 2025 | arXiv preprint               | 0      | (link)                      |                                    |
| Fast Isotropic Li-Ion Diffusion in Zeolitic Imidazolate Framework Glass Electrolytes for Batteries                                                       | Yong Li; Tao Du; Timothée Jamin et al.                      | 2026 | arXiv preprint               | 0      | (link)                      |                                    |
| kALDo 2.0: Scalable Thermal Transport from First Principles and Machine Learning Potentials                                                              | Giuseppe Barbalinardo; Zekun Chen; Dylan Folkner et al.     | 2026 | arXiv preprint               | 0      | (link)                      |                                    |
| Neural-Network Assisted Study of Nitrogen Atom Dynamics on Amorphous Solid Water. I. Adsorption & Desorption                                             | Germán Molpeceres; Viktor Zaverkin; Johannes Kästner        | 2020 | arXiv preprint               | 0      | 10.1093/mnras/staa2891      |                                    |
| Simulations of High Temperature Decomposition of Metal-Organic Frameworks to form Amorphous Catalysts                                                    | Connor W. Edwards; Oliver M. Linder-Patton; Jack D. Evans   | 2026 | arXiv preprint               | 0      | (link)                      |                                    |
| Thermal conductivity of glasses above the plateau: first-principles theory and applications                                                              | Michele Simoncelli; Francesco Mauri; Nicola Marzari         | 2022 | arXiv preprint               | 0      | (link)                      |                                    |
| Simulating alternating bias assisted annealing of amorphous oxide tunnel junctions                                                                       | Alexander C. Tyner; Alexander V. Balatsky                   | 2025 | arXiv preprint               | 0      | (link)                      |                                    |
| Origin of negative thermal expansion and pressure induced amorphization in zirconium tungstate from machine-learning potential                           | Ri He; Hongyu Wu; Yi Lu et al.                              | 2022 | arXiv preprint               | 0      | 10.1103/PhysRevB.106.174101 |                                    |

| Title                                                                                                                                                                                          | Authors                                                     | Year | Venue                                 | Cited | DOI                               | OpenAlex<br>2-yr mean<br>citedness |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|------|---------------------------------------|-------|-----------------------------------|------------------------------------|
| Harnessing Structural Disorder: Unraveling Hydrogen Evolution in Monolayer Amorphous Carbon via First-Principles Simulations and Machine-Learned Potentials                                    | Sreehari M S; Ashutosh Krishna Amaram; Raghavan Ranganathan | 2026 | arXiv preprint                        | 0     | (link)                            |                                    |
| Thermal transport of glasses via machine learning driven simulations                                                                                                                           | Paolo Pegolo; Federico Grasselli                            | 2024 | arXiv preprint                        | 0     | (link)                            |                                    |
| Machine learning Hamiltonian enables scalable and accurate defect calculations: The case of oxygen vacancies in amorphous SiO <sub>2</sub>                                                     | Zhenxing Dai; Zhong Yang; Mingjue Ni et al.                 | 2026 | arXiv preprint                        | 0     | (link)                            |                                    |
| Synthesizability, hardness, and stacking order in multicomponent transition metal carbides from machine-learned potentials                                                                     | Xin Liu; Anirudh Raju Natarajan                             | 2026 | arXiv preprint                        | 0     | (link)                            |                                    |
| Efficient and accurate simulation of vitrification in multi-component metallic liquids with neural-network potentials                                                                          | Rui Su; Jieyi Yu; Pengfei Guan et al.                       | 2022 | arXiv preprint                        | 0     | (link)                            |                                    |
| Atomic cluster expansion potential for the Si-H system                                                                                                                                         | Louise A. M. Rosset; Volker L. Deringer                     | 2025 | arXiv preprint                        | 0     | (link)                            |                                    |
| Mechanisms of temperature-dependent thermal transport in amorphous silica from machine-learning molecular dynamics                                                                             | Ting Liang; Penghua Ying; Ke Xu et al.                      | 2023 | Physical Review B 108, 184203 (2023)  | 0     | 10.1103/PhysRevB.108.184203       |                                    |
| Atomic-scale factors that control the rate capability of nanostructured amorphous Si for high-energy-density batteries                                                                         | Nongnuch Artrith; Alexander Urban; Yan Wang et al.          | 2019 | arXiv preprint                        | 0     | (link)                            |                                    |
| CO Adsorption Sites on Interstellar Water Ices Explored with Machine Learning Potentials. Binding energy distributions and snowline                                                            | Giulia M. Bovolenta; Germán Molpeceres; Kenji Furuya et al. | 2025 | A&A 703, A172 (2025)                  | 0     | 10.1051/0004-6361/202555836       |                                    |
| AI-Aided Mapping of the Structure-Composition-Conductivity Relationships of Glass-Ceramic Lithium Thiophosphate Electrolytes                                                                   | Haoyue Guo; Qian Wang; Alexander Urban et al.               | 2022 | arXiv preprint                        | 0     | (link)                            |                                    |
| Nine-element machine-learned interatomic potentials for multiphase refractory alloys                                                                                                           | Jesper Byggmästar; Tiago Lopes; Zheyong Fan et al.          | 2026 | arXiv preprint                        | 0     | (link)                            |                                    |
| Reaction dynamics on amorphous solid water surfaces using interatomic machine learned potentials. Microscopic energy partition revealed from the P + H -> PH reaction                          | Germán Molpeceres; Viktor Zaverkin; Kenji Furuya et al.     | 2023 | A&A 673, A51 (2023)                   | 0     | 10.1051/0004-6361/202346073       |                                    |
| Experiment-driven atomistic materials modeling: A case study combining X-ray photoelectron spectroscopy and machine learning potentials to infer the structure of oxygen-rich amorphous carbon | Tigany Zarrouk; Rina Ibragimova; Albert P. Bartók et al.    | 2024 | J. Am. Chem. Soc. 146, 14645 (2024)   | 0     | (link)                            |                                    |
| When More Data Hurts: Optimizing Data Coverage While Mitigating Diversity Induced Underfitting in an Ultra-Fast Machine-Learned Potential                                                      | Jason B. Gibson; Tesia D. Janicki; Ajinkya C. Hire et al.   | 2024 | arXiv preprint                        | 0     | (link)                            |                                    |
| Large Scale Structure Prediction of Near-Stoichiometric Magnesium Oxide Based on a Machine-Learned Interatomic Potential: Novel Crystalline Phases and Oxygen-Vacancy Ordering                 | Hossein Tahmasbi; Stefan Goedecker; S. Alireza Ghasemi      | 2021 | Phys. Rev. Materials 5, 083806 (2021) | 0     | 10.1103/PhysRevMaterials.5.083806 |                                    |

| Title                                                                                                                                                           | Authors                                                            | Year | Venue                          | Cited | DOI                        | OpenAlex<br>2-yr mean<br>citedness |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|------|--------------------------------|-------|----------------------------|------------------------------------|
| Extracting Atomic Environments for Machine Learning Interatomic Potentials                                                                                      | Jared C. Stimac; Fei Zhou; Kyle Bushick et al.                     | 2026 | arXiv preprint                 | 0     | (link)                     |                                    |
| Particle Swarm Based Hyper-Parameter Optimization for Machine Learned Interatomic Potentials                                                                    | Suresh Kondati Natarajan; Miguel A. Caro                           | 2020 | arXiv preprint                 | 0     | (link)                     |                                    |
| An Accurate and Transferable Machine Learning Potential for Carbon                                                                                              | Patrick Rowe; Volker L Deringer; Piero Gasparotto et al.           | 2020 | arXiv preprint                 | 0     | 10.1063/5.0005084          |                                    |
| Structure-property relations of silicon oxycarbides studied using a machine learning interatomic potential                                                      | Niklas Leimeroth; Jochen Rohrer; Karsten Albe                      | 2024 | arXiv preprint                 | 0     | (link)                     |                                    |
| SiC-TGAP: A machine learning interatomic potential for radiation damage simulations in 3C-SiC                                                                   | Ali Hamedani; Andrea E. Sand                                       | 2025 | arXiv preprint                 | 0     | (link)                     |                                    |
| Liquid anomalies and Fragility of Supercooled Antimony                                                                                                          | Flavio Giuliani; Francesco Guidarelli Mattioli; Yuhuan Chen et al. | 2025 | arXiv preprint                 | 0     | (link)                     |                                    |
| Unveiling hole-facilitated amorphisation in pressure-induced phase transformation of silicon                                                                    | Tong Zhao; Shulin Zhong; Yuxin Sun et al.                          | 2024 | arXiv preprint                 | 0     | (link)                     |                                    |
| Critical role of phase-dependent properties in modeling photothermal sintering of LiCoO <sub>2</sub> cathodes                                                   | Yang Hu; Benoit Sklénard; Wouter Vels et al.                       | 2026 | arXiv preprint                 | 0     | (link)                     |                                    |
| Unifying the description of hydrocarbons and hydrogenated carbon materials with a chemically reactive machine learning interatomic potential                    | Rina Ibragimova; Mikhail S. Kuklin; Tigany Zarrouk et al.          | 2024 | arXiv preprint                 | 0     | (link)                     |                                    |
| Unconventional crystallization pathway bypassing the intermediate cubic phase in phase-change superlattices                                                     | Bai-Qian Wang; Nian-Ke Chen; Yao-Jie Wang et al.                   | 2026 | arXiv preprint                 | 0     | (link)                     |                                    |
| Modeling phase separation in polymer-derived silicon carbonitride ceramics through extended machine learning molecular dynamics                                 | Fabien Mortier; Sylvian Cadars; Olivier Masson et al.              | 2026 | arXiv preprint                 | 0     | (link)                     |                                    |
| Understanding phase transitions of $\alpha$ -quartz under dynamic compression conditions by machine-learning driven atomistic simulations                       | Linus C. Erhard; Christoph Otzen; Jochen Rohrer et al.             | 2024 | npj Comput Mater 11, 58 (2025) | 0     | 10.1038/s41524-025-01542-4 |                                    |
| Simulation of the crystallization kinetics of Ge <sub>2</sub> Sb <sub>2</sub> Te <sub>5</sub> nanoconfined in superlattice geometries for phase change memories | Debdipto Acharya; Omar Abou El Kheir; Simone Marcorini et al.      | 2025 | arXiv preprint                 | 0     | (link)                     |                                    |
| Local Structure Dictates Ionic Transport and Mechanical Properties in Glassy Solid Electrolytes for Lithium Batteries                                           | Yong Li; Tao Du; Rasmus Christensen et al.                         | 2026 | arXiv preprint                 | 0     | (link)                     |                                    |
| Machine learning exploration of binding energy distributions of H <sub>2</sub> O at astrochemically relevant dust grain surfaces                                | Anant Vaishnav; Niels M. Mikkelsen; Mie Andersen                   | 2026 | arXiv preprint                 | 0     | (link)                     |                                    |
| Atomic cluster expansion for quantum-accurate large-scale simulations of carbon                                                                                 | Minaam Qamar; Matous Mrovec; Yury Lysogorskiy et al.               | 2022 | arXiv preprint                 | 0     | (link)                     |                                    |
| A high-efficiency neuroevolution potential for tobermorite and calcium silicate hydrate systems with ab initio accuracy                                         | Xiao Xu; Shijie Wang; Haifeng Qin et al.                           | 2025 | arXiv preprint                 | 0     | (link)                     |                                    |
| How Realistic are Idealized Copper Surfaces? A Machine Learning Study of Rough Copper-Water Interfaces                                                          | Linus C. Erhard; Johannes Schörghuber; Aleix Comas-Vives et al.    | 2025 | arXiv preprint                 | 0     | (link)                     |                                    |

| Title                                                                                                                                                             | Authors                                                                   | Year | Venue                                     | Cited | DOI                        | OpenAlex<br>2-yr mean<br>citedness |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|------|-------------------------------------------|-------|----------------------------|------------------------------------|
| Softening of Vibrational Modes and Anharmonicity Induced Thermal Conductivity Reduction in a-Si:H at High Temperatures                                            | Zhuo Chen; Yuejin Yuan; Yanzhou Wang et al.                               | 2025 | arXiv preprint                            | 0     | (link)                     |                                    |
| How to Train a Shallow Ensemble                                                                                                                                   | Moritz Schäfer; Matthias Kellner; Johannes Kästner et al.                 | 2026 | arXiv preprint                            | 0     | (link)                     |                                    |
| Atomistic Structure Generation and Neural-Network Screening of Hard Carbons to Identify High-Capacity Sodium Storage                                              | Harry Mclean; Aiden Daniel Emery; Theodore Thomas Walton et al.           | 2026 | arXiv preprint                            | 0     | (link)                     |                                    |
| Nuclear Quantum Effects in the Acetylene:Ammonia Plastic Co-crystal                                                                                               | Atul C. Thakur; Richard C. Remsing                                        | 2023 | arXiv preprint                            | 0     | (link)                     |                                    |
| The transformation mechanisms among cuboctahedra, Ino's decahedra and icosahedra structures of magic-size gold nanoclusters                                       | Ehsan Rahmatizad Khejehpasha; Mohammad Ismaeil Safa; Nasrin Eyvazi et al. | 2026 | arXiv preprint                            | 0     | (link)                     |                                    |
| Advances in modeling complex materials: The rise of neuroevolution potentials                                                                                     | Penghua Ying; Cheng Qian; Rui Zhao et al.                                 | 2025 | Chem. Phys. Rev. 6, 011310 (2025)         | 0     | 10.1063/5.0259061          |                                    |
| NEP89: Universal neuroevolution potential for inorganic and organic materials across 89 elements                                                                  | Ting Liang; Ke Xu; Eric Lindgren et al.                                   | 2025 | Nature Computational Science (2026)       | 0     | 10.1038/s43588-026-01009-6 |                                    |
| Reconstructions and Dynamics of $\beta$ -Lithium Thiophosphate Surfaces                                                                                           | Hanna Türk; Davide Tisi; Michele Ceriotti                                 | 2025 | PRX Energy 4(3), 033010 (2025)            | 0     | 10.1103/5hf9-hlj6          |                                    |
| Single-model uncertainty quantification in neural network potentials does not consistently outperform model ensembles                                             | Aik Rui Tan; Shingo Urata; Samuel Goldman et al.                          | 2023 | arXiv preprint                            | 0     | 10.1038/s41524-023-01180-8 |                                    |
| Revisit Many-body Interaction Heat Current and Thermal Conductivity Calculation in Moment Tensor Potential/LAMMPS Interface                                       | Siu Ting Tai; Chen Wang; Ruihuan Cheng et al.                             | 2024 | arXiv preprint                            | 0     | 10.1021/acs.jctc.4c01659   |                                    |
| Million-atom simulation of the set process in phase change memories at the real device scale                                                                      | Omar Abou El Kheir; Marco Bernasconi                                      | 2025 | arXiv preprint                            | 0     | (link)                     |                                    |
| Cluster expansion constructed over Jacobi-Legendre polynomials for accurate force fields                                                                          | Michelangelo Domina; Urvesh Patil; Matteo Cobelli et al.                  | 2022 | arXiv preprint                            | 0     | (link)                     |                                    |
| Nucleation mechanism for the direct graphite-to-diamond phase transition                                                                                          | Rustam Z. Khaliullin; Hargai Eshet; Thomas D. Kuhne et al.                | 2011 | arXiv preprint                            | 0     | 10.1038/nmat3078           |                                    |
| Structure and solidification of the (Fe <sub>0.75</sub> B <sub>0.15</sub> Si <sub>0.1</sub> ) <sub>100-x</sub> Tax (x=0-2) melts: experiment and machine learning | I. V. Sterkhova; L. V. Karmaeva; V. I. Ladyanov et al.                    | 2022 | arXiv preprint                            | 0     | 10.1016/j.jpcs.2022.111143 |                                    |
| An experimentally validated end-to-end framework for operando modeling of intrinsically complex metallosilicates                                                  | Jong Hyun Jung; Tom Schächtel; Yongliang Ou et al.                        | 2025 | arXiv preprint                            | 0     | (link)                     |                                    |
| Accelerated Discovery of Crystalline Materials with Record Ultralow Lattice Thermal Conductivity via a Universal Descriptor                                       | Xingchen Shen; Jiongzhi Zheng; Michael Marek Koza et al.                  | 2025 | arXiv preprint                            | 0     | 10.1038/s41467-025-67333-z |                                    |
| Cluster Models for Next-Generation, Machine-Learning-Based Energy Functions for Molecular Simulations                                                             | JingChun Wang; Meenu Upadhyay; Eric D. Boittier et al.                    | 2025 | arXiv preprint                            | 0     | (link)                     |                                    |
| Tuning the lattice thermal conductivity in van-der-Waals structures through rotational (dis)ordering                                                              | Fredrik Eriksson; Erik Fransson; Christopher Linderälv et al.             | 2023 | ACS Nano 17, 25565 (2023)                 | 0     | 10.1021/acsnano.3c09717    |                                    |
| Sub-micrometer phonon mean free paths in metal-organic frameworks revealed by machine-learning molecular dynamics simulations                                     | Penghua Ying; Ting Liang; Ke Xu et al.                                    | 2023 | ACS Appl.Mater.Interfaces, 2023,15, 36412 | 0     | 10.1021/acsami.3c07770     |                                    |

| Title                                                                                                                        | Authors                                            | Year | Venue                                 | Cited | DOI                           | OpenAlex<br>2-yr mean<br>citedness |
|------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|------|---------------------------------------|-------|-------------------------------|------------------------------------|
| Unconventional mechanical and thermal behaviors of MOF CALF-20                                                               | Dong Fan; Supriyo Naskar; Guillaume Maurin         | 2023 | Nature Communication 15, 3251 (2024)  | 0     | 10.1038/s41467-024-47695-6    |                                    |
| Heat transport in superionic materials via machine-learned molecular dynamics                                                | Wenjiang Zhou; Benrui Tang; Zheyong Fan et al.     | 2025 | arXiv preprint                        | 0     | (link)                        |                                    |
| Machine learning for predicting ultralow thermal conductivity and high ZT in complex thermoelectric materials                | Yuzhou Hao; Yuting Zuo; Jiongzhi Zheng et al.      | 2024 | arXiv preprint                        | 0     | 10.1021/acsami.4c09043        |                                    |
| Incipient ionic conductors: Ion-constrained lattices achieving superionic-like thermal conductivity by extreme anharmonicity | Yongheng Li; Chunqiu Lu; Bin Wei et al.            | 2025 | arXiv preprint                        | 0     | 10.1002/adma.202513381        |                                    |
| Understanding correlations in BaZrO <sub>3</sub> : Structure and dynamics on the nano-scale                                  | Erik Fransson; Petter Rosander; Paul Erhart et al. | 2023 | Chemistry of Materials 36, 514 (2024) | 0     | 10.1021/acs.chemmater.3c02548 |                                    |

## Block QC (439 unique)

| Title                                                                                                    | Authors                                                 | Year | Venue                                                                                     | Cited | DOI                            | OpenAlex<br>2-yr mean<br>citedness |
|----------------------------------------------------------------------------------------------------------|---------------------------------------------------------|------|-------------------------------------------------------------------------------------------|-------|--------------------------------|------------------------------------|
| Hyperuniform states of matter                                                                            | Salvatore Torquato                                      | 2018 | Physics Reports                                                                           | 485   | 10.1016/j.physrep.2018.03.001  |                                    |
| Localization and delocalization of light in photonic moiré lattices                                      | Peng Wang; Yuanlin Zheng; Xianfeng Chen et al.          | 2019 | Nature                                                                                    | 482   | 10.1038/s41586-019-1851-6      |                                    |
| Delocalization of a disordered bosonic system by repulsive interactions                                  | B. Deissler; Matteo Zaccanti; G. Roati et al.           | 2010 | Nature Physics                                                                            | 273   | 10.1038/nphys1635              |                                    |
| Topological triple phase transition in non-Hermitian Floquet quasicrystals                               | Sebastian Weidemann; Mark Kremer; Stefano Longhi et al. | 2022 | Nature                                                                                    | 225   | 10.1038/s41586-021-04253-0     |                                    |
| Disorder-Enhanced Transport in Photonic Quasicrystals                                                    | Liad Levi; Mikael C. Rechtsman; Barak Freedman et al.   | 2011 | Science                                                                                   | 205   | 10.1126/science.1202977        |                                    |
| Generalized Aubry-André self-duality and mobility edges in non-Hermitian quasiperiodic lattices          | Tong Liu; Hao Guo; Yong Pu et al.                       | 2020 | Physical review. B./Physical review. B                                                    | 148   | 10.1103/physrevb.102.024205    |                                    |
| Topological Phase Transitions and Mobility Edges in Non-Hermitian Quasicrystals                          | Quan Lin; Tianyu Li; Lei Xiao et al.                    | 2022 | Physical Review Letters                                                                   | 136   | 10.1103/physrevlett.129.113601 |                                    |
| Localization and delocalization of light in photonic moiré lattices                                      | Peng Wang; Yuanlin Zheng; Xianfeng Chen et al.          | 2020 | LA Referencia (Red Federada de Repositorios Institucionales de Publicaciones Científicas) | 121   | (link)                         |                                    |
| Bose-Einstein condensates in optical quasicrystal lattices                                               | Laurent Sanchez-Palencia; L. Santos                     | 2005 | Physical Review A                                                                         | 117   | 10.1103/physreva.72.053607     |                                    |
| Observing Localization in a 2D Quasicrystalline Optical Lattice                                          | Matteo Sbroscia; Konrad Viebahn; Edward Carter et al.   | 2020 | Physical Review Letters                                                                   | 105   | 10.1103/physrevlett.125.200604 |                                    |
| Power dependent soliton location and stability in complex photonic structures                            | Yannis Kominis; Kyriakos Hizanidis                      | 2008 | Optics Express                                                                            | 73    | 10.1364/oe.16.012124           |                                    |
| Localized modes in photonic quasicrystals with Penrose-type lattice                                      | Andrea Villa; Stéfan Enoch; G. Tayeb et al.             | 2006 | Optics Express                                                                            | 60    | 10.1364/oe.14.010021           |                                    |
| Octonacci photonic quasicrystals                                                                         | E.R. Brandão; Carlos H. Costa; M.S. Vasconcelos et al.  | 2015 | Optical Materials                                                                         | 56    | 10.1016/j.optmat.2015.04.051   |                                    |
| Observation of localization of light in linear photonic quasicrystals with diverse rotational symmetries | Peng Wang; Qidong Fu; V. V. Konotop et al.              | 2024 | Nature Photonics                                                                          | 54    | 10.1038/s41566-023-01350-6     |                                    |
| Localized photonic band edge modes and orbital angular momenta of light in a golden-angle spiral         | Seng Fatt Liew; Heeso Noh; Jacob Trevino et al.         | 2011 | Optics Express                                                                            | 47    | 10.1364/oe.19.023631           |                                    |
| Photonic quasicrystal single-cell cavity mode                                                            | Sun-Kyung Kim; Jeehye Lee; Se-Heon Kim et al.           | 2005 | Applied Physics Letters                                                                   | 45    | 10.1063/1.1852716              | 3.881 (2026)                       |
| Intrinsic photonic wave localization in a three-dimensional icosahedral quasicrystal                     | Seung-Yeol Jeon; Hyungho Kwon; Kahyun Hur               | 2017 | Nature Physics                                                                            | 43    | 10.1038/nphys4002              |                                    |

| Title                                                                                                                                                                 | Authors                                                             | Year | Venue                                         | Cited | DOI                              | OpenAlex<br>2-yr mean<br>citedness |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|------|-----------------------------------------------|-------|----------------------------------|------------------------------------|
| Mode localization and band-gap formation in defect-free photonic quasicrystals                                                                                        | Khaled Mnaymneh; Robert C. Gauthier                                 | 2007 | Optics Express                                | 42    | 10.1364/oe.15.005089             |                                    |
| Experimental Observation of Intrinsic Light Localization in Photonic Icosahedral Quasicrystals                                                                        | Artem D. Sinelnik; Ivan I. Shishkin; Xiaochang Yu et al.            | 2020 | Advanced Optical Materials                    | 39    | 10.1002/adom.202001574           | 1.574 (2026)                       |
| Observation of Topological Corner State Arrays in Photonic Quasicrystals                                                                                              | Aoqian Shi; Yiwei Peng; Jiapei Jiang et al.                         | 2024 | Laser & Photonics Review                      | 35    | 10.1002/lpor.202300956           |                                    |
| Dephasing-Induced Mobility Edges in Quasicrystals                                                                                                                     | Stefano Longhi                                                      | 2024 | Physical Review Letters                       | 34    | 10.1103/physrevlett.132.236301   |                                    |
| Enhanced Soliton Transport in Quasiperiodic Lattices with Introduced Aperiodicity                                                                                     | Andrey A. Sukhorukov                                                | 2006 | Physical Review Letters                       | 33    | 10.1103/physrevlett.96.113902    |                                    |
| Exact mobility edges and topological phase transition in two-dimensional non-Hermitian quasicrystals                                                                  | Zhihao Xu; X. C. Xia; Shu Chen                                      | 2021 | Science China Physics Mechanics and Astronomy | 31    | 10.1007/s11433-021-1802-4        |                                    |
| Quasiperiodic photonic crystal fiber [Invited]                                                                                                                        | Exian Liu; Jianjun Liu                                              | 2023 | Chinese Optics Letters                        | 30    | 10.3788/col202321.040203         | 0.40203 (2026)                     |
| Luminescence properties of a Fibonacci photonic quasicrystal                                                                                                          | V. Passias; N. V. Valappil; Z. Shi et al.                           | 2009 | Optics Express                                | 28    | 10.1364/oe.17.006636             |                                    |
| Non-Hermitian Delocalization in a Two-Dimensional Photonic Quasicrystal                                                                                               | Zhaoyang Zhang; Shun Liang; I. Septembre et al.                     | 2024 | Physical Review Letters                       | 27    | 10.1103/physrevlett.132.263801   |                                    |
| Simultaneous large band gaps and localization of electromagnetic and elastic waves in defect-free quasicrystals                                                       | Tianbao Yu; Zhong Wang; Wenxing Liu et al.                          | 2016 | Optics Express                                | 27    | 10.1364/oe.24.007951             |                                    |
| Parity-Induced Thermalization Gap in Disordered Ring Lattices                                                                                                         | Yao Wang; Jun Gao; Xiaoling Pang et al.                             | 2019 | Physical Review Letters                       | 26    | 10.1103/physrevlett.122.013903   |                                    |
| Photonic band gaps and localization in two-dimensional metallic quasicrystals                                                                                         | Mehmet Bayındır; Ertugrul Cubukcu; İrfan Bulu et al.                | 2001 | Europhysics Letters (EPL)                     | 25    | 10.1209/epl/i2001-00485-9        | 2.066 (2026)                       |
| Dielectric refractive index dependence of the focusing properties of a dielectric-cylinder-type decagonal photonic quasicrystal flat lens and its photon localization | Jianjun Liu; Exian Liu; Zhigang Fan et al.                          | 2015 | Applied Physics Express                       | 23    | 10.7567/apex.8.112003            | 0.112003 (2026)                    |
| Defect-free localized modes and coupled-resonator acoustic waveguides constructed in two-dimensional phononic quasicrystals                                           | Mingda Zhang; Wei Zhong; Xiangdong Zhang                            | 2012 | Journal of Applied Physics                    | 22    | 10.1063/1.4721372                |                                    |
| Reentrant delocalization transition in one-dimensional photonic quasicrystals                                                                                         | Sachin Vaidya; Christina Jörg; Kyle Linn et al.                     | 2023 | Physical Review Research                      | 21    | 10.1103/physrevresearch.5.033170 |                                    |
| Light Localization in Local Isomorphism Classes of Quasicrystals                                                                                                      | Chaney Lin; Paul J. Steinhardt; Salvatore Torquato                  | 2018 | Physical Review Letters                       | 20    | 10.1103/physrevlett.120.247401   |                                    |
| Effects of graphene on light transmission spectra in Dodecanacci photonic quasicrystals                                                                               | E.F. Silva; M.S. Vasconcelos; Carlos H. Costa et al.                | 2019 | Optical Materials                             | 20    | 10.1016/j.optmat.2019.109450     |                                    |
| Simultaneous localization of photons and phonons within the transparency bands of LiNbO <sub>3</sub> photonic quasicrystals                                           | Zhong Wang; Wenxing Liu; Tianbao Yu et al.                          | 2016 | Optics Express                                | 20    | 10.1364/oe.24.023353             |                                    |
| Spin waves in planar quasicrystal of Penrose tiling                                                                                                                   | Justyna Gruszecka; Michał Mieszcak; W. Kłos; Rychły-Szymon Jarosław | 2017 | Journal of Magnetism and Magnetic Materials   | 20    | 10.1016/j.jmmm.2017.03.029       |                                    |
| Non-Hermitian topological mobility edges and transport in photonic quantum walks                                                                                      | Stefano Longhi                                                      | 2022 | Optics Letters                                | 18    | 10.1364/ol.460484                |                                    |
| Exciton-polaritonic quasicrystalline and aperiodic structures                                                                                                         | Alexander N. Poddubny; Laura Piloizzi; M. M. Voronov et al.         | 2009 | Physical Review B                             | 18    | 10.1103/physrevb.80.115314       |                                    |
| Light Localisation and Lasing: Random and Quasi-random Photonic Structures                                                                                            | Mher Ghulinyan; Lorenzo Pavesi                                      | 2014 | Medical Entomology and Zoology                | 17    | (link)                           |                                    |



| Title                                                                                                                                     | Authors                                                       | Year | Venue                                                                                | Cited | DOI                                      | OpenAlex<br>2-yr mean<br>citedness |
|-------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|------|--------------------------------------------------------------------------------------|-------|------------------------------------------|------------------------------------|
| Anderson localization and Brewster anomalies in photonic disordered quasiperiodic lattices                                                | E. Reyes-Gómez; Alexys Bruno-Alfonso; S. B. Cavalcanti et al. | 2011 | Physical Review E                                                                    | 15    | 10.1103/physreve.84.036604               |                                    |
| Bose-Einstein condensates in quasiperiodic lattices: Bosonic Josephson junction, self-trapping, and multimode dynamics                    | Henrique C. Prates; Dmitry A. Zezyulin; V. V. Konotop         | 2022 | Physical Review Research                                                             | 15    | 10.1103/physrevresearch.4.033219         |                                    |
| Fundamental and vortex gap solitons in quasiperiodic photonic lattices                                                                    | Changming Huang; Liangwei Dong; Hanying Deng et al.           | 2021 | Optics Letters                                                                       | 15    | 10.1364/ol.443051                        |                                    |
| Stationary and traveling solitons via local dissipation in Bose-Einstein condensates in ring optical lattices                             | Russell Campbell; Gian-Luca Oppo                              | 2016 | Physical Review A                                                                    | 14    | 10.1103/physreva.94.043626               |                                    |
| A study of optical reflectance and localization modes of 1-D Fibonacci photonic quasicrystals using different graded dielectric materials | Bipin K. Singh; Praveen C. Pandey                             | 2014 | Journal of Modern Optics                                                             | 13    | 10.1080/09500340.2014.914587             |                                    |
| Localized modes in defect-free two-dimensional circular photonic crystals                                                                 | Wei Zhong; Xiangdong Zhang                                    | 2010 | Physical Review A                                                                    | 13    | 10.1103/physreva.81.013805               |                                    |
| Study of Optical Propagation in Hybrid Periodic/Quasiregular Structures Based on Porous Silicon                                           | Jose Escorcia-García; M.E. Mora-Ramos                         | 2009 | PIERS Online                                                                         | 13    | 10.2529/piers080906010703                |                                    |
| Acousto-optic interactions for terahertz waves using phoxonic quasicrystals                                                               | Zhong Wang; Tianbao Yu; Tongbiao Wang et al.                  | 2018 | Journal of Physics D Applied Physics                                                 | 12    | 10.1088/1361-6463/aaa98c                 |                                    |
| Resonant transmission of electromagnetic waves through two-dimensional photonic quasicrystals                                             | Yair Neve-Oz; Therese Pollok; Sven Burger et al.              | 2010 | Journal of Applied Physics                                                           | 10    | 10.1063/1.3329542                        |                                    |
| Pseudospectral Renormalization Method for Solitons in Quasicrystal Lattice with the Cubic-Quintic Nonlinearity                            | Nalan Antar                                                   | 2014 | Journal of Applied Mathematics                                                       | 10    | 10.1155/2014/848153                      |                                    |
| Robust Anderson transition in non-Hermitian photonic quasicrystals                                                                        | Stefano Longhi                                                | 2024 | Optics Letters                                                                       | 9     | 10.1364/ol.517182                        |                                    |
| The square Thue–Morse tiling for photonic application                                                                                     | Luigi Moretti; Vito Moccia                                    | 2008 | The Philosophical Magazine A Journal of Theoretical Experimental and Applied Physics | 9     | 10.1080/14786430802047076                |                                    |
| Topological wave phenomena in photonic time quasicrystals                                                                                 | Xiang Ni; Shixiong Yin; Huanan Li et al.                      | 2025 | Physical review. B/Physical review. B                                                | 8     | 10.1103/physrevb.111.125421              |                                    |
| Tunable defect modes through the (YBCO-Yttria) based on Octonacci photonic quasicrystals                                                  | Youssef Trabelsi; Francis Segovia-Chaves; Naim Ben Ali        | 2022 | Results in Physics                                                                   | 8     | 10.1016/j.rinp.2022.106176               |                                    |
| Spectral properties of two coupled Fibonacci chains                                                                                       | Anouar Moustaj; Malte Röntgen; Christian V. Morfonios et al.  | 2023 | New Journal of Physics                                                               | 8     | 10.1088/1367-2630/acf0e0                 |                                    |
| Interface states in two-dimensional quasicrystals with broken inversion symmetry                                                          | Danilo Beli; Matheus I. N. Rosa; Luca Lomazzi et al.          | 2025 | Physical Review Applied                                                              | 7     | 10.1103/physrevapplied.23.024039         |                                    |
| Radiative heat transfer mediated by topological phonon polaritons in a family of quasiperiodic nanoparticle chains                        | Boxiang Wang; Changying Zhao                                  | 2023 | International Journal of Heat and Mass Transfer                                      | 7     | 10.1016/j.ijheatmasstransfer.2023.124163 |                                    |
| Diffraction and pseudospectra in non-Hermitian quasiperiodic lattices                                                                     | Ananya Ghatak; Dimitrios Kaltsas; Manas Kulkarni et al.       | 2024 | Physical review. E                                                                   | 6     | 10.1103/physreve.110.064228              |                                    |
| Enhanced Light Scattering Using a Two-Dimensional Quasicrystal-Decorated 3D-Printed Nature-Inspired Bio-photonic Architecture             | Partha Kumbhakar; Ashim Pramanik; Shashank Mishra et al.      | 2023 | The Journal of Physical Chemistry C                                                  | 6     | 10.1021/acs.jpcc.3c00513                 |                                    |
| Stability of quasicrystalline ultracold fermions to dipolar interactions                                                                  | Paolo Mollignini                                              | 2025 | Physical Review Research                                                             | 6     | 10.1103/szdc-61nl                        |                                    |
| One-dimensional photonic quasicrystals                                                                                                    | Mher Ghulinyan                                                | 2014 | Cambridge University Press eBooks                                                    | 5     | 10.1017/cbo9781139835002.006             |                                    |
| Observation of Topological Corner State Arrays in Photonic Quasicrystals                                                                  | Aoqian Shi; Yiwei Peng; Jiawei Jiang et al.                   | 2022 | arXiv (Cornell University)                                                           | 5     | 10.48550/arxiv.2209.05751                |                                    |

| Title                                                                                                                            | Authors                                                         | Year | Venue                                                                                | Cited | DOI                            | OpenAlex<br>2-yr mean<br>citedness |
|----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|------|--------------------------------------------------------------------------------------|-------|--------------------------------|------------------------------------|
| Observation of Light Localization at the Edges of Quasicrystal Waveguide Arrays                                                  | Sergey K. Ivanov; S. A. Zhuravitskii; N. S. Kostyuchenko et al. | 2025 | Physical Review Letters                                                              | 4     | 10.1103/physrevlett.134.113803 |                                    |
| Optical properties of 1D photonic time quasicrystals                                                                             | Anny Araújo; Carlos Costa; S. Azevedo et al.                    | 2025 | Journal of the Optical Society of America B                                          | 4     | 10.1364/josab.542001           |                                    |
| Symmetry-Induced Light Confinement in a Photonic Quasicrystal-Based Mirrorless Cavity                                            | Gianluigi Zito; Giulia Rusciano; Antonio Sasso et al.           | 2016 | Crystals                                                                             | 4     | 10.3390/cryst6090113           | 13.465 (2026)                      |
| Resonant modes of 12-fold symmetric defect free photonic quasicrystal                                                            | Minfeng Chen; Yunjing Li; Yuh-Jen Cheng et al.                  | 2014 | Optics Express                                                                       | 4     | 10.1364/oe.22.002007           |                                    |
| Tunability in the terahertz range of a one-dimensional photonic quasicrystal containing an InSb semiconductor                    | Francis Segovia-Chaves; Hussein A. Elsayed; Ahmed Mehaney       | 2021 | Optik                                                                                | 4     | 10.1016/j.ijleo.2021.167675    |                                    |
| Diffraction propagation of light wave and image distortionless transmission in ten-fold symmetric photonic quasicrystal lattices | Wentao Jin; Meng Song; Wen-Cheng Hu et al.                      | 2021 | Optics Communications                                                                | 4     | 10.1016/j.optcom.2021.127329   |                                    |
| Properties of defect modes in two-dimensional organic octagonal quasicrystals at low dielectric contrast                         | Yuanyuan Cai; Zhi Wang; Xiao Chen et al.                        | 2018 | Optics Communications                                                                | 4     | 10.1016/j.optcom.2018.06.010   |                                    |
| Isotropic photonic bandgap in Penrose textured metallic microcavity                                                              | J. Y. Zhang; Hoi Lam Tam; W.H. Wong et al.                      | 2006 | Solid State Communications                                                           | 4     | 10.1016/j.ssc.2006.02.035      |                                    |
| Anomalous optical Anderson localization in mixed one dimensional photonic quasicrystals                                          | Cunding Liu; Mingdong Kong; Bincheng Li                         | 2015 | Optics Express                                                                       | 3     | 10.1364/oe.23.0a1297           |                                    |
| Interaction of fast electron beam with photonic quasicrystals                                                                    | Wei Zhong; Jinying Xu; Xiangdong Zhang                          | 2009 | Optics Express                                                                       | 3     | 10.1364/oe.17.013270           |                                    |
| Programmable 2D photonic quasicrystals with real-time reconfigurability                                                          | Peisheng Sun; Xuesong Hu; Ruonan Li et al.                      | 2025 | Optics Letters                                                                       | 3     | 10.1364/ol.568662              |                                    |
| Shear based gap control in 2D photonic quasicrystals of dielectric cylinders                                                     | Ángel Andueza; Joaquín Sevilla; Jesús Pérez-Conde et al.        | 2021 | Optics Express                                                                       | 3     | 10.1364/oe.427235              |                                    |
| Properties of microcavities in two-dimensional photonic quasicrystals with octagonal rotational symmetry                         | D. M. Beggs; M. A. Kalitchevski; R. A. Abram                    | 2007 | Journal of Modern Optics                                                             | 3     | 10.1080/09500340601102441      |                                    |
| Simultaneous localization of photons and phonons in defect-free dodecagonal photonic quasicrystals                               | Bihang Xu; Zhong Wang; Yixiang Tan et al.                       | 2018 | Modern Physics Letters B                                                             | 3     | 10.1142/s0217984918500963      |                                    |
| Interface states in two-dimensional quasicrystals with broken inversion symmetry                                                 | Danilo Beli; Matheus I. N. Rosa; Luca Lomazzi et al.            | 2023 | arXiv (Cornell University)                                                           | 3     | 10.48550/arxiv.2304.09628      |                                    |
| Observation of Topological Corner State Arrays in Photonic Quasicrystals (Laser Photonics Rev. 18(7)/2024)                       | Aoqian Shi; Yiwei Peng; Jiawei Jiang et al.                     | 2024 | Laser & Photonics Review                                                             | 2     | 10.1002/lpor.202470041         |                                    |
| Negative refraction and localized states of a classical wave in high-symmetry quasicrystals                                      | Xiangdong Zhang; Wei Zhong; Zhifang Feng et al.                 | 2010 | The Philosophical Magazine A Journal of Theoretical Experimental and Applied Physics | 2     | 10.1080/14786435.2010.512576   |                                    |
| Light wave propagation and diffraction inhibition in three-dimensional photonic quasicrystal lattices                            | Meng Song; Jin Wentao; Shaochun Fu et al.                       | 2023 | Results in Physics                                                                   | 2     | 10.1016/j.rinp.2023.106884     |                                    |
| Probing Structural Correlated Disorder with Photonic Transport                                                                   | Ying-Yue Yang; Yi-Jun Chang; Yong-Heng Lu et al.                | 2024 | ACS Photonics                                                                        | 2     | 10.1021/acsphotonics.3c00132   | 5.360 (2026)                       |
| Topological states in Penrose-square photonic crystals                                                                           | Qichen Zhang; Jianzhi Chen; Dongyang Liu et al.                 | 2024 | Journal of the Optical Society of America A                                          | 2     | 10.1364/josaa.520606           |                                    |
| About optical localization in photonic quasicrystals                                                                             | Faris Mohammed; Alexander Quandt                                | 2016 | Optical and Quantum Electronics                                                      | 1     | 10.1007/s11082-016-0648-1      |                                    |
| Three- and four-material photonic quasicrystal                                                                                   | Chittaranjan Nayak; Jitendra K. Behera; Rajesh Moharana et al.  | 2025 | Optical Engineering                                                                  | 1     | 10.1117/1.oe.64.7.077101       |                                    |
| Mode localization in defect-free GaN-based bottom-up photonic quasicrystal lasers                                                | Tzeng-Tsong Wu; Chih-Cheng Chen; Hao Chen et al.                | 2012 |                                                                                      | 1     | 10.1109/islc.2012.6348389      |                                    |
| Fast Light and Focusing in 2D Photonic Quasicrystals                                                                             | Yair Neve-Oz; Therese Pollok; Sven Burger et al.                | 2009 | PIERS Online                                                                         | 1     | 10.2529/piers081223080021      |                                    |

| Title                                                                                                   | Authors                                                      | Year | Venue                                                                                                                            | Cited | DOI                            | OpenAlex<br>2-yr mean<br>citedness |
|---------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|------|----------------------------------------------------------------------------------------------------------------------------------|-------|--------------------------------|------------------------------------|
| Emission enhancement in hybrid Tamm plasmon/photonic quasicrystal structure                             | Konstantin M. Morozov; . . . ; Aleksei V. Belonovskii et al. | 2019 | SN Applied Sciences                                                                                                              | 1     | 10.1007/s42452-019-1375-6      |                                    |
| Formation of nonlinear modes in one-dimensional quasiperiodic lattices with a mobility edge             | Dmitry A. Zezyulin; G. L. Alfimov                            | 2024 | Physical Review A                                                                                                                | 1     | 10.1103/physreva.110.063304    |                                    |
| Coupling photonic quasicrystals with a one-dimensional critical high-temperature superconducting cavity | Francis Segovia-Chaves; Youssef Trabelsi; T.A. Taha          | 2023 | Optik                                                                                                                            | 1     | 10.1016/j.ijleo.2023.170847    |                                    |
| Novel hybrid organic/inorganic 2D photonic quasicrystals with 8-fold and 12-fold diffraction symmetries | Lucia Petti; Massimo Rippa; Rossella Capasso et al.          | 2012 | Proceedings of SPIE, the International Society for Optical Engineering/Proceedings of SPIE                                       | 1     | 10.1117/12.923183              |                                    |
| Quantum Diffusion in a Photonic Fibonacci Chain: From Localization to Ballistic Dynamics                | Jiankun Zhu; Yao Qin; Yuxiang Guo et al.                     | 2026 | Physical Review Letters                                                                                                          | 1     | 10.1103/tjm4-gjx8              |                                    |
| Vortex solitons in disclination quasicrystals                                                           | Hua Zhong; Yaroslav Kartashov; Yongdong Li et al.            | 2026 | Reports on Progress in Physics                                                                                                   | 1     | 10.1088/1361-6633/ae73b5       |                                    |
| A tribute to Sydney G. Davison                                                                          | Aart W. Kleyn                                                | 2003 | Progress in Surface Science                                                                                                      | 1     | 10.1016/j.progsurf.2003.09.003 |                                    |
| Stability of quasicrystalline ultracold fermions to dipolar interactions                                | Paolo Molignini                                              | 2024 | arXiv (Cornell University)                                                                                                       | 1     | 10.48550/arxiv.2403.04830      |                                    |
| Engineering Aperiodic Spiral Order in Nanophotonics: Fundamentals and Device Applications               | Luca Dal Negro; Nate Lawrence; Jacob Trevino                 | 2015 |                                                                                                                                  | 1     | 10.1201/b19365-5               |                                    |
| Colloquium: Synthetic quantum matter in non-standard geometries                                         | Tobias Graß; Dario Bercioux; Utso Bhattacharya et al.        | 2024 | arXiv (Cornell University)                                                                                                       | 1     | 10.48550/arxiv.2407.06105      |                                    |
| Localized resonant states and transmission in a two-dimensional photonic quasicrystal                   | Yair Neve-Oz; Therese Pollok; Sven Burger et al.             | 2011 | International Conference on Applied Mathematics                                                                                  | 0     | (link)                         |                                    |
| Photonic band gaps and localization in two-dimensional metallic quasicrystals                           | Enrique Maciá                                                | 2016 |                                                                                                                                  | 0     | (link)                         |                                    |
| Multiple -wavelength Fabry-Perot filters with localized modes in symmetric photonics quasicrystals      | Liu Cunding; Mingdong Kong; Bincheng Li                      | 2016 |                                                                                                                                  | 0     | 10.1364/oic.2016.fa.9          |                                    |
| Finite-difference methods used to model photonic wave localization in 3D quasicrystals                  | Aditi Risbud                                                 | 2017 | MRS Bulletin                                                                                                                     | 0     | 10.1557/mrs.2017.39            |                                    |
| Diffraction and pseudospectra in non-Hermitian quasiperiodic lattices                                   | Ananya Ghatak; Dimitrios Kaltsas; Manas Kulkarni et al.      | 2024 | arXiv (Cornell University)                                                                                                       | 0     | 10.48550/arxiv.2410.09185      |                                    |
| Reentrant delocalization transition in one-dimensional photonic quasicrystals                           | Sachin Vaidya; Christina Jörg; Kyle Linn et al.              | 2022 | arXiv (Cornell University)                                                                                                       | 0     | 10.48550/arxiv.2211.06047      |                                    |
| Appearance of Mobility Edge in Self-Dual Quasiperiodic Lattices                                         | Gang Wang; Nianbei Li; Tsuneyoshi Nakayama                   | 2013 | arXiv (Cornell University)                                                                                                       | 0     | 10.48550/arxiv.1312.0844       |                                    |
| Non-Hermitian delocalization in a 2D photonic quasicrystal                                              | Zhaoyang Zhang; Shun Liang; I. Septembre et al.              | 2023 | arXiv (Cornell University)                                                                                                       | 0     | 10.48550/arxiv.2312.09322      |                                    |
| Vortex and corner solitons in Stampfli-tiling dodecagonal quasiperiodic lattices                        | Boquan Ren; Yongli Qu; Milivoj R. Belić et al.               | 2025 | Chaos Solitons & Fractals                                                                                                        | 0     | 10.1016/j.chaos.2025.115243    |                                    |
| Localization in photonic crystals                                                                       | Faris Siedahmended Mohammed Osman                            | 2017 | University of the Witwatersrand, Johannesburg Institutional Repository on DSpace (University of the Witwatersrand, Johannesburg) | 0     | (link)                         |                                    |
| Optical properties of 1D-photonic time quasicrystals                                                    | Claudionor Bezerra; Anny Araújo; S. Azevedo et al.           | 2024 |                                                                                                                                  | 0     | 10.1364/opticaopen.26973478    |                                    |
| Optical properties of 1D-photonic time quasicrystals                                                    | Claudionor Bezerra; Anny Araújo; S. Azevedo et al.           | 2024 |                                                                                                                                  | 0     | 10.1364/opticaopen.26973478.v1 |                                    |
| Reentrant delocalization transition in one-dimensional photonic quasicrystals                           | Kyle Linn; Sachin Vaidya; Christina Jörg et al.              | 2023 |                                                                                                                                  | 0     | 10.1364/cleo_fs.2023.ftu4d.8   |                                    |

| Title                                                                                                                             | Authors                                                     | Year | Venue                                                                                            | Cited | DOI                            | OpenAlex<br>2-yr mean<br>citedness |
|-----------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|------|--------------------------------------------------------------------------------------------------|-------|--------------------------------|------------------------------------|
| Robust Anderson transition in non-Hermitian photonic quasicrystals                                                                | Stefano Longhi                                              | 2024 |                                                                                                  | 0     | 10.1364/opticaopen.25134767    |                                    |
| Robust Anderson transition in non-Hermitian photonic quasicrystals                                                                | Stefano Longhi                                              | 2024 |                                                                                                  | 0     | 10.1364/opticaopen.25134767.v1 |                                    |
| Photonic Crystal/Quasicrystal Nanolasers and Spontaneous Emission Control                                                         | Kengo Nozaki; Toshihiko Baba                                | 2006 | The Review of Laser Engineering                                                                  | 0     | 10.2184/ljsj.34.756            |                                    |
| Formation of nonlinear modes in one-dimensional quasiperiodic lattices with a mobility edge                                       | Dmitry A. Zezyulin; G. L. Alfimov                           | 2024 | arXiv (Cornell University)                                                                       | 0     | 10.48550/arxiv.2411.13936      |                                    |
| Linear light localization in two-dimensional fractional diffraction quasicrystals                                                 | Zeyun Shi; Lu Qin; Pengfei Li et al.                        | 2025 | Optics Letters                                                                                   | 0     | 10.1364/ol.566565              |                                    |
| Analytical mobility edge in nonreciprocal quasiperiodic lattices with next-nearest-neighbor hopping                               | Wenmin Wang; Xiaosen Yang; Xianqi Tong                      | 2026 | arXiv (Cornell University)                                                                       | 0     | 10.48550/arxiv.2607.10996      |                                    |
| Spectral Distribution of one-dimensional Photonic Quasicrystals: The Role of Irrational Numbers                                   | Hui Quan; Wei Si; Kai Jiang                                 | 2026 | arXiv (Cornell University)                                                                       | 0     | (link)                         |                                    |
| Super-ballistic diffusion induced by nonlinear interactions in a one-dimensional quasiperiodic lattice                            | J. K. Dong; Long-Hua Gu; Luchen Zhang et al.                | 2024 | Physics Letters A                                                                                | 0     | 10.1016/j.physleta.2024.129549 |                                    |
| Analytical mobility edge in nonreciprocal quasiperiodic lattices with next-nearest-neighbor hopping                               | Wenmin Wang; Xiaosen Yang; Xianqi Tong                      | 2026 | arXiv (Cornell University)                                                                       | 0     | (link)                         |                                    |
| Spectral Distribution of one-dimensional Photonic Quasicrystals: The Role of Irrational Numbers                                   | Hui Quan; Wei Si; Kai Jiang                                 | 2026 | arXiv (Cornell University)                                                                       | 0     | 10.48550/arxiv.2601.06482      |                                    |
| Time-Domain Interferometric Measurement of Photonic Band Structure in a One-Dimensional Quasicrystal                              | Toshiaki Hattori; Noriaki Tsurumachi; Sakae Kawato et al.   | 1994 | Ultrafast Phenomena                                                                              | 0     | 10.1364/up.1994.md.14          |                                    |
| Families of localized modes of Bose-Einstein condensates enabled by incommensurate optical lattice and photon-atom interactions   | Pedro S. Gil; V. V. Kono-top                                | 2026 | Physical Review Research                                                                         | 0     | 10.1103/5hcq-tfjz              |                                    |
| Families of localized modes of Bose-Einstein condensates enabled by incommensurate optical lattice and photon-atom interactions   | Pedro S. Gil; V. V. Kono-top                                | 2026 | arXiv (Cornell University)                                                                       | 0     | 10.48550/arxiv.2602.17780      |                                    |
| Micro-cylinder mode in photonic quasicrystal observed by near-field optical microscopy                                            | Zheyu Fang; Tao Dai; Bei Zhang et al.                       | 2007 | Proceedings of SPIE, the International Society for Optical Engineering/Proceedings of SPIE       | 0     | 10.1117/12.767590              |                                    |
| Optical QuasiCrystals - Properties and Dynamics                                                                                   | Demetrios N. Christodoulides; Jason W. Fleischer            | 2005 |                                                                                                  | 0     | (link)                         |                                    |
| Photonic band gap effect and localization in two-dimensional Penrose lattice                                                      | Mehmet Bayındır; Ertugrul Cubukcu; İrfan Bulu et al.        | 2002 |                                                                                                  | 0     | 10.1109/qels.2001.961940       |                                    |
| Photonic band gap effect and localization in two-dimensional Penrose lattice                                                      | Mehmet Bayındır; Ertugrul Cubukcu; İrfan Bulu et al.        | 2001 |                                                                                                  | 0     | 10.1109/cleo.2001.948207       |                                    |
| Critical states and anomalous wave transport in an aperiodic polariton monotile                                                   | Valtýr Kári Daníelsson; Helgi Sigurðsson                    | 2026 | arXiv (Cornell University)                                                                       | 0     | 10.48550/arxiv.2605.29023      |                                    |
| Critical states and anomalous wave transport in an aperiodic polariton monotile                                                   | Valtýr Kári Daníelsson; Helgi Sigurðsson                    | 2026 | arXiv (Cornell University)                                                                       | 0     | (link)                         |                                    |
| Dephasing-induced mobility edges in quasicrystals                                                                                 | Stefano Longhi                                              | 2024 | arXiv (Cornell University)                                                                       | 0     | 10.48550/arxiv.2405.07198      |                                    |
| Study of Localization-Delocalization Transition of Light in Photonic Moiré Lattices Fabricated with Saturable Nonlinear Materials | Trong Dat Ngo; Chinh Cuong Duong; Luong Thien Nguyen et al. | 2024 | Diffusion and defect data, solid state data. Part B, Solid state phenomena/Solid state phenomena | 0     | 10.4028/p-ttxk2p               | 0.68 (2026)                        |

| Title                                                                                                                                                                                                                                                                                  | Authors                                                                 | Year | Venue                                                                            | Cited              | DOI                              | OpenAlex<br>2-yr mean<br>citedness |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|------|----------------------------------------------------------------------------------|--------------------|----------------------------------|------------------------------------|
| Non-diffusion transport in decoherent non-Hermitian quasicrystals                                                                                                                                                                                                                      | Yudong Ren; Rui Zhao; Kangpeng Ye et al.                                | 2025 | arXiv (Cornell University)                                                       | 0                  | 10.48550/arxiv.2505.04205        |                                    |
| Simulating non-Hermitian quasicrystals with single-photon quantum walks                                                                                                                                                                                                                | Quan Lin; Tianyu Li; Lei Xiao et al.                                    | 2021 | arXiv (Cornell University)                                                       | 0                  | 10.48550/arxiv.2112.15024        |                                    |
| Floquet Harper-Hofstadter Butterflies and non-Hermitian phase transition in quasicrystals                                                                                                                                                                                              | Mark Kremer; Sebastian Weidemann; Stefano Longhi et al.                 | 2021 | Conference on Lasers and Electro-Optics                                          | 0                  | 10.1364/cleo_qels.2021.2620n.2   |                                    |
| Advances in classical and quantum wave dynamics on quasiperiodic lattices : a dissertation submitted for the degree of Doctor of Philosophy in Physics, Centre for Theoretical Chemistry and Physics, New Zealand Institute for Advanced Study, Massey University, Albany, New Zealand | Carlo Danieli                                                           | 2016 | Massey Research Online (Massey University)                                       | 0                  | (link)                           |                                    |
| Exploring Light-Induced Phases of 2D Materials in a Modulated 1D Quasicrystal                                                                                                                                                                                                          | Yifei Bai; Anna R. Dardia; Toshihiko Shimasaki et al.                   | 2026 | Physical Review X                                                                | 0                  | 10.1103/37vk-qz71                |                                    |
| Exploring light-induced phases of 2D materials in a modulated 1D quasicrystal                                                                                                                                                                                                          | Yifei Bai; Anna R. Dardia; Toshihiko Shimasaki et al.                   | 2025 | arXiv (Cornell University)                                                       | 0                  | 10.48550/arxiv.2506.11984        |                                    |
| Multicomponent Photonic Crystals with Inhomogeneous Scatterers                                                                                                                                                                                                                         | Alexander B. Khanikaev; Mikhail V. Rybin; M. F. Limonov                 | 2012 | Series in optics and optoelectronics                                             | 0                  | 10.1201/b12175-11                |                                    |
| Time-resolved light propoagation at the band-edge states of 1D Fibonacci quasicrystals                                                                                                                                                                                                 | Luca Dal Negro; Claudio J. Otón; Z. Gaburro et al.                      | 2002 | MRS Proceedings                                                                  | 0                  | 10.1557/proc-722-16.9            |                                    |
| Exploring Aperiodic Order in Photonic Time Crystals                                                                                                                                                                                                                                    | Marino Coppolaro; Massimo Moccia; Giuseppe Castaldi et al.              | 2025 | arXiv (Cornell University)                                                       | 0                  | 10.48550/arxiv.2505.20039        |                                    |
| Strong light confinement of tunable resonances in low symmetric quasicrystal through orientational variations                                                                                                                                                                          | Döne Yilmaz; Mediha Tugun; Hamza Kurt                                   | 2018 |                                                                                  | 0                  | 10.1117/12.2307331               |                                    |
| Topological States in 1D Photonic Quasi-crystals                                                                                                                                                                                                                                       | M.S. Vasconcelos; E.L. Albuquerque                                      | 2014 | MRS Proceedings                                                                  | 0                  | 10.1557/opl.2014.839             |                                    |
| Exploration of localization physics with atomic, molecular, and optical platforms                                                                                                                                                                                                      | Jun Gao; Hang Li; Val Zwiller et al.                                    | 2026 | Journal of Physics Condensed Matter                                              | 0                  | 10.1088/1361-648x/ae6aee         |                                    |
| Photonic Band Structure of Fibonacci Superlattices with Metamaterials. Part I: The Algebraic Method of Calculation for Lossless Layers                                                                                                                                                 | Karol Tarnowski; W. Salejda                                             | 2009 | Photonics Letters of Poland                                                      | 0                  | 10.4302/plp.2009.3.09            |                                    |
| Quasicrystalline Structures for Electromagnetic Wave Manipulation                                                                                                                                                                                                                      | Mikhail V. Rybin                                                        | 2024 |                                                                                  | 0                  | 10.1109/piers62282.2024.10618664 |                                    |
| Boundary modes in quasiperiodic elastic structures                                                                                                                                                                                                                                     | Matheus I. N. Rosa; Raj Kumar Pal; José Roberto de França Arruda et al. | 2018 | Chinese Optics Letters                                                           | 0                  | 10.3788/col/2023/21/054 (2026)   |                                    |
| Spectral Properties of Two Coupled Fibonacci Chains                                                                                                                                                                                                                                    | Anouar Moustaj; Malte Röntgen; Christian V. Morfonios et al.            | 2022 | arXiv (Cornell University)                                                       | 0                  | 10.48550/arxiv.2208.05178        |                                    |
| Aubry-André-Harper momentum-state chain in curved spacetime                                                                                                                                                                                                                            | Yiyi MAO; Hanning DAI                                                   | 2024 | Acta Physica Sinica                                                              | 0                  | 10.7498/aps.74.202405(2026)      |                                    |
| Efficient local classification of parity-based material topology                                                                                                                                                                                                                       | Stephan Wong; Ichitaro Yamazaki; Chris Siefert et al.                   | 2026 | arXiv (Cornell University)                                                       | 0                  | 10.48550/arxiv.2601.13598        |                                    |
| Efficient local classification of parity-based material topology                                                                                                                                                                                                                       | Stephan Wong; Ichitaro Yamazaki; Chris Siefert et al.                   | 2026 | arXiv (Cornell University)                                                       | 0                  | (link)                           |                                    |
| Colloidal monolayers on quasiperiodic laser fields                                                                                                                                                                                                                                     | Jules Mikhael                                                           | 2010 | OPUS Publication Server of the University of Stuttgart (University of Stuttgart) | 0                  | 10.18419/opus-4924               |                                    |
| Topological critical states and anomalous electronic transmittance in one dimensional quasicrystals                                                                                                                                                                                    | Junmo Jeon; SungBin Lee                                                 | 2020 | Phys. Rev. Research                                                              | 3, 013168 (2021)   | 10.1103/PhysRevResearch.3.013168 |                                    |
| Non-Fermi-Liquid Behavior in Metallic Quasicrystals with Local Magnetic Moments                                                                                                                                                                                                        | Eric C. Andrade; Anuradha Jagannathan; Eduardo Miranda et al.           | 2014 | Phys. Rev. Lett.                                                                 | 115, 036403 (2015) | 10.1103/PhysRevLett.115.036403   |                                    |

| Title                                                                                                                                  | Authors                                                    | Year | Venue                                                                                                                                               | Cited | DOI                              | OpenAlex<br>2-yr mean<br>citedness |
|----------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------|-------|----------------------------------|------------------------------------|
| Emergent Localization in Dodecagonal Bilayer Quasicrystals                                                                             | Moon Jip Park; Hee Seung Kim; SungBin Lee                  | 2018 | Phys. Rev. B 99, 245401 (2019)                                                                                                                      | 0     | 10.1103/PhysRevB.99.245401       |                                    |
| Quantum critical state in a magnetic quasicrystal                                                                                      | Kazuhiko Deguchi; Shuya Matsukawa; Noriaki K. Sato et al.  | 2012 | Nature Materials (Advance Online Publication 7 October 2012)                                                                                        | 0     | 10.1038/Nmat3432                 |                                    |
| Quantum-geometric origin of superfluid weight in quasicrystals with critical states                                                    | Kazuma Saito; Ryo Okugawa; Yusuke Kato et al.              | 2026 | arXiv preprint                                                                                                                                      | 0     | (link)                           |                                    |
| Quantum metric and localization in a quasicrystal                                                                                      | Quentin Marsal; Patric Holmval; Annica M. Black-Schaffer   | 2025 | arXiv preprint                                                                                                                                      | 0     | (link)                           |                                    |
| Critical analysis of the re-entrant localization transition in a one-dimensional dimerized quasiperiodic lattice                       | Shilpi Roy; Sourav Chattopadhyay; Tapan Mishra et al.      | 2021 | arXiv preprint                                                                                                                                      | 0     | 10.1103/PhysRevB.105.214203      |                                    |
| Localization and topological transitions in non-Hermitian quasiperiodic lattices                                                       | Ling-Zhi Tang; Guo-Qing Zhang; Ling-Feng Zhang et al.      | 2021 | Phys. Rev. A 103, 033325 (2021)                                                                                                                     | 0     | 10.1103/PhysRevA.103.033325      |                                    |
| Non-Abelian generalization of non-Hermitian quasicrystal: PT-symmetry breaking, localization, entanglement and topological transitions | Longwen Zhou                                               | 2023 | Phys. Rev. B 108, 014202 (2023)                                                                                                                     | 0     | 10.1103/PhysRevB.108.014202      |                                    |
| New Classes of Quasicrystals and Marginal Critical States                                                                              | Nobuhisa Fujita; Komajiro Niizeki                          | 2000 | Phys. Rev. Lett. 85, 4924 (2000)                                                                                                                    | 0     | 10.1103/PhysRevLett.85.4924      |                                    |
| Localization and spectral structure in two-dimensional quasicrystal potentials                                                         | Zhaoxuan Zhu; Shengjie Yu; Dean Johnstone et al.           | 2023 | Phys. Rev. A 109, 013314 (2024)                                                                                                                     | 0     | (link)                           |                                    |
| Localization transitions in non-Hermitian quasiperiodic lattice                                                                        | Aruna Prasad Acharya; Sanjoy Datta                         | 2023 | arXiv preprint                                                                                                                                      | 0     | (link)                           |                                    |
| Field-Tunable Valley Coupling and Localization in a Dodecagonal Semiconductor Quasicrystal                                             | Zhida Liu; Qiang Gao; Yanxing Li et al.                    | 2024 | arXiv preprint                                                                                                                                      | 0     | (link)                           |                                    |
| Localization of Electronic Wave Functions on Quasiperiodic Lattices                                                                    | Thomas Rieth; Uwe Grimm; Michael Schreiber                 | 1998 | Proceedings of the 6th International Conference on Quasicrystals, Eds. S. Takeuchi and T. Fujiwara (World Scientific, Singapore, 1998), pp. 639-642 | 0     | 10.1088/0953-8984/10/4/008       |                                    |
| Appearance of Mobility Edge in Self-Dual Quasiperiodic Lattices                                                                        | Gang Wang; Nianbei Li; Tsuneyoshi Nakayama                 | 2013 | arXiv preprint                                                                                                                                      | 0     | (link)                           |                                    |
| Spin-wave localization on phasonic defects in one-dimensional magnonic quasicrystal                                                    | Szymon Mieszczyk; Maciej Krawczyk; Jarosław W. Klos        | 2022 | arXiv preprint                                                                                                                                      | 0     | 10.1103/PhysRevB.106.064430      |                                    |
| Localization transitions of correlated particles in non-reciprocal quasicrystals                                                       | Lei Wang; Juan Kang; Ni Liu et al.                         | 2024 | arXiv preprint                                                                                                                                      | 0     | (link)                           |                                    |
| Critical states and anomalous mobility edges in two-dimensional diagonal quasicrystals                                                 | Callum W. Duncan                                           | 2023 | Phys. Rev. B 109, 014210 (2024)                                                                                                                     | 0     | 10.1103/PhysRevB.109.014210      |                                    |
| Anomalous localization and multifractality in a kicked quasicrystal                                                                    | Toshihiko Shimasaki; Max Prichard; H. Esat Kondakci et al. | 2022 | arXiv preprint                                                                                                                                      | 0     | (link)                           |                                    |
| Fractal Spectrum of the Aubry-Andre Model                                                                                              | Ang-Kun Wu                                                 | 2021 | arXiv preprint                                                                                                                                      | 0     | (link)                           |                                    |
| Quantum Antiferromagnetism in Quasicrystals                                                                                            | Stefan Wessel; Anuradha Jagannathan; Stephan Haas          | 2002 | Phys. Rev. Lett. 90, 0177205 (2003).                                                                                                                | 0     | 10.1103/PhysRevLett.90.177205    |                                    |
| Boundary-dependent Self-dualities, Winding Numbers and Asymmetrical Localization in non-Hermitian Quasicrystals                        | Xiaoming Cai                                               | 2020 | Phys. Rev. B 103, 014201 (2021)                                                                                                                     | 0     | 10.1103/PhysRevB.103.014201      |                                    |
| Engineering mobility in quasiperiodic lattices with exact mobility edges                                                               | Zhenbo Wang; Yu Zhang; Li Wang et al.                      | 2023 | Phys. Rev. B 108, 174202 (2023)                                                                                                                     | 0     | 10.1103/PhysRevB.108.174202      |                                    |
| Extended-localized transition in diffusive quasicrystals                                                                               | Zhoufei Liu; Pei-Chao Cao; Ying Li et al.                  | 2022 | Phys. Rev. Appl. 21, 064035 (2024)                                                                                                                  | 0     | 10.1103/PhysRevApplied.21.064035 |                                    |
| Interplay of non-Hermitian skin effects and Anderson localization in non-reciprocal quasiperiodic lattices                             | Hui Jiang; Li-Jun Lang; Chao Yang et al.                   | 2019 | Phys. Rev. B 100, 054301 (2019)                                                                                                                     | 0     | 10.1103/PhysRevB.100.054301      |                                    |

| Title                                                                                                                         | Authors                                                       | Year | Venue                                      | Cited | DOI                              | OpenAlex<br>2-yr mean<br>citedness |
|-------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|------|--------------------------------------------|-------|----------------------------------|------------------------------------|
| Composite-State Localization Beyond the External Landscape in Non-Hermitian Quasicrystals                                     | Tian Zhou; Xue-Bin Wang; Zhongmin Yang et al.                 | 2026 | arXiv preprint                             | 0     | (link)                           |                                    |
| Non-Hermitian delocalization in a 2D photonic quasicrystal                                                                    | Zhaoyang Zhang; Shun Liang; Ismael Septembre et al.           | 2023 | arXiv preprint                             | 0     | (link)                           |                                    |
| Nonlinear topological Toda quasicrystal                                                                                       | Motohiko Ezawa                                                | 2022 | J. Phys. Soc. Jpn. 91, 084703 (2022)       | 0     | 10.7566/JPSJ.91.084703           |                                    |
| Theory of Localized States in Quasiperiodic Lattices                                                                          | Jin-Rong Chen; Xin-Yu Guo; Shi-Ping Ding et al.               | 2025 | arXiv preprint                             | 0     | (link)                           |                                    |
| Emergent extended states in an unbounded quasiperiodic lattice                                                                | Jia-Ming Zhang; Shan-Zhong Li; Shi-Liang Zhu et al.           | 2025 | arXiv preprint                             | 0     | (link)                           |                                    |
| Nonmonotonic crossover and scaling behaviors in a disordered 1D quasicrystal                                                  | Anuradha Jagannathan; Piyush Jeena; Marco Tarzia              | 2018 | Phys. Rev. B 99, 054203 (2019)             | 0     | 10.1103/PhysRevB.99.054203       |                                    |
| Growing Perfect Decagonal Quasicrystals by Local Rules                                                                        | Hyeong-Chai Jeong                                             | 2007 | Phys. Rev. Lett. 98, 135501 (2007)         | 0     | 10.1103/PhysRevLett.98.135501    |                                    |
| Renormalization-Group Theory of 1D quasiperiodic lattice models with commensurate approximants                                | Miguel Gonçalves; Bruno Amorim; Eduardo V. Castro et al.      | 2022 | Phys. Rev. B 108, L100201 (2023)           | 0     | 10.1103/PhysRevB.108.L100201     |                                    |
| Dephasing-induced distinct mobility edges in a dimerized off-diagonal quasicrystal                                            | Ming-Jie Tao; Yi-Ting Wang; Jing Li et al.                    | 2026 | arXiv preprint                             | 0     | (link)                           |                                    |
| Spatially Localized Quasicrystals                                                                                             | P. Subramanian; A. J. Archer; E. Knobloch et al.              | 2017 | New J. Phys. 20 (2018) 122002              | 0     | 10.1088/1367-2630/aaf3bd         |                                    |
| Floquet engineering of topological localization transitions and mobility edges in one-dimensional non-Hermitian quasicrystals | Longwen Zhou                                                  | 2021 | Phys. Rev. Research 3, 033184 (2021)       | 0     | 10.1103/PhysRevResearch.3.033184 |                                    |
| Dimerization induced mobility edges and multiple reentrant localization transitions in non-Hermitian quasicrystals            | Wenqian Han; Longwen Zhou                                     | 2021 | Phys. Rev. B 105, 054204 (2022)            | 0     | 10.1103/PhysRevB.105.054204      |                                    |
| First-order Quantum Phase Transitions and Localization in the 2D Haldane Model with Non-Hermitian Quasicrystal Boundaries     | Xianqi Tong; Su-Peng Kou                                      | 2023 | arXiv preprint                             | 0     | (link)                           |                                    |
| How Do Quasicrystals Grow?                                                                                                    | Aaron S. Keys; Sharon C. Glotzer                              | 2007 | Phys. Rev. Lett. 99, pp. 235503 (2007)     | 0     | 10.1103/PhysRevLett.99.235503    |                                    |
| One-dimensional $L\{\acute{e}\}$ vy Quasicrystal                                                                              | Pallabi Chatterjee; Ranjan Modak                              | 2022 | J. Phys.: Condens. Matter 35 505602 (2023) | 0     | 10.1088/1361-648X/acf9d4         |                                    |
| Re-entrant topological phase transition in a non-Hermitian quasiperiodic lattice                                              | Ashirbad Padhan; Soumya Ranjan Padhi; Tapan Mishra            | 2023 | arXiv preprint                             | 0     | (link)                           |                                    |
| A patchy-particle 3-dimensional octagonal quasicrystal                                                                        | Akie Kowaguchi; Savan Mehta; Jonathan P. K. Doye et al.       | 2024 | J. Chem. Phys. 163, 244904 (2025)          | 0     | 10.1063/5.0292922                |                                    |
| Perpendicular space accounting of localized states in a quasicrystal                                                          | Murod Mirzhalilov; M. Ö. Oktel                                | 2020 | Phys. Rev. B 102, 064213 (2020)            | 0     | 10.1103/PhysRevB.102.064213      |                                    |
| Purely local growth of a quasicrystal                                                                                         | Thomas Fernique; Ilya Galanov                                 | 2022 | arXiv preprint                             | 0     | (link)                           |                                    |
| Mott transition and correlation effects on strictly localized states in an octagonal quasicrystal                             | Efe Yelesti; Onur Erten; M. O. Oktel                          | 2025 | Phys. Rev. B 112, 144204 (2025)            | 0     | 10.1103/hxhj-7tvt                |                                    |
| Phononic Topological States in 1D Quasicrystals                                                                               | J. R. M. Silva; M. S. Vasconcelos; D. H. A. L. Anselmo et al. | 2019 | arXiv preprint                             | 0     | 10.1088/1361-648X/ab312a         |                                    |
| Emergent multi-loop nested point gap in a non-Hermitian quasiperiodic lattice                                                 | Yi-Qi Zheng; Shan-Zhong Li; Zhi Li                            | 2024 | Phys. Rev. B 111, 104204 (2025)            | 0     | 10.1103/PhysRevB.111.104204      |                                    |
| Spin susceptibility in quasicrystal superconductor with spin-orbit interactions                                               | Yasuhiro Tada                                                 | 2025 | arXiv preprint                             | 0     | (link)                           |                                    |
| Localization of Pairs in One-Dimensional Quasicrystals with Power-Law Hopping                                                 | G. A. Domínguez-Castro; R. Paredes                            | 2022 | arXiv preprint                             | 0     | 10.1103/PhysRevB.106.134208      |                                    |
| Hybrid Quasicrystals, Transport and Localization in Products of Minimal Sets                                                  | Tulio O. Carvalho; Cesar R. de Oliveira                       | 2007 | arXiv preprint                             | 0     | 10.1007/s10955-007-9299-8        |                                    |

| Title                                                                                                                                                | Authors                                                           | Year | Venue                                                                                                                                                                        | Cited | DOI                             | OpenAlex<br>2-yr mean<br>citedness |
|------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|---------------------------------|------------------------------------|
| Localization, Symmetry Breaking and Topological Transitions in non-Hermitian Quasicrystals                                                           | Aruna Prasad Acharya; Aditi Chakrabarty; Deepak Kumar Sahu et al. | 2021 | Phys. Rev. B 105, 014202 (2022)                                                                                                                                              | 0     | 10.1103/PhysRevB.105.014202     |                                    |
| Localized States in Local Isomorphism Classes of Pentagonal Quasicrystals                                                                            | M. Ö. Oktel                                                       | 2022 | Phys. Rev. B 106, 024201 (2022)                                                                                                                                              | 0     | 10.1103/PhysRevB.106.024201     |                                    |
| Exploring Metallic-Insulating Transition and Thermodynamic Applications of Fibonacci Quasicrystals                                                   | He-Guang Xu; Shujie Cheng                                         | 2024 | arXiv preprint                                                                                                                                                               | 0     | (link)                          |                                    |
| Self-induced Bose glass phase in quantum cluster quasicrystals                                                                                       | Matheus Grossklags; Matteo Ciardi; Vinicius Zampronio et al.      | 2023 | Results in Physics, 65, 107991, (2024)                                                                                                                                       | 0     | 10.1016/j.rinp.2024.107991      |                                    |
| Dislocations in quasicrystals and their interaction with cluster-like obstacles                                                                      | R. Mikulla; P. Gumbsch; H. -R. Trebin                             | 1997 | arXiv preprint                                                                                                                                                               | 0     | (link)                          |                                    |
| Computational Self-Assembly of a Six-Fold Chiral Quasicrystal                                                                                        | Nydia Roxana Varela-Rosales; Michael Engel                        | 2024 | arXiv preprint                                                                                                                                                               | 0     | (link)                          |                                    |
| Controlling light with nonlinear quasicrystal metasurfaces                                                                                           | Yutao Tang; Junhong Deng; King Fai Li et al.                      | 2019 | arXiv preprint                                                                                                                                                               | 0     | (link)                          |                                    |
| Intrinsic vortex pinning in superconducting quasicrystals                                                                                            | Yuki Nagai                                                        | 2021 | Phys. Rev. B 106, 064506 (2022)                                                                                                                                              | 0     | 10.1103/PhysRevB.106.064506     |                                    |
| Driving-induced multiple $\mathcal{PT}$ -symmetry breaking transitions and reentrant localization transitions in non-Hermitian Floquet quasicrystals | Longwen Zhou; Wenqian Han                                         | 2022 | Phys. Rev. B 106, 054307 (2022)                                                                                                                                              | 0     | 10.1103/PhysRevB.106.054307     |                                    |
| Hyperuniform electron distributions controlled by electron interactions in quasicrystal                                                              | Shiro Sakai; Ryotaro Arita; Tomi Ohtsuki                          | 2021 | Phys. Rev. B 105, 054202 (2022)                                                                                                                                              | 0     | 10.1103/PhysRevB.105.054202     |                                    |
| Stripe order in quasicrystals                                                                                                                        | Rafael M. P. Teixeira; Eric C. Andrade                            | 2025 | Eur. Phys. J. B 98, 188 (2025)                                                                                                                                               | 0     | 10.1140/epjb/s10051-025-01040-y |                                    |
| Topological Anderson insulator phase in a quasicrystal lattice                                                                                       | Rui Chen; Dong-Hui Xu; Bin Zhou                                   | 2019 | Phys. Rev. B 100, 115311 (2019)                                                                                                                                              | 0     | 10.1103/PhysRevB.100.115311     |                                    |
| Anomalous Localization and Mobility Edges in Non-Hermitian Quasicrystals with Disordered Imaginary Gauge Fields                                      | Guolin Nan; Zhijian Li; Feng Mei et al.                           | 2026 | arXiv preprint                                                                                                                                                               | 0     | (link)                          |                                    |
| Bose-Einstein Condensation and quasicrystals                                                                                                         | Moorad Alexanian; Vanik E. Mkrtchian                              | 2021 | Armenian Journal of Physics, 2021, vol. 14, issue 2, pp. 128-131                                                                                                             | 0     | (link)                          |                                    |
| Modelling Quasicrystal Growth                                                                                                                        | Uwe Grimm; Dieter Joseph                                          | 1999 | Quasicrystals - An Introduction to Structure, Physical Properties and Applications, edited by J.-B. Suck, M. Schreiber and P. H"aussler (Springer, Berlin, 2002) pp. 199-218 | 0     | (link)                          |                                    |
| Quantum Cluster Quasicrystals                                                                                                                        | Guido Pupillo; Primoz Zihlerl; Fabio Cinti                        | 2019 | Phys. Rev. B 101, 134522 (2020)                                                                                                                                              | 0     | 10.1103/PhysRevB.101.134522     |                                    |
| Nonlinearity-Induced Thouless Pumping in Quasiperiodic Lattices                                                                                      | Xiao-Xiao Hu; Dun Zhao; Hong-Gang Luo                             | 2026 | arXiv preprint                                                                                                                                                               | 0     | (link)                          |                                    |
| Strictly localized states in the octagonal Ammann-Beenker quasicrystal                                                                               | M. Ö. Oktel                                                       | 2021 | Phys. Rev. B 104, 014204 (2021)                                                                                                                                              | 0     | 10.1103/PhysRevB.104.014204     |                                    |
| Family of self-dual quasicrystals with critical Phases                                                                                               | Wenzhi Wang; Wei Yi; Tianyu Li                                    | 2025 | W. Wang, W. Yi and T. Li, Phys. Rev. B 112, L140201 (2025)                                                                                                                   | 0     | 10.1103/yntg-6yh2               |                                    |
| Length scale formation in the Landau levels of quasicrystals                                                                                         | Junmo Jeon; Moon Jip Park; SungBin Lee                            | 2021 | arXiv preprint                                                                                                                                                               | 0     | 10.1103/PhysRevB.105.045146     |                                    |
| Asymmetric transfer matrix analysis of Lyapunov exponents in one-dimensional non-reciprocal quasicrystals                                            | Shan-Zhong Li; Enhong Cheng; Shi-Liang Zhu et al.                 | 2024 | Phys. Rev. B 110, 134203 (2024)                                                                                                                                              | 0     | 10.1103/PhysRevB.110.134203     |                                    |
| The Fibonacci quasicrystal: case study of hidden dimensions and multifractality                                                                      | Anuradha Jagannathan                                              | 2020 | arXiv preprint                                                                                                                                                               | 0     | 10.1103/RevModPhys.93.045001    |                                    |
| Universal self-assembly of one-component three-dimensional dodecagonal quasicrystals                                                                 | Roman Ryltsev; Nikolay Chtchelkatchev                             | 2017 | arXiv preprint                                                                                                                                                               | 0     | (link)                          |                                    |
| Growth modes of quasicrystals                                                                                                                        | C. V. Achim; M. Schmiedeburg; H. Löwen                            | 2014 | Phys. Rev. Lett. 112, 255501 (2014)                                                                                                                                          | 0     | 10.1103/PhysRevLett.112.255501  |                                    |



| Title                                                                                                                                                         | Authors                                                             | Year | Venue                                | Cited | DOI                              | OpenAlex<br>2-yr mean<br>citedness |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|------|--------------------------------------|-------|----------------------------------|------------------------------------|
| Non-Hermitian quasicrystal in dimerized lattices                                                                                                              | Longwen Zhou; Wenqian Han                                           | 2021 | arXiv preprint                       | 0     | 10.1088/1674-1056/ac1efc         |                                    |
| On the characterisation of a hitherto unreported icosahedral quasicrystal phase in additively manufactured aluminium alloy AA7075                             | Shravan K. Kairy; Oumaima Gharbi; Juan Nicklaus et al.              | 2018 | arXiv preprint                       | 0     | 10.1007/s11661-018-5025-1        |                                    |
| Conventional superconductivity in quasicrystals                                                                                                               | Ronaldo N. Araújo; Eric C. Andrade                                  | 2019 | Phys. Rev. B 100, 014510 (2019)      | 0     | 10.1103/PhysRevB.100.014510      |                                    |
| Reentrant delocalization transition in one-dimensional photonic quasicrystals                                                                                 | Sachin Vaidya; Christina Jörg; Kyle Linn et al.                     | 2022 | arXiv preprint                       | 0     | (link)                           |                                    |
| A Database of 2,500 Quasicrystal Cells                                                                                                                        | Tony Robbin; George Francis; Kurt Baumann                           | 2018 | arXiv preprint                       | 0     | (link)                           |                                    |
| Non-Hermitian butterfly spectra in a family of quasiperiodic lattices                                                                                         | Li Wang; Zhenbo Wang; Shu Chen                                      | 2024 | Phys. Rev. B 110, L060201 (2024)     | 0     | 10.1103/PhysRevB.110.L060201     |                                    |
| Noncoplanar ferromagnetism and local crystalline-electric-field anisotropy in the quasicrystal approximant Au <sub>70</sub> Si <sub>17</sub> Tb <sub>13</sub> | Takanobu Hiroto; Taku J Sato; Huibo Cao et al.                      | 2019 | arXiv preprint                       | 0     | 10.1088/1361-648X/ab997d         |                                    |
| Topological Hofstadter Insulators in a Two-Dimensional Quasicrystal                                                                                           | Duc-Thanh Tran; Alexandre Dauphin; Nathan Goldman et al.            | 2014 | Phys. Rev. B 91, 085125 (2015)       | 0     | 10.1103/PhysRevB.91.085125       |                                    |
| Interface states in two-dimensional quasicrystals with broken inversion symmetry                                                                              | Danilo Beli; Matheus I. N. Rosa; Luca Lomazzi et al.                | 2023 | arXiv preprint                       | 0     | (link)                           |                                    |
| Reversible phasonic control of a quantum phase transition in a quasicrystal                                                                                   | Toshihiko Shimasaki; Yifei Bai; H. Esat Kondakci et al.             | 2023 | arXiv preprint                       | 0     | (link)                           |                                    |
| Interaction tuned pattern-selective superconductivity: Application to the dodecagonal quasicrystal                                                            | Junmo Jeon; SungBin Lee                                             | 2025 | arXiv preprint                       | 0     | (link)                           |                                    |
| The Aubry-Andre Anderson model: Magnetic impurities coupled to a fractal spectrum                                                                             | Ang-Kun Wu; Daniel Bauernfeind; Xiaodong Cao et al.                 | 2022 | arXiv preprint                       | 0     | 10.1103/PhysRevB.106.165123      |                                    |
| Energy levels and their correlations in quasicrystals                                                                                                         | A. Jagannathan; F. Piechon                                          | 2006 | arXiv preprint                       | 0     | 10.1080/14786430701196990        |                                    |
| Correlation-induced phase transitions and mobility edges in an interacting non-Hermitian quasicrystal                                                         | Tian Qian; Yongjian Gu; Longwen Zhou                                | 2023 | Phys. Rev. B 109, 054204 (2024)      | 0     | 10.1103/PhysRevB.109.054204      |                                    |
| Wigner distribution, Wigner entropy and Quantum Refrigerator of a One-Dimensional Off-diagonal Quasicrystal                                                   | Shan Suo; Ao Zhou; Yanting Chen et al.                              | 2026 | arXiv preprint                       | 0     | (link)                           |                                    |
| Localization transition, spectrum structure and winding numbers for one-dimensional non-Hermitian quasicrystals                                               | Yanxia Liu; Qi Zhou; Shu Chen                                       | 2020 | Phys. Rev. B 104, 024201 (2021)      | 0     | 10.1103/PhysRevB.104.024201      |                                    |
| Bloch-like Electronic Wave Functions in Two-Dimensional Quasicrystals                                                                                         | Shahar Even-Dar Mandel; Ron Lifshitz                                | 2008 | arXiv preprint                       | 0     | (link)                           |                                    |
| Quasicrystals in pattern formation, Part I: Local existence and basic properties                                                                              | Ian Melbourne; Jens Rademacher; Bob Rink et al.                     | 2024 | arXiv preprint                       | 0     | (link)                           |                                    |
| Transition pathways connecting crystals and quasicrystals                                                                                                     | Jianyuan Yin; Kai Jiang; An-Chang Shi et al.                        | 2020 | arXiv preprint                       | 0     | 10.1073/pnas.2106230118          |                                    |
| Computation and visualization of photonic quasicrystal spectra via Blochs theorem                                                                             | Alejandro W. Rodriguez; Alexander P. McCauley; Yehuda Avniel et al. | 2007 | Physical Review B, 77 104201, 2008   | 0     | 10.1103/PhysRevB.77.104201       |                                    |
| Atomistic mechanisms of dynamics in a two-dimensional dodecagonal quasicrystal                                                                                | Kun Zhao; Matteo Baggioli; Wen-Sheng Xu et al.                      | 2024 | J. Chem. Phys. 162, 234503 (2025)    | 0     | 10.1063/5.0270291                |                                    |
| Magnons in a Quasicrystal: Propagation, Localization and Extinction of Spin Waves in Fibonacci Structures                                                     | Filip Lisiecki; Justyna Rychły; Piotr Kuświk et al.                 | 2018 | Phys. Rev. Applied 11, 054061 (2019) | 0     | 10.1103/PhysRevApplied.11.054061 |                                    |
| Al-Cu-Fe alloys: the relationship between the quasicrystal and its melt                                                                                       | L. V. Kamaeva; R. E. Ryltsev; V. I. Lad'yanov et al.                | 2019 | arXiv preprint                       | 0     | (link)                           |                                    |

| Title                                                                                                                | Authors                                                            | Year | Venue                                         | Cited | DOI                              | OpenAlex<br>2-yr mean<br>citedness |
|----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|------|-----------------------------------------------|-------|----------------------------------|------------------------------------|
| Existence of quasicrystals and universal stable sampling and interpolation in LCA groups                             | E. Agora; J. Antezana; C. Cabrelli et al.                          | 2016 | arXiv preprint                                | 0     | (link)                           |                                    |
| How Quasicrystals Remember: Hierarchical Memory Under Cyclic Shear                                                   | Edwin A. Bedolla-Montiel; Marjolein Dijkstra                       | 2026 | arXiv preprint                                | 0     | (link)                           |                                    |
| Electronic properties of ternary quasicrystals in one dimension                                                      | Nobuhisa Fujita; Komajiro Niizeki                                  | 2001 | Phys. Rev. B 64 (2001)                        | 0     | 10.1103/PhysRevB.64.144207       |                                    |
| One-Dimensional Quasicrystals from Incommensurate Charge Order                                                       | Felix Flicker; Jasper van Wezel                                    | 2015 | Phys. Rev. Lett. 115, 236401 (2015)           | 0     | 10.1103/PhysRevLett.115.236401   |                                    |
| Magnetic ground state of a prototype quasicrystal approximant: a candidate for octahedral spin ice physics           | Leonie Woodland; Farid Labib; Dmitry Khalyavin et al.              | 2026 | arXiv preprint                                | 0     | (link)                           |                                    |
| Valence Change Driven by Constituent Element Substitution in the Mixed-Valence Quasicrystal and Approximant Au-Al-Yb | Shuya Matsukawa; Katsumasa Tanaka; Mika Nakayama et al.            | 2014 | J. Phys. Soc. Jpn. 83 (2014) 034705 (5 Pages) | 0     | 10.7566/JPSJ.83.034705           |                                    |
| Quantum Entanglement of Non-Hermitian Quasicrystals                                                                  | Li-Mei Chen; Yao Zhou; Shuai A. Chen et al.                        | 2021 | Phys. Rev. B 105, L121115 (2022)              | 0     | 10.1103/PhysRevB.105.L121115     |                                    |
| Nature of Protected Zero Energy States in Penrose Quasicrystals                                                      | Ezra Day-Roberts; Rafael M. Fernandes; Alex Kamenev                | 2020 | Phys. Rev. B 102, 064210 (2020)               | 0     | 10.1103/PhysRevB.102.064210      |                                    |
| Emergence of multiple localization transitions in a one-dimensional quasiperiodic lattice                            | Ashirbad Padhan; Mrinal Kanti Giri; Suman Mondal et al.            | 2021 | Phys. Rev. B 105, L220201 (2022)              | 0     | 10.1103/PhysRevB.105.L220201     |                                    |
| Origin of Quantum Criticality in Yb-Al-Au Approximant Crystal and Quasicrystal                                       | Shinji Watanabe; Kazumasa Miyake                                   | 2016 | J. Phys. Soc. Jpn. 85 (2016) 063703           | 0     | 10.7566/JPSJ.85.063703           |                                    |
| Fractal Surface States in Three-Dimensional Topological Quasicrystals                                                | Zhu-Guang Chen; Cunchong Lou; Kaige Hu et al.                      | 2024 | Phys. Rev. Research 6, 043054 (2024)          | 0     | 10.1103/PhysRevResearch.6.043054 |                                    |
| Host-atom-driven transformation of a honeycomb oxide into a dodecagonal quasicrystal                                 | Martin Haller; Julia Hewelt; V. Y. M. Rajesh Chirala et al.        | 2025 | Phys. Rev. Lett. 136, 156201 (2026)           | 0     | 10.1103/ws7j-tvty                |                                    |
| Closing of gaps and gap labeling and passage from molecular states to critical states in a 2D quasicrystal           | Anuradha Jagannathan                                               | 2023 | arXiv preprint                                | 0     | 10.1103/PhysRevB.108.115109      |                                    |
| Ground states of a family of frustrated spin models for quasicrystals and their approximants                         | Anuradha Jagannathan                                               | 2025 | arXiv preprint                                | 0     | 10.1103/ht6g-5dwl                |                                    |
| Phonon transmittance of one dimensional quasicrystals                                                                | Junmo Jeon; SungBin Lee                                            | 2020 | arXiv preprint                                | 0     | (link)                           |                                    |
| Site-selective correlations in interacting "flat-band" quasicrystals                                                 | Yuxi Zhang; Richard T. Scalettar; Rafael M. Fernandes              | 2024 | arXiv preprint                                | 0     | 10.1103/77l3-krmj                |                                    |
| Density Distribution in Soft Matter Crystals and Quasicrystals                                                       | Priya Subramanian; Daniel J. Ratliff; Alastair M. Rucklidge et al. | 2020 | Phys. Rev. Lett. 126, 218003 (2021)           | 0     | 10.1103/PhysRevLett.126.218003   |                                    |
| Quantum state transfer and maximal entanglement between distant qubits using a minimal quasicrystal pump             | Arnob Kumar Ghosh; Rubén Seoane Souto; Vahid Azimi-Mousolou et al. | 2025 | Phys. Rev. B 112, 205427 (2025)               | 0     | 10.1103/8zys-w2v4                |                                    |
| Scale-free localization versus Anderson localization in unidirectional quasiperiodic lattices                        | Yu Zhang; Luhong Su; Shu Chen                                      | 2025 | Phys. Rev. B 111, L140201 (2025)              | 0     | 10.1103/PhysRevB.111.L140201     |                                    |
| Quantum interferences in quasicrystals                                                                               | Stephan Roche                                                      | 1999 | arXiv preprint                                | 0     | (link)                           |                                    |
| Cut-and-Project Density Functional Theory for Quasicrystals                                                          | Gavin N. Nop; Jonathan D. H. Smith; Thomas Koschny et al.          | 2026 | arXiv preprint                                | 0     | (link)                           |                                    |
| Fractalized magnon transport on the quasicrystal with enhanced stability                                             | Junmo Jeon; Se Kwon Kim; SungBin Lee                               | 2022 | arXiv preprint                                | 0     | 10.1103/PhysRevB.106.134431      |                                    |
| Common quantum phase transition in quasicrystals and heavy-fermion metals                                            | V. R. Shaginyan; A. Z. Msezane; K. G. Popov et al.                 | 2013 | Phys. Rev. B 87, 245122 (2013)                | 0     | 10.1103/PhysRevB.87.245122       |                                    |
| High-harmonic spectroscopy of mobility edges in one-dimensional quasicrystals                                        | H. K. Avetissian; B. R. Avchyan; A. Brown et al.                   | 2025 | arXiv preprint                                | 0     | (link)                           |                                    |
| Anomalous electronic conductance in quasicrystals                                                                    | Stephan Roche                                                      | 1999 | arXiv preprint                                | 0     | 10.1103/PhysRevB.61.6048         |                                    |

| Title                                                                                                           | Authors                                                             | Year | Venue                                                                | Cited | DOI                             | OpenAlex<br>2-yr mean<br>citedness |
|-----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|------|----------------------------------------------------------------------|-------|---------------------------------|------------------------------------|
| The Aubry-André model as the hobbyhorse for understanding localization phenomenon                               | G. A. Domínguez-Castro; R. Paredes                                  | 2018 | arXiv preprint                                                       | 0     | 10.1088/1361-6404/ab1670        |                                    |
| Tensor network method for real-space topology in quasicrystal Chern mosaics                                     | Tiago V. C. Antão; Yitao Sun; Adolfo O. Fumega et al.               | 2025 | Phys. Rev. Lett. 136, 156601 (2026)                                  | 0     | 10.1103/hhdf-xpwg               |                                    |
| Interlayer Decoupling in 30° Twisted Bilayer Graphene Quasicrystal                                              | Bing Deng; Binbin Wang; Ning Li et al.                              | 2020 | ACS Nano (2020)                                                      | 0     | 10.1021/acs.nano.9b07094        | (2026)                             |
| Spin waves in one-dimensional bi-component quasicrystals                                                        | J. Rychlý; J. W. Klos; M. Mruczkiewicz et al.                       | 2014 | Phys. Rev. B 92, 054414 (2015)                                       | 0     | 10.1103/PhysRevB.92.054414      |                                    |
| Equilibrium configurations for generalized Frenkel-Kontorova models on quasicrystals                            | Rodrigo Treviño                                                     | 2017 | arXiv preprint                                                       | 0     | 10.1007/s00220-019-03557-7      |                                    |
| Localization, multifractality, and many-body localization in periodically kicked quasiperiodic lattices         | Yu Zhang; Bozhen Zhou; Haiping Hu et al.                            | 2022 | Phys. Rev. B 106, 054312 (2022)                                      | 0     | 10.1103/PhysRevB.106.054312     |                                    |
| A parametric study of the lensing properties of dodecagonal photonic quasicrystals                              | E. Di Gennaro; D. Morello; C. Miletto et al.                        | 2007 | arXiv preprint                                                       | 0     | 10.1016/j.photonics.2007.12.001 |                                    |
| Exact anomalous mobility edges in one-dimensional non-Hermitian quasicrystals                                   | Xiang-Ping Jiang; Weilei Zeng; Yayun Hu et al.                      | 2024 | arXiv preprint                                                       | 0     | (link)                          |                                    |
| Identification and properties of topological states in the bulk of quasicrystals                                | Frode Balling-Ansø; Jeppe Lykke Krogh; Ella Elisabeth Lassen et al. | 2025 | Phys. Rev. B 113, 075134 (2026)                                      | 0     | 10.1103/56vs-zwr8               |                                    |
| Relationship between Structure and Dynamics of an Icosahedral Quasicrystal using Unsupervised Machine Learning  | Edwin A. Bedolla-Montiel; Susana Marín-Aguilar; Marjolein Dijkstra  | 2025 | arXiv preprint                                                       | 0     | (link)                          |                                    |
| Quantum bridge states and sub-dimensional transports in quasicrystals                                           | Junmo Jeon; SungBin Lee                                             | 2022 | arXiv preprint                                                       | 0     | (link)                          |                                    |
| 1D quasicrystals and topological markers                                                                        | Joseph Sykes; Ryan Barnett                                          | 2022 | arXiv preprint                                                       | 0     | 10.1088/2633-4356/ac75a6        |                                    |
| Fragile magnetic order in metallic quasicrystals                                                                | Ronaldo N. Araújo; Carlo C. Bellinati; Eric C. Andrade              | 2023 | Phys. Rev. B 110, 165131 (2024)                                      | 0     | 10.1103/PhysRevB.110.165131     |                                    |
| Quantum Fluctuations and Excitations in Antiferromagnetic Quasicrystals                                         | Stefan Wessel; Igor Milat                                           | 2004 | Phys. Rev. B 71, 104427 (2005).                                      | 0     | 10.1103/PhysRevB.71.104427      |                                    |
| Exploring light-induced phases of 2D materials in a modulated 1D quasicrystal                                   | Yifei Bai; Anna R. Dardia; Toshihiko Shimasaki et al.               | 2025 | arXiv preprint                                                       | 0     | (link)                          |                                    |
| Emergent Nonperturbative Universal Floquet Localization                                                         | Soumadip Pakrashi; Atanu Rajak; Sambuddha Sanyal                    | 2026 | arXiv preprint                                                       | 0     | (link)                          |                                    |
| Localization transition in weakly-interacting Bose superfluids in one-dimensional quasiperiodic lattices        | Samuel Lellouch; Laurent Sanchez-Palencia                           | 2014 | Physical Review A, American Physical Society, 2014, 90, pp.061602(R) | 0     | 10.1103/PhysRevA.90.061602      |                                    |
| Twisted Multilayer Graphene: Superperiodicity and quasicrystals                                                 | Pedro Alcázar Guerrero                                              | 2026 | arXiv preprint                                                       | 0     | (link)                          |                                    |
| Gapless superconductivity and its real-space topology in quasicrystals                                          | Kazuma Saito; Masahiro Hori; Ryo Okugawa et al.                     | 2024 | Physical Review Research 7, L022077 (2025)                           | 0     | 10.1103/nb7l-f44r               |                                    |
| Robust flat bands in twisted trilayer graphene quasicrystals                                                    | Chen-Yue Hao; Zhen Zhan; Pierre A. Pantaleón et al.                 | 2024 | arXiv preprint                                                       | 0     | (link)                          |                                    |
| Non-diffusion transport in decoherent non-Hermitian quasicrystals                                               | Yudong Ren; Rui Zhao; Kangpeng Ye et al.                            | 2025 | arXiv preprint                                                       | 0     | (link)                          |                                    |
| Consistent Simulation of Fibonacci Anyon Braiding within a Qubit Quasicrystal Inflation Code                    | Marcelo M. Amaral                                                   | 2025 | arXiv preprint                                                       | 0     | (link)                          |                                    |
| Holographic Foliations: Self-Similar Quasicrystals from Hyperbolic Honeycombs                                   | Latham Boyle; Justin Kulp                                           | 2024 | Phys. Rev. D 111, 046001 (2025)                                      | 0     | 10.1103/PhysRevD.111.046001     |                                    |
| Evidence of local effects in anomalous refraction and focusing properties of dodecagonal photonic quasicrystals | Emiliano Di Gennaro; Carlo Miletto; Salvatore Savo et al.           | 2008 | arXiv preprint                                                       | 0     | 10.1103/PhysRevB.77.193104      |                                    |

| Title                                                                                                                              | Authors                                                                      | Year | Venue                                                                               | Cited | DOI                            | OpenAlex<br>2-yr mean<br>citedness |
|------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|------|-------------------------------------------------------------------------------------|-------|--------------------------------|------------------------------------|
| Plasmonic quasicrystals for designable ultra broadband transmission enhancement and second harmonic generation                     | Sachin Kasture; Ajith P R; V J Yallapragada et al.                           | 2013 | arXiv preprint                                                                      | 0     | (link)                         |                                    |
| Which wavenumbers determine the thermodynamic stability of soft matter quasicrystals?                                              | D. J. Ratliff; A. J. Archer; P. Subramanian et al.                           | 2019 | Phys. Rev. Lett. 123, 148004 (2019)                                                 | 0     | 10.1103/PhysRevLett.123.148004 |                                    |
| Space-Time Quasicrystal Structures and Inflationary and Late Time Evolution Dynamics in Accelerating Cosmology                     | Sergiu I. Vacaru                                                             | 2018 | Clas. Quant. Grav. (2018) no. 24, 245009                                            | 35 0  | 10.1088/1361-6382/aaec93       |                                    |
| Quasiperiodic pairing in graphene quasicrystals                                                                                    | Rasoul Ghadimi; Bohm-Jung Yang                                               | 2024 | Nano Lett. 2025, 25, 5, 1808 1815                                                   | 0     | 10.1021/acs.nanolett.4c04386   |                                    |
| Electronic states of a disordered 2D quasiperiodic tiling: from critical states to Anderson localization                           | Anuradha Jagannathan; Marco Tarzia                                           | 2022 | arXiv preprint                                                                      | 0     | 10.1103/PhysRevB.107.054206    |                                    |
| Binary icosahedral quasicrystals of hard spheres in spherical confinement                                                          | Da Wang; Tonnishtha Dasgupta; Ernest B. van der Wee et al.                   | 2019 | arXiv preprint                                                                      | 0     | 10.1038/s41567-020-1003-9      |                                    |
| Topological metal-insulator transitions in one-dimensional non-Hermitian quasicrystals: beyond PT-symmetry                         | Guangjie Zhang; Bing Shao; Longwen Zhou et al.                               | 2026 | arXiv preprint                                                                      | 0     | (link)                         |                                    |
| Asymptotic estimates of large gaps between directions in certain planar quasicrystals                                              | Gustav Hammarhjelm; Andreas Strömbergsson; Shucheng Yu                       | 2024 | arXiv preprint                                                                      | 0     | (link)                         |                                    |
| Phasons and the Plastic Deformation of Quasicrystals                                                                               | Maurice Kleman                                                               | 2002 | Eur. phys. J. B 31 (2003) 315-325                                                   | 0     | 10.1140/epjb/e2003-00037-3     |                                    |
| Starobinsky Inflation and Dark Energy and Dark Matter Effects from Quasicrystal Like Spacetime Structures                          | R. Aschheim; L. Bubuianu; Fang Fang et al.                                   | 2016 | Annals of Physics (2018) 120-138                                                    | 394 0 | 10.1016/j.aop.2018.04.033      |                                    |
| Raman study of lattice dynamics in quasicrystals                                                                                   | Yu. S. Ponosov; N. I. Shchegolikhina; A. F. Prekul                           | 2009 | arXiv preprint                                                                      | 0     | (link)                         |                                    |
| Localization Transitions in a Half-Filled Helical Aubry-André Model                                                                | Taylan Yildiz; B. Tanatar; Balázs Hetényi                                    | 2026 | arXiv preprint                                                                      | 0     | 10.1103/c5wk-3ym5              |                                    |
| Reentrant Localization Transitions in a Topological Anderson Insulator: A Study of a Generalized Su-Schrieffer-Heeger Quasicrystal | Zhanpeng Lu; Yunbo Zhang; Zhihao Xu                                          | 2023 | Front. Phys. 20, 024204 (2025)                                                      | 0     | 10.15302/frontphys.2025.024204 |                                    |
| Anderson Localization and Swing Mobility Edge in Curved Spacetime                                                                  | Shan-Zhong Li; Xue-Jia Yu; Shi-Liang Zhu et al.                              | 2022 | Phys. Rev. B 108, 094209 (2023)                                                     | 0     | 10.1103/PhysRevB.108.094209    |                                    |
| Generic non-Hermitian mobility edges in a class of duality-breaking quasicrystals                                                  | Xiang-Ping Jiang; Mingdi Xu; Lei Pan                                         | 2024 | Results in Physics (2025) 108146                                                    | 70 0  | 10.1016/j.rinp.2025.108146     |                                    |
| Quantum transport in quasicrystals and complex metallic alloys                                                                     | Didier Mayou; Guy Tramblay De Laissardière                                   | 2007 | Quasicrystals, Elsevier, Amsterdam (Ed.) (2008) 209-265                             | 0     | (link)                         |                                    |
| Hybrid skin-topological effect induced by eight-site cells and arbitrary adjustment of the localization of topological edge states | Jianzhi Chen; Aoqian Shi; Yuchen Peng et al.                                 | 2024 | Chinese Physics Letters 41, 037103 (2024) Chinese Physics Letters 41, 037103 (2024) | 0     | 10.1088/0256-307X/41/3/037103  |                                    |
| Su-Schrieffer-Heeger quasicrystal: Topology, localization, and mobility edge                                                       | D. A. Miranda; T. V. C. Antão; N. M. R Peres                                 | 2024 | Phys. Rev. B 109, 195427, Published 21 May 2024                                     | 0     | 10.1103/PhysRevB.109.195427    |                                    |
| The scales of disorder in perfect quasicrystals                                                                                    | Alan Rodrigo Mendoza Sosa; Atahualpa S. Kraemer; Michael Schmiedeberg et al. | 2026 | arXiv preprint                                                                      | 0     | (link)                         |                                    |
| Entropic crystallization of geometrically frustrated magnets on 1/1 approximant Tsai-type quasicrystal                             | Oscar Novat; Ludovic D. C. Jaubert; Masafumi Udagawa                         | 2026 | arXiv preprint                                                                      | 0     | (link)                         |                                    |
| Unveiling exotic magnetic phase diagram of a non-Heisenberg quasicrystal approximant                                               | Farid Labib; Kazuhiro Nawa; Shintaro Suzuki et al.                           | 2023 | arXiv preprint                                                                      | 0     | (link)                         |                                    |
| Non-Fermi-Liquid Behaviors in Weakly Interacting Two-Dimensional Quasicrystals                                                     | Junmo Jeon; Shiro Sakai                                                      | 2026 | arXiv preprint                                                                      | 0     | (link)                         |                                    |

| Title                                                                                                               | Authors                                                           | Year | Venue                                                     | Cited        | DOI                            | OpenAlex<br>2-yr mean<br>citedness |
|---------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|------|-----------------------------------------------------------|--------------|--------------------------------|------------------------------------|
| Existence results in the linear dynamics of quasicrystals with phason diffusion and non-linear gyroscopic effects   | Luca Bisconti; Paolo Maria Mariano                                | 2015 | arXiv preprint                                            | 0            | (link)                         |                                    |
| Striking Similarities in Dynamics and Vibrations of 2D Quasicrystals and Supercooled Liquids                        | Edwin A. Bedolla-Montiel; Marjolein Dijkstra                      | 2025 | arXiv preprint                                            | 0            | (link)                         |                                    |
| Cholesteric shells: two-dimensional blue fog and finite quasicrystals                                               | Livio Nicola Carenza; Giuseppe Gonnella; Davide Marenduzzo et al. | 2021 | Physical Review Letters                                   | 0            | 10.1103/PhysRevLett.128.027801 |                                    |
| Mapping of strained graphene into one-dimensional Hamiltonians: quasicrystals and modulated crystals                | Gerardo G. Naumis; Pedro Roman-Taboada                            | 2014 | arXiv preprint                                            | 0            | 10.1103/PhysRevB.89.241404     |                                    |
| The fate of Wannier-Stark localization and skin effect in periodically driven non-Hermitian quasiperiodic lattices  | Aditi Chakrabarty; Sanjoy Datta                                   | 2024 | arXiv preprint                                            | 0            | (link)                         |                                    |
| Information Theoretic Signatures of Localization and Mobility Edges in Quasiperiodic Systems                        | Arpita Goswami                                                    | 2026 | arXiv preprint                                            | 0            | (link)                         |                                    |
| Quasilattice-conserved optimization of the atomic structure of decagonal Al-Co-Ni quasicrystals                     | Xiao-Tian Li; Xiao-Bao Yang; Yu-Jun Zhao                          | 2014 | arXiv preprint                                            | 0            | 10.1088/0256-307X/32/3/036102  |                                    |
| Quasiperiodic Skin Criticality in an Exactly Solvable Non-Hermitian Quasicrystal                                    | Zhangyuan Chen; Muhammad Idrees; Ying Yang et al.                 | 2026 | Phys. Rev. B                                              | 113, 0       | 235423(2026)                   | 10.1103/wlsw-zyq9                  |
| Localization characteristics of two-dimensional quasicrystals consisting of metal nanoparticles                     | Jian-Wen Dong; Kin Hung Fung; C. T. Chan et al.                   | 2009 | Phys. Rev. B                                              | 80, 155118   | (2009)                         | 10.1103/PhysRevB.80.155118         |
| Symmetry-breaking dynamics of the finite-size Lipkin-Meshkov-Glick model near ground state                          | Yi Huang; Tongcang Li; Zhang-qi Yin                               | 2017 | Phys. Rev. A                                              | 97, 012115   | (2018)                         | 10.1103/PhysRevA.97.012115         |
| Entanglement phase transitions in non-Hermitian quasicrystals                                                       | Longwen Zhou                                                      | 2023 | Phys. Rev. B                                              | 109, 024204  | (2024)                         | 10.1103/PhysRevB.109.024204        |
| Antiferromagnetism in two-dimensional quasicrystals                                                                 | Anuradha Jagannathan; Attila Szallas                              | 2008 | arXiv preprint                                            | 0            | (link)                         |                                    |
| First-principles evidence for conventional superconductivity in a quasicrystal approximant                          | Pedro N. Ferreira; Roman Lucrezi; Sangmin Lee et al.              | 2025 | arXiv preprint                                            | 0            | (link)                         |                                    |
| Quasicrystalline $30^\circ$ twisted bilayer graphene: Fractal patterns and electronic localization properties       | Kevin J. U. Vidarte; Caio Lewenkopf                               | 2024 | Frontiers in Carbon                                       | 3, 0         | 1496179 (2024)                 | 10.3389/frcrb.2024.1496179         |
| Bulk Localised Transport States in Infinite and Finite Quasicrystals via Magnetic Aperiodicity                      | Dean Johnstone; Matthew J. Colbrook; Anne E. B. Nielsen et al.    | 2021 | Phys. Rev. B                                              | 106, 045149  | (2022)                         | 10.1103/PhysRevB.106.045149        |
| Correlation-enhanced Friedel oscillations in amorphous alloys and quasicrystals                                     | Johann Kroha                                                      | 1998 | J. Noncryst. Solids                                       | 250-252, 865 | (1999)                         | 10.1016/S0022-3093(99)00194-5      |
| Monte Carlo study on critical exponents of the classical Heisenberg model in ferromagnetic icosahedral quasicrystal | Shinji Watanabe; Tsunetomo Yamada; Hiroyuki Takakura et al.       | 2025 | Phys. Rev. Research                                       | 7, 0         | (2025) 043113                  | 10.1103/4zb8-2zjq                  |
| Catalytic mechanism of LENR in quasicrystals based on localized anharmonic vibrations and phasons                   | Volodymyr Dubinko; Denis Laptev; Klee Irwin                       | 2016 | arXiv preprint                                            | 0            | (link)                         |                                    |
| A general approach to the exact localized transition points of 1D mosaic disorder models                            | Yanxia Liu                                                        | 2022 | arXiv preprint                                            | 0            | (link)                         |                                    |
| Optical reflectivity as a simple diagnostic method for testing structural quality of icosahedral quasicrystals      | Valerie Brien; Anne Dauscher; F. Machizaud                        | 2020 | Journal of Applied Physics, American Institute of Physics | 0            | 10.1063/1.2234558              |                                    |

| Title                                                                                                                                   | Authors                                                             | Year | Venue                                       | Cited | DOI                         | OpenAlex<br>2-yr mean<br>citedness |
|-----------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|------|---------------------------------------------|-------|-----------------------------|------------------------------------|
| Magnetic properties of icosahedral quasicrystals and their cubic approximants in the Cd-Mg-RE (RE = Gd, Tb, Dy, Ho, Er, and Tm) systems | Farid Labib; Daisuke Okuyama; Nobuhisa Fujita et al.                | 2020 | arXiv preprint                              | 0     | 10.1088/1361-648X/ab9343    |                                    |
| Delone measures of finite local complexity and applications to spectral theory of one-dimensional continuum models of quasicrystals     | Steffen Klassert; Daniel Lenz; Peter Stollmann                      | 2010 | arXiv preprint                              | 0     | (link)                      |                                    |
| Self-duality of One-dimensional Quasicrystals with Spin-Orbit Interaction                                                               | Deepak Kumar Sahu; Aruna Prasad Acharya; Debajyoti Choudhuri et al. | 2021 | Phys. Rev. B 104, 054202 (2021)             | 0     | 10.1103/PhysRevB.104.054202 |                                    |
| Real-space Formalism for the Euler Class and Fragile Topology in Quasicrystals and Amorphous Lattices                                   | Dexin Li; Citian Wang; Huaqing Huang                                | 2023 | SciPost Phys. 17, 086 (2024)                | 0     | (link)                      |                                    |
| Logarithmically Correlated Landscapes and Localization in Non-Hermitian Quasicrystals                                                   | Xianqi Tong; Qifeng Ding; Xiaosen Yang                              | 2026 | arXiv preprint                              | 0     | (link)                      |                                    |
| Long-range Magnetic Order in Models for Rare Earth Quasicrystals                                                                        | Stefanie Thiem; J. T. Chalker                                       | 2015 | Phys. Rev. B 92, 224409 (2015)              | 0     | 10.1103/PhysRevB.92.224409  |                                    |
| Exact mobility edges, $\mathcal{PT}$ -symmetry breaking and skin effect in one-dimensional non-Hermitian quasicrystals                  | Yanxia Liu; Yucheng Wang; Xiong-Jun Liu et al.                      | 2020 | Phys. Rev. B 103, 014203 (2021)             | 0     | 10.1103/PhysRevB.103.014203 |                                    |
| Anomalous electronic transport in Quasicrystals and related Complex Metallic Alloys                                                     | Guy Trambly de Laisardière; Didier Mayou                            | 2013 | C. R. Physique 15 (2014) 70-81              | 0     | 10.1016/j.crhy.2013.09.010  |                                    |
| Macroscopically degenerate localized zero-energy states of quasicrystalline bilayer systems in strong coupling limit                    | Hyunsoo Ha; Bohm-Jung Yang                                          | 2021 | Phys. Rev. B 104, 165112 (2021)             | 0     | 10.1103/PhysRevB.104.165112 |                                    |
| Quantum transport in a one-dimensional quasicrystal with mobility edges                                                                 | Yan Xing; Lu Qi; Xuedong Zhao et al.                                | 2022 | Physical Review A 105, 032443 (2022)        | 0     | 10.1103/PhysRevA.105.032443 |                                    |
| Super Hamiltonian in superspace for incommensurate superlattices and quasicrystals                                                      | Manuel Valiente; Callum W. Duncan; Nikolaj T. Zinner                | 2019 | arXiv preprint                              | 0     | (link)                      |                                    |
| Hierarchical Structures of Quantum Geometric Spectrum in Quasicrystals: A Renormalization-Group Study                                   | Jundi Wang; Yuxiao Chen; Huaqing Huang                              | 2025 | Physical Review Letters (2026)              | 0     | 10.1103/v4ky-6v45           |                                    |
| Topological delocalization transitions and mobility edges in the nonreciprocal Maryland model                                           | Longwen Zhou; Yongjian Gu                                           | 2021 | J. Phys.: Condens. Matter 34, 115402 (2022) | 0     | 10.1088/1361-648X/ac4530    |                                    |
| Exact non-Hermitian mobility edges in one-dimensional quasicrystal lattice with exponentially decaying hopping and its dual lattice     | Yanxia Liu; Yongjian Wang; Zuohuan Zheng et al.                     | 2020 | Phys. Rev. B 103, 134208 (2021)             | 0     | 10.1103/PhysRevB.103.134208 |                                    |
| Spectral and Diffusive Properties of Silver-Mean Quasicrystals in 1,2, and 3 Dimensions                                                 | V. Z. Cerovski; M. Schreiber; U. Grimm                              | 2004 | Phys. Rev. B 72, 054203 (2005)              | 0     | 10.1103/PhysRevB.72.054203  |                                    |
| Decoding flat bands from compact localized states                                                                                       | Yuge Chen; Juntao Huang; Kun Jiang et al.                           | 2022 | arXiv preprint                              | 0     | 10.1016/j.scib.2023.11.032  |                                    |
| Magnetic and transport properties of i- $\text{Sr}$ -Cd icosahedral quasicrystals ( $\text{Sr} = \text{Y}, \text{Gd-Tm}$ )              | Tai Kong; Sergey L. Bud'ko; Anton Jesche et al.                     | 2014 | arXiv preprint                              | 0     | 10.1103/PhysRevB.90.014424  |                                    |
| Mobility Crossover in Two-Dimensional Berry Crystals                                                                                    | Zixuan Chai; Si-Yuan Chen; Chenzheng Yu et al.                      | 2024 | arXiv preprint                              | 0     | (link)                      |                                    |
| Exact complex mobility edges and flagellate spectra for non-Hermitian quasicrystals with exponential hoppings                           | Li Wang; Jiaqi Liu; Zhenbo Wang et al.                              | 2024 | Phys. Rev. B 110, 144205 (2024)             | 0     | 10.1103/PhysRevB.110.144205 |                                    |
| Non-Hermitian mobility edges in one-dimensional quasicrystals with parity-time symmetry                                                 | Yanxia Liu; Xiang-Ping Jiang; Junpeng Cao et al.                    | 2020 | Phys. Rev. B 101, 174205 (2020)             | 0     | 10.1103/PhysRevB.101.174205 |                                    |

| Title                                                                                                                                        | Authors                                                     | Year | Venue                                               | Cited | DOI                            | OpenAlex<br>2-yr mean<br>citedness |
|----------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|------|-----------------------------------------------------|-------|--------------------------------|------------------------------------|
| The local atomic quasicrystal structure of the icosahedral Mg <sub>25</sub> Y <sub>11</sub> Zn <sub>64</sub> alloy                           | S. Bruehne; E. Uhrig; C. Gross et al.                       | 2004 | J. Phys.: Condens. Matter 17 (2005) 1561-1572       | 0     | 10.1088/0953-8984/17/10/011    |                                    |
| Classifying extended, localized and critical states in quasiperiodic lattices via unsupervised learning                                      | Bohan Zheng; Siyu Zhu; Xingping Zhou et al.                 | 2024 | arXiv preprint                                      | 0     | (link)                         |                                    |
| Strictly Localized States on the Socolar Dodecagonal Lattice                                                                                 | M. Akif Keskiner; M. Ö Oktel                                | 2022 | Phys. Rev. B 106, 064207 (2022)                     | 0     | (link)                         |                                    |
| Fate of Majorana zero modes, critical states and non-conventional real-complex transition in non-Hermitian quasiperiodic lattices            | Tong Liu; Shujie Cheng; Hao Guo et al.                      | 2020 | Phys. Rev. B 103, 104203 (2021)                     | 0     | 10.1103/PhysRevB.103.104203    |                                    |
| Two-body interaction induced phase transitions and intermediate phases in nonreciprocal non-Hermitian quasicrystals                          | Yalun Zhang; Longwen Zhou                                   | 2024 | Phys. Rev. B 111, 064204 (2025)                     | 0     | 10.1103/PhysRevB.111.064204    |                                    |
| Conduction in quasiperiodic and quasi-random lattices: Fibonacci, Riemann, and Anderson models                                               | Vipin Kerala Varma; Sebastiano Pilati; Vladimir E. Kravtsov | 2016 | Phys. Rev. B 94, 214204 (2016)                      | 0     | 10.1103/PhysRevB.94.214204     |                                    |
| Lattice Bounded Distance Equivalence for 1D Delone Sets with Finite Local Complexity                                                         | Petr Ambrož; Zuzana Masáková; Edit Pelantová                | 2020 | arXiv preprint                                      | 0     | (link)                         |                                    |
| Interlaced wire medium with quasicrystal lattice                                                                                             | Eugene A. Koreshein; Mikhail V. Rybin                       | 2022 | arXiv preprint                                      | 0     | 10.1103/PhysRevB.105.024307    |                                    |
| Local growth of icosahedral quasicrystalline tilings                                                                                         | Connor Hann; Joshua E. S. Socolar; Paul J. Steinhardt       | 2016 | Phys. Rev. B 94, 014113 (2016)                      | 0     | 10.1103/PhysRevB.94.014113     |                                    |
| Mobility rings in a non-Hermitian non-Abelian quasiperiodic lattice                                                                          | Rui-Jie Chen; Guo-Qing Zhang; Zhi Li et al.                 | 2025 | Phys. Rev. A 112, 013320 (2025)                     | 0     | 10.1103/zfrn-53lz              |                                    |
| Anyon Quasilocalization in a Quasicrystalline Toric Code                                                                                     | Soumya Sur; Mohammad Saad; Adhip Agarwala                   | 2025 | arXiv preprint                                      | 0     | (link)                         |                                    |
| Breakdown of the correspondence between the real-complex and delocalization-localization transitions in non-Hermitian quasicrystals          | Wen Chen; Shujie Cheng; Ji Lin et al.                       | 2022 | arXiv preprint                                      | 0     | 10.1103/PhysRevB.106.144208    |                                    |
| N-independent Localized Krylov Bogoliubov-de Gennes Method: Ultrafast Numerical Approach to Large-scale Inhomogeneous Superconductors        | Yuki Nagai                                                  | 2020 | J. Phys. Soc. Jpn. 89, 074703 (2020)                | 0     | 10.7566/JPSJ.89.074703         |                                    |
| Discrete quasiperiodic sets with predefined local structure                                                                                  | Nicolae Cotfas                                              | 2004 | Journal of Geometry and Physics 56 (2006) 2415-2428 | 0     | 10.1016/j.geomphys.2005.12.008 |                                    |
| Growth of a One Dimensional Quasiperiodic Covering with Locally Determined Decorations                                                       | Hyeong-Chai Jeong                                           | 2008 | arXiv preprint                                      | 0     | (link)                         |                                    |
| Local 3D real space atomic structure of the simple icosahedral Ho <sub>11</sub> Mg <sub>15</sub> Zn <sub>74</sub> quasicrystal from PDF data | S. Bruehne; E. Uhrig; C. Gross et al.                       | 2003 | Cryst. Res. Technol. 38(12) (2003) 1023-1036.       | 0     | (link)                         |                                    |
| Mobility edge and intermediate phase in one-dimensional incommensurate lattice potentials                                                    | Xiao Li; S. Das Sarma                                       | 2018 | Phys. Rev. B 101, 064203 (2020)                     | 0     | 10.1103/PhysRevB.101.064203    |                                    |
| Multifractal-enriched mobility edges and emergent quantum phases in Rydberg atomic arrays                                                    | Shan-Zhong Li; Yi-Cai Zhang; Yucheng Wang et al.            | 2025 | Sci. China-Phys. Mech. Astron. 69, 217212 (2026)    | 0     | 10.1007/s11433-025-2774-2      |                                    |
| Local Rules for Computable Planar Tilings                                                                                                    | Thomas Fernique; Mathieu Sablik                             | 2012 | EPTCS 90, 2012, pp. 133-141                         | 0     | 10.4204/EPTCS.90.11            |                                    |
| Local spectroscopy of moiré-induced electronic structure in gate-tunable twisted bilayer graphene                                            | Dillon Wong; Yang Wang; Jeil Jung et al.                    | 2015 | Phys. Rev. B 92, 155409 (2015)                      | 0     | 10.1103/PhysRevB.92.155409     |                                    |
| Topological zero-modes of the spectral localizer of trivial metals                                                                           | Selma Franca; Adolfo G. Grushin                             | 2023 | Phys. Rev. B 109, 195107 (2024)                     | 0     | 10.1103/PhysRevB.109.195107    |                                    |
| Hidden dualities in 1D quasiperiodic lattice models                                                                                          | Miguel Gonçalves; Bruno Amorim; Eduardo V. Castro et al.    | 2021 | SciPost Phys. 13, 046 (2022)                        | 0     | 10.21468/SciPostPhys.13.3.046  |                                    |

| Title                                                                                                         | Authors                                               | Year | Venue                                                                            | Cited | DOI                             | OpenAlex<br>2-yr mean<br>citedness |
|---------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|------|----------------------------------------------------------------------------------|-------|---------------------------------|------------------------------------|
| An efficient saddle search method for ordered phase transitions involving translational invariance            | Gang Cui; Kai Jiang; Tiejun Zhou                      | 2023 | arXiv preprint                                                                   | 0     | (link)                          |                                    |
| Invariable mobility edge in a quasiperiodic lattice                                                           | Tong Liu; Yufei Zhu; Shujie Cheng et al.              | 2021 | arXiv preprint                                                                   | 0     | 10.1088/1674-1056/ac140e        |                                    |
| Interplay of Quasiperiodic Criticality and the Non-Hermitian Skin Effect                                      | Zhangyuan Chen; Xianqi Tong; Xiaosen Yang             | 2026 | arXiv preprint                                                                   | 0     | (link)                          |                                    |
| Computing with almost periodic functions                                                                      | R. V. Moody; M. Nesterenko; J. Patera                 | 2008 | arXiv preprint                                                                   | 0     | (link)                          |                                    |
| Quantum criticality and Kibble-Zurek scaling in the Aubry-André-Stark model                                   | En-Wen Liang; Ling-Zhi Tang; Dan-Wei Zhang            | 2024 | Phys. Rev. B 110, 024207 (2024)                                                  | 0     | 10.1103/PhysRevB.110.024207     |                                    |
| Single-particle and many-body analyses of a quasiperiodic integrable system after a quench                    | Kai He; Lea F. Santos; Tod M. Wright et al.           | 2013 | Phys. Rev. A 87, 063637 (2013)                                                   | 0     | 10.1103/PhysRevA.87.063637      |                                    |
| Coexistence of topological Anderson insulator and multifractal critical phase in a non-Hermitian quasicrystal | Qi-Bo Zeng; Rong Lü                                   | 2026 | Phys. Rev. B 113, 224203 (2026)                                                  | 0     | 10.1103/n9rf-zk9t               |                                    |
| Quasicrystalline Spin Liquid                                                                                  | Sunghoon Kim; Mohammad Saad; Dan Mao et al.           | 2024 | Phys. Rev. B 110, 214438, 2024                                                   | 0     | 10.1103/PhysRevB.110.214438     |                                    |
| Disordered, Quasicrystalline and Crystalline Phases of Densely Packed Tetrahedra                              | Amir Haji-Akbari; Michael Engel; Aaron S. Keys et al. | 2010 | Nature 462, 773–777 (2009)                                                       | 0     | 10.1038/nature08641             |                                    |
| Photoinduced Phase Transition in Two-Band model on Penrose Tiling                                             | Ken Inayoshi; Yuta Murakami; Akihisa Koga             | 2021 | J. Phys.: Conf. Ser. 2164, 012050 (2022)                                         | 0     | 10.1088/1742-6596/2164/1/012050 |                                    |
| Quasicrystalline electronic states in $30^\circ$ rotated twisted bilayer graphene                             | Pilkyung Moon; Mikito Koshino; Young-Woo Son          | 2019 | Phys. Rev. B 99, 165430 (2019)                                                   | 0     | 10.1103/PhysRevB.99.165430      |                                    |
| Observation of subdiffusion of a disordered interacting system                                                | E. Lucioni; B. Deissler; L. Tanzi et al.              | 2010 | Phys. Rev. Lett. 106, 230403 (2011)                                              | 0     | 10.1103/PhysRevLett.106.230403  |                                    |
| Correlation function of weakly interacting bosons in a disordered lattice                                     | B. Deissler; E. Lucioni; M. Modugno et al.            | 2010 | New J. Phys. 13 (2011) 023020                                                    | 0     | 10.1088/1367-2630/13/2/023020   |                                    |
| Theory of spin Bott index for quantum spin Hall states in non-periodic systems                                | Huaqing Huang; Feng Liu                               | 2018 | Phys. Rev. B 98, 125130 (2018)                                                   | 0     | 10.1103/PhysRevB.98.125130      |                                    |
| Time-quasiperiodic topological superconductors with Majorana Multiplexing                                     | Yang Peng; Gil Refael                                 | 2018 | Phys. Rev. B 98, 220509 (2018)                                                   | 0     | 10.1103/PhysRevB.98.220509      |                                    |
| Hubbard Models for Quasicrystalline Potentials                                                                | Emmanuel Gottlob; Ulrich Schneider                    | 2022 | Phys. Rev. B 107, 144202 (2023)                                                  | 0     | 10.1103/PhysRevB.107.144202     |                                    |
| Affine Dihedral Subgroups of Higher Dimensional Cubic Lattices $\mathbb{Z}^n$ and Quasicrystallography        | Nazife Ozdes Koca; Mehmet Koca; Ramazan Koc et al.    | 2023 | Sultan Qaboos University Journal For Science 2025 : Vol. 30: Iss. 2, 102-116     | 0     | 10.53539/2414-536X.1405         |                                    |
| Magnetic states induced by electron-electron interactions in a plane quasiperiodic tiling                     | A. Jagannathan; H. J. Schulz                          | 1996 | arXiv preprint                                                                   | 0     | 10.1103/PhysRevB.55.8045        |                                    |
| Amplitude-dependent edge states and discrete breathers in nonlinear modulated phononic lattices               | Matheus I. N. Rosa; Michael J. Leamy; Massimo Ruzzene | 2022 | arXiv preprint                                                                   | 0     | (link)                          |                                    |
| Superconducting properties of Fibonacci chains with enhanced superconducting pairing at the boundaries        | Quanyong Zhu; Guo-Qiao Zha; A. A. Shanenko et al.     | 2024 | Physical Review B 112, 134503 (2025)                                             | 0     | 10.1103/j8tj-82ty               |                                    |
| Chemical localization                                                                                         | V. F. Gantmakher                                      | 2002 | UFN 172(11), 1283 (2002) [English translation in Physics-Uspekhi 45(11), (2002)] | 0     | 10.1070/PU2002v045n11ABEH001246 |                                    |
| Sub-diffusive electronic states in octagonal tiling                                                           | G. Trambly de Laisardière; C. Oguey; D. Mayou         | 2016 | Journal of Physics: Conf. Series 809 (2017) 012020                               | 0     | 10.1088/1742-6596/809/1/012020  |                                    |
| Phase transitions and bunching of correlated particles in a non-Hermitian quasicrystal                        | Stefano Longhi                                        | 2023 | Phys. Rev. B 108, 075121 (2023)                                                  | 0     | 10.1103/PhysRevB.108.075121     |                                    |
| Higher-dimensional Hofstadter butterfly on Penrose lattice                                                    | Rasoul Ghadimi; Takanori Sugimoto; Takami Tohyama     | 2022 | arXiv preprint                                                                   | 0     | 10.1103/PhysRevB.106.L201113    |                                    |
| Lyapunov exponent for pure and random Fibonacci chains                                                        | M. T. Velinho; I. R. Pimentel                         | 2000 | Phys. Rev. B 61, 1043 (2000)                                                     | 0     | 10.1103/PhysRevB.61.1043        |                                    |



| Title                                                                                                                                      | Authors                                                               | Year | Venue                                                     | Cited | DOI                              | OpenAlex<br>2-yr mean<br>citedness |
|--------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|------|-----------------------------------------------------------|-------|----------------------------------|------------------------------------|
| Exploring Aperiodic Order in Photonic Time Crystals                                                                                        | Marino Coppolaro; Massimo Moccia; Giuseppe Castaldi et al.            | 2025 | arXiv preprint                                            | 0     | (link)                           |                                    |
| Ground state of a two dimensional quasiperiodic antiferromagnet                                                                            | A. Jagannathan                                                        | 2004 | arXiv preprint                                            | 0     | 10.1103/PhysRevB.71.115101       |                                    |
| On the Spectra of Sieved Schrödinger Operators                                                                                             | Jake Fillman; Alexandro Luna                                          | 2025 | arXiv preprint                                            | 0     | (link)                           |                                    |
| Complete Structural Determination of Mesosctructural Dodecagonal Quasicrystalline Particles                                                | Xueliang Zhang; Xi Wang; Nobuhisa Fujita et al.                       | 2026 | arXiv preprint                                            | 0     | (link)                           |                                    |
| Antiferromagnetic order in the Hubbard Model on the Penrose Lattice                                                                        | Akihisa Koga; Hirokazu Tsunetsugu                                     | 2017 | Phys. Rev. B 96, 214402 (2017)                            | 0     | 10.1103/PhysRevB.96.214402       |                                    |
| Mott transition for a Lieb-Liniger gas in a shallow quasiperiodic potential: Delocalization induced by disorder                            | Hepeng Yao; Luca Tanzi; Laurent Sanchez-Palencia et al.               | 2024 | Phys. Rev. Lett. 133, 0123401 (2024)                      | 0     | 10.1103/PhysRevLett.133.123401   |                                    |
| Understanding shape entropy through local dense packing                                                                                    | Greg van Anders; Daphne Klotsa; N. Khalid Ahmed et al.                | 2013 | PNAS 111, E4812-E4821 (2014)                              | 0     | 10.1073/pnas.1418159111          |                                    |
| Approximating Metal-Insulator Transitions                                                                                                  | C. Danieli; K. Rayanov; B. Pavlov et al.                              | 2014 | arXiv preprint                                            | 0     | 10.1142/S0217979215500368        |                                    |
| Engineering bands of extended electronic states in a class of topologically disordered and quasiperiodic lattices                          | Biplab Pal; Arunava Chakrabarti                                       | 2014 | Physics Letters A, Volume 378, Issue 37, Page 2782 (2014) | 0     | 10.1016/j.physleta.2014.07.034   |                                    |
| Combined energy – diffraction data refinement of decagonal AlNiCo                                                                          | M. Mihalkovič; C. L. Henley; M. Widom                                 | 2003 | arXiv preprint                                            | 0     | 10.1016/j.jnoncrysol.2003.11.034 |                                    |
| Electronic and optical properties of strained graphene and other strained 2D materials: a review                                           | Gerardo G. Naumis; Salvador Barraza-Lopez; Maurice Oliva-Leyva et al. | 2016 | Rep. Prog. Phys. 80, 096501 (2017)                        | 0     | 10.1088/1361-6633/aa74ef         |                                    |
| Fate of Majorana zero modes by a modified real-space-Pfaffian method and mobility edges in a one-dimensional quasiperiodic lattice         | Shujie Cheng; Yufei Zhu; Gao Xianlong et al.                          | 2021 | arXiv preprint                                            | 0     | (link)                           |                                    |
| Stochastic Flips on Dimer Tilings                                                                                                          | Thomas Fernique; Damien Regnault                                      | 2011 | arXiv preprint                                            | 0     | (link)                           |                                    |
| Cleaved surface of i-AlPdMn quasicrystals: Influence of the local temperature elevation at the crack tip on the fracture surface roughness | Laurent Ponson; Daniel Bonamy; Luc Barbier                            | 2006 | Physical Review B 74, 020111 (2006/11/2006) 184205        | 0     | 10.1103/PhysRevB.74.184205       |                                    |
| Resonant metallic states in driven quasiperiodic lattices                                                                                  | L. Morales-Molina; E. Dörmner; C. Danieli et al.                      | 2014 | arXiv preprint                                            | 0     | (link)                           |                                    |
| On the spectrum of 1D quantum Ising quasicrystal                                                                                           | W. N. Yessen                                                          | 2011 | arXiv preprint                                            | 0     | (link)                           |                                    |
| Emergent periodic and quasiperiodic lattices on surfaces of synthetic Hall tori and synthetic Hall cylinders                               | Yangqian Yan; Shao-Liang Zhang; Sayan Choudhury et al.                | 2018 | Phys. Rev. Lett. 123, 0260405 (2019)                      | 0     | 10.1103/PhysRevLett.123.260405   |                                    |
| Observation of an aperiodic polariton monotile                                                                                             | Sergey Alyatkin; Yaroslav V. Kartashov; Kirill Sitnik et al.          | 2026 | arXiv preprint                                            | 0     | (link)                           |                                    |
| Unraveling Multifractality and Mobility Edges in Quasiperiodic Aubry-André-Harper Chains through High-Harmonic Generation                  | Marlena Dziurawiec; Jessica O. de Almeida; Mohit Lal Bera et al.      | 2023 | arXiv preprint                                            | 0     | 10.1103/PhysRevB.110.014209      |                                    |
| Phase Diagram of Hard Tetrahedra                                                                                                           | Amir Haji-Akbari; Michael Engel; Sharon C. Glotzer                    | 2011 | J. Chem. Phys. 135, 194101 (2011)                         | 0     | 10.1063/1.3651370                |                                    |
| Superconductivity on a Quasiperiodic Lattice: Extended-to-Localized Crossover of Cooper Pairs                                              | Shiro Sakai; Nayuta Take-mori; Akihisa Koga et al.                    | 2016 | Phys. Rev. B 95, 024509 (2017)                            | 0     | 10.1103/PhysRevB.95.024509       |                                    |
| Subdiffusion of nonlinear waves in quasiperiodic potentials                                                                                | Marco Larcher; Tetyana V. Lapyteva; Joshua D. Bodyfelt et al.         | 2012 | New Journal of Physics 14, 103036 (2012)                  | 0     | 10.1088/1367-2630/14/10/103036   |                                    |
| Diffraction and pseudospectra in non-Hermitian quasiperiodic lattices                                                                      | Ananya Ghatak; Dimitrios H. Kaltsas; Manas Kulka-rni et al.           | 2024 | arXiv preprint                                            | 0     | (link)                           |                                    |
| Discrete quasiperiodic sets with predefined covering cluster                                                                               | Nicolae Cotfas                                                        | 2005 | Philosophical Magazine 86, 895-900 (2006)                 | 0     | 10.1080/14786430500263413        |                                    |

| Title                                                                         | Authors                                                           | Year | Venue             | Cited | DOI                              | OpenAlex<br>2-yr mean<br>citedness |
|-------------------------------------------------------------------------------|-------------------------------------------------------------------|------|-------------------|-------|----------------------------------|------------------------------------|
| Localization control born of intertwined quasiperiodicity and non-Hermiticity | Jeon, Junmo; Lee, Sung-Bin                                        | 2022 | SciPost Phys.Core | 0     | 10.21468/SciPostPhysCore.6.4.077 |                                    |
| Topological Pumping over a Photonic Fibonacci Quasicrystal                    | Verbin, Mor; Zilberberg, Oded; Lahini, Yoav et al.                | 2014 | Phys.Rev.B        | 0     | 10.1103/PhysRevB.91.064201       |                                    |
| Anderson localization in Bose-Einstein condensates                            | Modugno, Giovanni                                                 | 2010 | Rept.Prog.Phys.   | 0     | 10.1088/0034-4885/73/10/102401   |                                    |
| Support Vector Machine Kernels as Quantum Propagators                         | Kuo, Nan-Hong; Wong, Renata                                       | 2025 | INSPIRE record    | 0     | (link)                           |                                    |
| Anomalous localization in a kicked quasicrystal                               | Shimasaki, Toshihiko; Prichard, Max; Kondakci, H. Esat et al.     | 2022 | Nature Phys.      | 0     | 10.1038/s41567-023-02329-4       |                                    |
| Colloquium: Synthetic quantum matter in non-standard geometries               | Grass, Tobias; Bercioux, Dario; Bhattacharya, Utso et al.         | 2024 | Rev.Mod.Phys.     | 0     | 10.1103/RevModPhys.97.011001     |                                    |
| Observing the two-dimensional Bose glass in an optical quasicrystal           | Yu, Jr-Chiun; Bhave, Shaurya; Reeve, Lee et al.                   | 2023 | Nature            | 0     | 10.1038/s41586-024-07875-2       |                                    |
| Phasonic Spectroscopy of a Quantum Gas in a Quasicrystalline Lattice          | Rajagopal, Shankari V.; Shimasaki, Toshihiko; Dotti, Peter et al. | 2019 | Phys.Rev.Lett.    | 0     | 10.1103/PhysRevLett.123.223201   |                                    |
| Non-Hermitian physics                                                         | Ashida, Yuto; Gong, Zongping; Ueda, Masahito                      | 2020 | Adv.Phys.         | 0     | 10.1080/00018732.2021.1876991    |                                    |
| Cavity QED with Quantum Gases: New Paradigms in Many-Body Physics             | Mivehvar, Farokh; Piazza, Francesco; Donner, Tobias et al.        | 2021 | Adv.Phys.         | 0     | 10.1080/00018732.2021.1969727    |                                    |

## Source overlap

| Source   | Retrieved | Found only here | Filtered venues |
|----------|-----------|-----------------|-----------------|
| openalex | 656       | 599             | 37              |
| arxiv    | 613       | 567             | 0               |
| inspire  | 17        | 10              | 0               |

'Found only here' counts unique records no other source returned – a measure of each database's marginal contribution.

## Distributions

### Publication year

| Year | Records |
|------|---------|
| 1994 | 1       |
| 1996 | 1       |
| 1997 | 1       |
| 1998 | 2       |
| 1999 | 3       |
| 2000 | 2       |
| 2001 | 3       |
| 2002 | 5       |
| 2003 | 3       |
| 2004 | 5       |
| 2005 | 4       |
| 2006 | 6       |
| 2007 | 9       |
| 2008 | 7       |
| 2009 | 8       |
| 2010 | 10      |
| 2011 | 9       |
| 2012 | 9       |
| 2013 | 9       |
| 2014 | 23      |
| 2015 | 25      |
| 2016 | 64      |
| 2017 | 56      |
| 2018 | 84      |
| 2019 | 99      |
| 2020 | 89      |
| 2021 | 79      |
| 2022 | 87      |
| 2023 | 94      |
| 2024 | 152     |

| Year | Records |
|------|---------|
| 2025 | 154     |
| 2026 | 123     |

## Top venues

| Venue                                    | Records | OpenAlex 2-yr mean citedness |
|------------------------------------------|---------|------------------------------|
| arXiv preprint                           | 354     |                              |
| arXiv (Cornell University)               | 57      |                              |
| Advanced Energy Materials                | 34      | 18.63 (2026)                 |
| Advanced Materials                       | 23      | 22.086 (2026)                |
| Advanced Functional Materials            | 19      | 14.975 (2026)                |
| ACS Applied Materials & Interfaces       | 19      | 8.382 (2026)                 |
| The Journal of Physical Chemistry C      | 16      |                              |
| Nature Energy                            | 15      |                              |
| ACS Energy Letters                       | 15      | 16.712 (2026)                |
| The Journal of Chemical Physics          | 13      |                              |
| Journal of the American Chemical Society | 12      |                              |
| Nano Energy                              | 12      |                              |
| Journal of Materials Chemistry A         | 11      |                              |
| Optics Express                           | 11      |                              |
| Science                                  | 10      |                              |

## Most frequent authors

| Author               | Records |
|----------------------|---------|
| Michaël Grätzel      | 21      |
| Marco Bernasconi     | 19      |
| Shaik M. Zakeeruddin | 15      |
| Zheyong Fan          | 14      |
| Nam-Gyu Park         | 13      |
| Omar Abou El Kheir   | 13      |
| Shu Chen             | 11      |
| Volker L. Deringer   | 10      |
| Stefano Longhi       | 10      |
| Un-Gi Jong           | 10      |
| Chol-Jun Yu          | 10      |
| Longwen Zhou         | 10      |
| Anders Hagfeldt      | 9       |
| Michael D. McGehee   | 9       |
| Henry J. Snaith      | 9       |

## Journal metrics

Metric: OpenAlex 2-yr mean citedness (openalex\_2yr), from lit/journals/metrics.json (journals.py). Values are kept per year; OpenAlex figures are snapshots taken in the fetch year (the API serves only current values); the evolution table shows every year on file for venues that appear in this record set.

## Venues in this set by OpenAlex 2-yr mean citedness

| Venue                                   | Records | OpenAlex 2-yr mean citedness | Year | Q |
|-----------------------------------------|---------|------------------------------|------|---|
| Chemical Reviews                        | 1       | 53.886                       | 2026 |   |
| Chemical Society Reviews                | 1       | 42.455                       | 2026 |   |
| Energy & Environmental Science          | 1       | 30.475                       | 2026 |   |
| Advanced Materials                      | 23      | 22.086                       | 2026 |   |
| Advanced Energy Materials               | 34      | 18.63                        | 2026 |   |
| ACS Nano                                | 3       | 16.894                       | 2026 |   |
| ACS Energy Letters                      | 15      | 16.712                       | 2026 |   |
| Angewandte Chemie International Edition | 6       | 15.006                       | 2026 |   |
| Advanced Functional Materials           | 19      | 14.975                       | 2026 |   |
| Energy Storage Materials                | 1       | 13.488                       | 2026 |   |
| Cement and Concrete Research            | 1       | 13.05                        | 2026 |   |
| Chem                                    | 2       | 11.7                         | 2026 |   |
| Applied Physics Reviews                 | 1       | 11.309                       | 2026 |   |
| Chemical Engineering Journal            | 3       | 10.953                       | 2026 |   |
| Acta Materialia                         | 3       | 10.171                       | 2026 |   |
| ACS Materials Letters                   | 1       | 9.392                        | 2026 |   |
| Artificial Intelligence Chemistry       | 1       | 9.034                        | 2026 |   |
| Advanced Science                        | 6       | 8.903                        | 2026 |   |
| Current Opinion in Chemical Engineering | 1       | 8.446                        | 2026 |   |
| ACS Applied Materials & Interfaces      | 19      | 8.382                        | 2026 |   |
| EcoMat                                  | 1       | 7.557                        | 2026 |   |
| Chemistry of Materials                  | 3       | 6.387                        | 2026 |   |
| Cell Reports Physical Science           | 1       | 5.933                        | 2026 |   |
| Chaos Solitons & Fractals               | 1       | 5.898                        | 2026 |   |

| Venue                  | Records | OpenAlex 2-yr mean citedness | Year | Q |
|------------------------|---------|------------------------------|------|---|
| Ceramics International | 1       | 5.869                        | 2026 |   |

## Filtered non-curated venues

| Venue                   | Records removed |
|-------------------------|-----------------|
| Figshare                | 15              |
| Zenodo                  | 11              |
| SSRN Electronic Journal | 9               |
| Open MIND               | 1               |
| Preprints.org           | 1               |

## Count history

Per-block totals across archived runs (counts\_history.csv); drift shows how the indexes – or your queries – changed.

| Block | 20260815T095908 | 20260828T091142 | 20260828T100120 | 20260828T160426 |
|-------|-----------------|-----------------|-----------------|-----------------|
| PSC   | -               | -               | 14,620          | 4,744           |
| MLP   | -               | 239             | 838             | 376             |
| NOV   | -               | 0               | 0               | 0               |
| QC    | -               | -               | 978             | 585             |

## Suggestions

- Block PSC: openalex 4,569 hits – a generic term is probably driving this; tighten a group or add a more specific one before reading.
- Block PSC: counts differ >20x across backends (175 to 4,569); grammar and coverage diverge, so do not compare these numbers – discover here, quote one source.
- Block NOV: zero hits on every backend. Either the intersection is genuinely empty (a finding – check the synonyms first) or one group is too narrow; try dropping one group and rerunning.
- Block QC: counts differ >20x across backends (17 to 414); grammar and coverage diverge, so do not compare these numbers – discover here, quote one source.
- 2 block/backend pair(s) hit the `--limit` cap (300); raise it (largest total 4,569) if you need the complete record set rather than the most-cited slice.
- No citation-grade backend (Scopus, Web of Science, NASA ADS) was in this search; add ADS (free token) or Scopus before quoting counts.
- Open-access status was not looked up; rerun with `--pdfs` (optionally `--pdf-blocks`) to collect legal OA PDF links via Unpaywall.
- The PRISMA flow's manual stages are empty: fill in prisma.json (screened / excluded / assessed / included) and rerun report.py to complete the diagram.
- Block PSC: total hits changed from 14,620 (run 20260828T100120) to 4,744; count drift is expected as indexes grow, but a large jump usually means the query changed – diff queries.json between the runs.
- Block MLP: total hits changed from 838 (run 20260828T100120) to 376; count drift is expected as indexes grow, but a large jump usually means the query changed – diff queries.json between the runs.
- Block QC: total hits changed from 978 (run 20260828T100120) to 585; count drift is expected as indexes grow, but a large jump usually means the query changed – diff queries.json between the runs.